

A Major Project Report on
**A Hybrid ViT with Confidence-Aware Clinical Interpretation for
Medical Imaging Diagnosis**



Submitted in partial fulfillment of the Requirements for Bachelor of Engineering in
Computer Engineering under United Technical College

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Project Work for the Degree of Bachelor's in Computer Engineering

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Supervised by Er. Prashant Poudel

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September, 2026

DEDICATION

This project is dedicated with profound gratitude to our exceptional teachers, whose unwavering guidance and mentorship have been instrumental in our successful completion of this endeavor. Through this journey, their wisdom and encouragement have served as a guiding light, inspiring us to strive for excellence.

We also extend our heartfelt appreciation to our parents, whose unwavering financial and moral support have sustained us throughout the development of this project. Their belief in our abilities, their sacrifices, and the valuable life lessons they've imparted have shown us that even the most daunting tasks can be conquered, one step at a time.

The unwavering faith of our dedicated educators and loving parents has been the foundation of our success, and for that we are ever grateful.

DECLARATION

We hereby declare that this study entitled “**A Hybrid ViT with Confidence-Aware Clinical Interpretation for Medical Imaging Diagnosis**” is based on our original work. Related works on the topic by other researchers have been duly acknowledged. We owe all the liabilities relating to the accuracy and authenticity of the data and any other information included hereunder.

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RECOMMENDATION

This is to certify that the project report entitled “**A Hybrid ViT with Confidence-Aware Clinical Interpretation for Medical Imaging Diagnosis**” has been prepared by **Aayush Sapkota, Sandip Lamsal, Suraj Jha, and Yuresh Gurung** under my continuous guidance and supervision, in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering, as prescribed by the curriculum of Pokhara University, Department of Computer Engineering, United Technical College.

Throughout the course of this project, the students have demonstrated a sound understanding of deep learning architectures, medical image analysis, and explainable artificial intelligence techniques. They have shown consistent dedication in the design, implementation, and evaluation of the Explainable Hybrid Vision Transformer (HViT) framework across seven diverse medical imaging datasets, and have honestly documented both the strengths and the current limitations of their system, including the areas identified for future improvement.

I have supervised this work from its proposal stage through to its final documentation and am satisfied that it fulfills the academic and technical requirements set forth for a major project at this level. I therefore recommend this project report to the Department of Computer Engineering for formal evaluation and final approval.

Signature:

Er. Prashant Poudel

Project Supervisor

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Date:

CERTIFICATE OF APPROVAL

This is to certify that the project report entitled "**A Hybrid ViT with Confidence-Aware Clinical Interpretation for Medical Imaging Diagnosis**", submitted by **Aayush Sapkota, Sandip Lamsal, Suraj Jha, and Yuresh Gurung** of the Department of Computer Engineering, United Technical College, has been examined by the undersigned panel of examiners. A formal presentation and viva voce examination, the panel finds the work to be of satisfactory academic and technical standard, and declares it successful in fulfilling the partial requirements for the degree of **Bachelor of Engineering in Computer Engineering**, under the affiliation of Pokhara University.

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We are also grateful to the Department of Computer Engineering and **United Technical College**, affiliated to **Pokhara University**, for providing us with the academic environment, laboratory resources, and computational support necessary to carry out this project.

We also thank our fellow classmates and peers for their constructive discussions and moral support during the course of this project, and our families for their unwavering patience, encouragement, and support throughout our academic journey.

Finally, we acknowledge that this project, while achieving strong classification performance across all seven datasets, has certain limitations, most notably in the explainability module, whose outputs were found to be inconsistent and not yet fully accurate. We view this not as a shortcoming to be hidden but as an honest finding that will guide our continued work on this system.

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ABSTRACT

The increasing availability of medical imaging data has highlighted the need for accurate and interpretable automated diagnostic systems. Traditional deep learning models, including Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), achieve high classification performance but often lack transparency, limiting their clinical adoption. This work proposes a Hybrid Vision Transformer (HViT) framework that combines CNNs for local feature extraction with ViTs for modeling global contextual relationships across multiple imaging modalities, including CT, MRI, X-ray, and microscopic images. The system employs a Confidence-Aware Clinical Interpretation (CACI) layer, which converts each raw prediction into five structured trust signals, a confidence band, a ranked differential diagnosis, a decision margin, a disease-severity band drawn from a clinical knowledge base, and a natural-language interpretation generated via a large language model surfaced through a dedicated Explainability Hub interface. CACI directly answers the questions a clinician asks of a diagnostic result: how confident is the model, what else could it be, how close was the decision, how serious is the finding, and what should happen next. A comprehensive preprocessing pipeline, including resizing, normalization, noise reduction, and data augmentation, ensures standardized input and robust learning. Experimental evaluation using standard metrics such as accuracy, precision, recall, and F1-score demonstrates that the HViT model achieves high diagnostic performance, while the CACI layer provides an accessible, structured interpretation of that performance for clinical users.

Keywords: *Hybrid Vision Transformer, Medical Image Analysis, CNN, Vision Transformer, CACI, Explainability Hub, Multi-modal Imaging, LLMs, etc.*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CACI	Confidence-Aware Clinical Interpretation
CNN	Convolutional Neural Network
CSRF	Cross- Site Request Forgery
CSS	Cascading Style Sheet
CT	Computed Tomography
DFD	Data Flow Diagram
EH	Explainability Hub
ER	Entity Relationship
FFN	Feed-Forward Network
GPU	Graphics Processing Unit
Grad-CAM++	Gradient-Weighted Class Activation Mapping
HTML	Hypertext Mark-up Language
H-Vit	Hybrid Vision Transformer
LLMs	Large Language Models
MHSA	Multi-Head Self Attention
MRI	Magnetic Resonance Imaging
SQL	Structured Query Language
Vit	Vision Transformer
XSS	Cross Site Scripting

CHAPTER 1: INTRODUCTION

Medical imaging constitutes a critical component of modern healthcare systems, providing indispensable tools for the early identification, diagnosis, and longitudinal monitoring of diseases such as cancer, pneumonia, and neurological disorders (Litjens, et al., 2017) (Ronneberger, et al., 2015). With the rapid growth of medical image data, Artificial Intelligence (AI) has become essential in assisting clinicians with faster and more accurate decision-making (Dosovitskiy, et al., 2021). Convolutional Neural Networks (CNNs) have been widely used for medical image analysis due to their ability to extract local spatial features, but they often struggle to capture long-range dependencies and global contextual relationships within complex images (Gonzalez & Woods, 2018). Vision Transformers (ViTs) have emerged as a powerful alternative, leveraging self-attention mechanisms to model global interactions and achieve competitive or superior performance compared to CNNs in tasks like classification, segmentation, and disease detection (He, et al., 2015) (Lindeberg, 1994).

Despite their accuracy, ViTs often operate as black boxes, providing little insight into why a particular decision was made, which limits their adoption in clinical practice (Perez & Wang, 2017) (Srinivas, et al., 2021). Interpretability techniques have therefore been developed to help clarify model predictions and enhance clinician trust, ranging from visual attribution methods to structured, confidence-based interpretation layers. The goal is to deliver not only accurate predictions but also meaningful, trustworthy explanations, making the system more transparent, reliable, and suitable for real-world medical applications (Vaswani, et al., 2017) (Selvaraju, et al., 2017).

1.1 Background

Deep learning has significantly advanced medical image analysis by enabling automatic detection and classification of diseases from imaging modalities such as X-ray, CT, and MRI. Convolutional Neural Networks (CNNs) have traditionally dominated this field due to their ability to capture local spatial features effectively, achieving high performance in tasks such as tumor detection and pneumonia classification (Gonzalez & Woods, 2018). However, CNNs often struggle to model long-range dependencies and global contextual relationships within images, which are critical for complex medical patterns (Litjens, et al., 2017) (Lindeberg, 1994).

To overcome these limitations, Vision Transformers (ViTs) were introduced as an alternative architecture capable of capturing global image context through self-attention mechanisms. ViTs and hybrid CNN-Transformer models such as TransUNet and hybrid ViT-CNN frameworks have demonstrated strong performance in medical imaging tasks including segmentation and classification (Srinivas, et al., 2021) (Vaswani, et al., 2017). Despite their improved accuracy, transformer-based models remain largely opaque, making it difficult for clinicians to understand the reasoning behind automated predictions and limiting their adoption in clinical environments (Abnar & Zuidema, 2020) (Chung, et al., 2025).

Interpretable AI techniques have therefore become essential for medical applications, whether delivered as visual attribution or as structured, confidence-based interpretation. Recent studies show that combining interpretability with transformer architectures improves trust and clinical reliability in diagnostic systems. Consequently, hybrid explainable Vision Transformer frameworks have emerged as a promising solution, integrating local feature learning, global context modeling, and interpretable outputs to support transparent and accurate medical diagnosis (Fu, et al., 2025).

1.2 Problem Statement

Although deep learning models, particularly Vision Transformers, have achieved high accuracy in medical image diagnosis, their decisions remain largely opaque and difficult to interpret. These complex architectures function as "black boxes," producing accurate predictions without revealing the clinical reasoning behind them, leaving clinicians unable to verify whether decisions are based on relevant pathological features or misleading patterns. This lack of explainability poses a critical barrier in healthcare, where medical professionals bear ethical and legal responsibility for diagnostic outcomes and cannot trust systems they do not understand. Radiologists and pathologists need transparency to confirm that model predictions align with established medical indicators, yet current high-performing models provide little insight into their decision-making processes. The inability to explain predictions undermines clinical trust, as even highly accurate systems are often rejected by practitioners who fear unpredictable failures or hidden biases that could lead to misdiagnosis. This trust deficit is reinforced by regulatory requirements, as health authorities demand explainability and accountability for any AI system deployed in patient care, making opaque models difficult to approve for clinical use. Without interpretability, automated

diagnostic systems cannot be effectively validated against expert knowledge, restricting their role to experimental tools rather than reliable decision-support partners in real-world healthcare. There is therefore a pressing need for a hybrid and explainable diagnostic framework that combines the predictive power of advanced deep learning models with meaningful, clinically interpretable visual explanations, enabling clinicians to understand, verify, and trust automated predictions while maintaining high diagnostic performance and ensuring patient safety (Lai, 2024) (Fu, et al., 2025).

1.3 Objectives

The primary objective of the proposed system is to assist clinicians in accurate and interpretable diagnosis of diseases by developing an Explainable Hybrid Vision Transformer (HViT) for multi-modal medical images. Specific goals include:

1. To develop a Hybrid Vision Transformer (HViT) for automated disease classification from multi-modal medical images (CT, MRI, X-ray, Microscopy).
2. To integrate CNNs for local feature extraction and Vision Transformers for global context modeling to improve diagnostic accuracy.
3. To design and implement a Confidence-Aware Clinical Interpretation (CACI) layer, comprising confidence banding, ranked differential diagnosis, decision-margin scoring, disease-severity classification, and an LLM-generated clinical narrative, surfaced through a dedicated Explainability Hub.
4. To evaluate model performance using standard metrics and explainability outputs to ensure reliability for clinical use.

1.4 Motivation and Significance

Motivation Despite significant advancements in deep learning for medical image analysis, the adoption of AI systems in clinical practice remains limited due to a lack of transparency and interpretability (Litjens, et al., 2017). Most high-performing models, including CNNs and Vision Transformers, function as black-box systems, making it difficult for clinicians to understand the reasoning behind automated diagnoses. This motivated the adoption of a Confidence-Aware Clinical Interpretation (CACI) approach: rather than requiring clinicians to visually interpret a model output, the system pairs each diagnosis with structured trust signals and a natural-language explanation, providing a more directly usable form of interpretability

The proposed HViT framework, paired with the CACI interpretation layer, holds significant value for both research and clinical applications. By presenting confidence, differential diagnosis, decision margin, and severity information alongside a plain-language explanation, the system enhances transparency and helps clinicians validate model predictions without requiring visual interpretation of attribution maps, increasing trust in AI-assisted decision-making. This interpretability is crucial for clinical adoption, regulatory approval, and ethical deployment of AI in healthcare (Chen & Lu, 2021) (Selvaraju, et al., 2024).

1.5 Scope of the Project

The scope of this project focuses on the design, implementation, and evaluation of a Hybrid Vision Transformer framework for medical image analysis, paired with a Confidence-Aware Clinical Interpretation (CACI) layer. The system is intended to process medical images such as X-ray, CT, or MRI scans for disease classification tasks. The project includes data preprocessing, model training, and performance evaluation using standard metrics such as accuracy, precision, recall and F1-score. The emphasis is placed on achieving a balance between high diagnostic accuracy and clinically usable interpretability (Barekatin & Glocker, 2025).

In addition to prediction performance, the project scope includes a dedicated Explainability Hub that surfaces confidence bands, ranked differentials, decision margins, severity bands, and LLM-generated clinical narratives for each diagnosis, supporting transparency and trust in automated diagnosis. The proposed framework is designed as a research-oriented prototype that can be extended to multiple imaging modalities and disease detection tasks in future work (Abnar & Zuidema, 2020) (Lai, 2024).

1.6 Limitations of the Study

Several limitations constrain the clinical readiness and generalizability of the proposed system:

1.6.1 No Cross-Validation

Performance metrics are based on a single train/test split per dataset, lacking variance estimates and statistical reliability. Without K-fold cross-validation, the study cannot provide confidence intervals or standard deviations for the reported metrics.

1.6.2 Class Imbalance Issues

Standard categorical cross-entropy loss fails to handle severe class imbalance (up to 58.3×),

leading to poor minority-class performance, particularly in FracAtlas (recall 0.29 for fractures), Diabetic Retinopathy, and HAM10000 (precision 0.55 for melanoma). This makes the system unsuitable for unsupervised clinical deployment on imbalanced datasets.

1.6.3 Clinical Validation of CACI Layer

The Confidence-Aware Clinical Interpretation (CACI) layer implemented in this project produces structured, reproducible clinical narratives across all seven datasets. However, the current implementation has not yet undergone formal clinician validation. A structured reader study with radiologists, evaluating CACI outputs on clinical relevance, interpretability, and actionable clarity using Likert-scale ratings, remains outstanding and is described in Section 6.2.4. Until this validation is completed, CACI outputs are treated as a structured decision-support aid rather than a replacement for clinical judgment. Additionally, CACI's narrative quality is bounded by the underlying classifier's calibration; miscalibrated probabilities on minority classes (e.g., FracAtlas fractures, HAM10000 melanoma) propagate directly into the interpretation quality.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Existing Systems

Existing medical image analysis systems widely use AI and deep learning models to automate disease detection and diagnosis from images such as X-rays, CT scans, and MRI. Convolutional Neural Networks (CNNs) have been the most commonly used models due to their effectiveness in learning local spatial features and achieving high accuracy. However, CNNs have limitations in capturing global context. To overcome this, recent systems adopt Vision Transformers (ViTs) and hybrid CNN–Transformer models, which use self-attention mechanisms to improve global feature representation and overall diagnostic performance. These hybrid approaches combine the local features extraction strength of CNNs with the long-range dependency modeling capability of Transformers, leading to more robust and reliable predictions. Additionally, these advanced models are increasingly being integrated into real-world clinical workflows to assist healthcare professionals in making faster and more accurate decisions. The ongoing evolution of these deep learning techniques continues to push the boundaries of automated medical diagnosis, offering promising avenues for improving healthcare delivery and patient outcomes.

2.1.1 Evaluating the Explainability of Vision Transformers in Medical Imaging

This study focused on understanding how Vision Transformers make decisions when applied to medical imaging datasets such as MRI and chest X-rays. Their research aimed to bridge the gap between high predictive performance and clinical interpretability. They proposed an attention regularization technique, which encourages the transformer to focus on anatomically relevant regions when making predictions. This approach generates attention maps that highlight areas in the image most influential to the model's decision, allowing clinicians to validate model reasoning against known pathological features. Experiments demonstrated that ViTs could focus on global structures effectively but might miss fine-grained details without a CNN backbone. While the study improved interpretability using attention regularization, the reliance solely on ViTs limited the model's ability to capture subtle local features such as small lesions, microcalcifications, or thin tissue structures, which are crucial in many medical diagnoses (Barekatain & Glocker, 2025).

2.1.2 Evaluating Visual Explanations of Attention Maps for Transformer-Based Medical Imaging

This research presents a hybrid explainability framework that combines Vision Transformer attention maps with gradient-based visualization techniques such as Grad-CAM to enhance the reliability and clinical relevance of model explanations in medical image classification tasks. By quantitatively comparing explanation outputs with expert-annotated regions of interest, the study demonstrates that integrating gradient information improves the localization and trustworthiness of visual explanations compared to attention maps alone. The findings highlight that transformer attention is effective for capturing global context but insufficient for accurately representing localized causal features, making hybrid interpretability approaches more effective. Nevertheless, the study is limited to classification tasks and does not evaluate its framework for segmentation, detection, or multi-modal medical imaging, reducing its applicability to complex real-world clinical scenarios involving multiple organs, diseases, or imaging modalities (Chung, et al., 2025).

2.1.3 Enhanced Tuberculosis Detection Using Vision Transformers & Grad-CAM

This paper proposes a Vision Transformer-based system for automated tuberculosis detection from chest X-ray images, emphasizing both high classification accuracy and model explainability. The ViT model leverages long-range contextual learning to capture diffuse infection patterns across lung regions, outperforming conventional CNN-based models in detecting widespread disease manifestations. To improve interpretability, Grad-CAM is applied to generate heatmaps that visually highlight infected lung areas, enabling clinicians to verify whether predictions align with medically relevant regions. While the approach demonstrates strong performance for tuberculosis screening, it suffers from limited generalizability due to its focus on a single disease and imaging modality. Furthermore, the absence of a CNN backbone restricts the model’s ability to detect small or subtle lesions, and the reliance on Grad-CAM alone provides limited insight into transformer-specific attention behavior (Vanitha, et al., 2025).

2.1.4 Hybrid CNN–ViT Lung Disease Detection

This study introduces a hybrid CNN–Vision Transformer architecture for lung disease classification using CT scans, combining CNN-based local feature extraction with transformer-based global contextual modeling. The CNN component captures fine-grained pathological details such as nodules and small lesions, while the Vision Transformer models long-range spatial relationships across the lung structure, resulting in improved multi-class classification performance compared to standalone CNN or ViT models. The authors incorporate Grad-CAM to

provide visual interpretability, enabling clinicians to understand which lung regions influenced the model’s decisions. However, the explainability framework is limited to gradient-based methods and does not analyze transformer attention maps, restricting deeper insight into model reasoning. Additionally, the model is evaluated only on lung CT datasets, limiting its scalability to multi-modal imaging or multi-organ diagnostic applications (Fu, et al., 2025).

2.1.5 CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels

CheXpert introduces a large-scale chest X-ray dataset containing over 224,000 labeled images across multiple thoracic disease categories, designed to support the development and benchmarking of deep learning models for radiological diagnosis. A major contribution of the dataset is the inclusion of uncertainty labels, which account for ambiguous or inconclusive radiology reports and enable researchers to design more robust training strategies under imperfect labeling conditions. The dataset has become a widely used benchmark for CNN- and Vision Transformer-based chest disease classification. However, CheXpert is limited to a single imaging modality (chest X-rays), lacks spatial lesion annotations required for evaluating explainability methods, and does not support multi-modal research involving CT or MRI, thereby constraining its usefulness for advanced explainable AI and multi-organ diagnostic studies (Irvin, et al., 2019).

2.1.6 TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation

TransUNet proposes a hybrid medical image segmentation framework that integrates convolutional neural networks with a Transformer encoder to effectively capture both fine-grained local textures and long-range global contextual information in CT and MRI images. By extending the U-Net architecture, the model achieves improved segmentation accuracy for complex anatomical structures, demonstrating superior performance in Dice score and boundary delineation compared to traditional CNN-based segmentation methods. The integration of Transformers enables better modeling of spatial dependencies across entire anatomical regions, enhancing robustness in challenging segmentation tasks. However, the architecture is primarily designed for segmentation rather than disease classification, lacks built-in explainability mechanisms such as Grad-CAM or attention visualization, and requires high computational and memory resources, which may limit its practicality for real-time or resource-constrained clinical deployment (Chen, et al., 2024).

Table 2.1 Comparison Table

Title	Key Features	Strengths	Weaknesses / Drawbacks
Evaluating the Explainability of ViTs in Medical Imaging (Barekatain & Glocker, 2025)	ViT + attention regularization, attention heatmaps (MRI, chest X-ray)	Improves interpretability; captures global anatomical patterns; clinically relevant attention	No CNN backbone; misses fine-grained lesions; attention $\neq$ true causal reasoning
Evaluating Visual Explanations of Attention Maps (Chung, et al., 2025)	ViT attention + Grad-CAM hybrid, ROI-based evaluation	More reliable, clinically meaningful, quantitatively validated explanations	Classification-only; no segmentation/detection; no multi-modal support
Enhanced TB Detection Using ViT & Grad-CAM (Vanitha, et al., 2025)	ViT + Grad-CAM, TB-specific chest X-ray classification	High accuracy; strong global context modeling; interpretable outputs	Single disease/modality; no CNN backbone; weak on subtle lesions; poor generalization
Hybrid CNN–ViT Lung Disease Detection (Fu, et al., 2025)	CNN (local) + ViT (global) + Grad-CAM, lung CT classification	Outperforms standalone CNN/ViT; strong local+global feature extraction	limited to lung CT; not multi-modal
CheXpert (Irvin, et al., 2019)	224,000 chest X-rays, multi-label, uncertainty-aware labeling	Large-scale benchmark; realistic clinical uncertainty modeling; widely adopted	Single modality (X-ray only); no lesion annotations

TransUNet (Chen & Lu, 2021)	CNN encoder + Transformer + U-Net backbone, CT/MRI segmentation	High segmentation accuracy; robust local+global anatomical modeling	Segmentation-only, not classification; no explainability hub; high compute cost
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2.2 Gaps in Existing Systems

While recent research has demonstrated promising advancements in Vision Transformer–based medical imaging and explainable AI, existing systems still suffer from several critical limitations that restrict their clinical applicability and reliability.

2.2.1 Limited Multi-Modal Data Utilization

Most existing systems focus on a single imaging modality. For example, TB detection models typically rely only on chest X-rays (Vanitha, et al., 2025), while datasets like CheXpert provide only X-ray images (Irvin, et al., 2019). Such single-modality reliance limits generalization to multi-modal clinical environments, where MRI, CT, and microscopic imagery are often required for comprehensive diagnostics. Models trained on one modality may fail to capture complementary anatomical or pathological information available in other modalities, restricting their practical deployment in hospitals with diverse imaging equipment.

2.2.2 Architectural Limitations in Feature Extraction

Vision Transformers excel at capturing global structural context but often underperform when detecting fine-grained local features such as micro-lesions, microcalcifications, or thin tissue structures (Barekatain & Glocker, 2025). This is because pure ViTs lack convolutional inductive biases that CNNs provide, which are critical for local texture and edge detection in medical images. As a result, subtle but clinically significant abnormalities may be missed, reducing diagnostic reliability, especially in early-stage disease detection.

2.2.3 Scalability and Computational Constraints

Transformer-based architectures like TransUNet (Chen & Lu, 2021) require high computational resources for training and inference. Large memory consumption and the need for GPUs limit deployment in resource-constrained clinical settings, such as rural hospitals or

smaller diagnostic centers. These systems may also have slower inference times, making them unsuitable for real-time or emergency diagnostics, which are critical in clinical workflows.

2.2.4 Underutilization of Structured, Multi-Signal Interpretability

Most existing systems rely on a single-method visual interpretability technique, such as Grad-CAM or raw attention visualization (Chung, et al., 2025), or provide no structured interpretation at all beyond a class label and a confidence score. Visual attribution methods, even when technically functional, are often incomplete, coarse, and difficult for a non-specialist to independently verify. This gap motivates an interpretation approach built around multiple structured, clinically meaningful signals, confidence, differential diagnosis, decision margin, and severity, rather than a single visual attribution method.

2.2.5 Limited Disease and Organ Coverage

Many systems target single-disease classification, e.g., tuberculosis (Ali, et al., 2025), or focus on a single organ such as lungs (Irvin, et al., 2019). This limitation prevents these models from being applied to multi-organ or multi-disease diagnostic workflows. In real-world clinical practice, patients often have complex, multi-system conditions, so models must generalize across multiple pathologies to be truly useful.

2.3 Significance of Proposed System

The proposed Explainable Hybrid Vision Transformer (HViT) framework is designed to overcome the identified gaps and enhance the performance, interpretability, and clinical usability of automated medical imaging systems through the following contributions.

2.3.1 Enhanced Diagnostic Accuracy

The proposed HViT model integrates CNN-based local feature extraction with Vision Transformer global context modeling, combining the strengths of both architectures. This allows detection of both subtle local lesions and large-scale anatomical patterns, which improves diagnostic accuracy over standalone CNNs or ViTs (Barekatain & Glocker, 2025); (Chen, et al., 2024). In practice, this means earlier and more reliable detection of diseases, which is critical for timely clinical intervention.

2.3.2 Multi-Modal Imaging Support

HViT is designed to process multi-modal inputs, including MRI, CT, X-ray, and microscopic imagery. This enables the model to generalize across different imaging types and provide a more comprehensive view of patient health (Irvin, et al., 2019). Multi-modal capability ensures that complementary information from different imaging techniques can improve diagnostic decisions, which is particularly valuable in complex cases.

2.3.3 Scalable and Computationally Efficient Design

Despite using transformers, HViT incorporates model optimization techniques such as lightweight CNN stems and patch embedding reduction. This decreases memory usage and computational load, allowing deployment in resource-limited clinical settings while maintaining high performance (Chen, et al., 2024). Efficient design ensures feasibility for real-time diagnostics in hospitals that lack extensive GPU infrastructure.

2.3.4 Improved Interpretability and Clinical Trust through CACI

The system combines five structured trust signals, a confidence band, a ranked differential diagnosis, a decision margin, a disease-severity band, and a six-section natural-language clinical interpretation generated via a large language model, surfaced through a dedicated Explainability Hub. This provides clinicians with a direct answer to the questions they actually ask of a diagnostic result (how confident is the model, what else could it be, how close was the decision, how serious is it, and what should happen next), rather than requiring visual interpretation of an attribution map. Improved, directly usable interpretability enhances clinician confidence and adoption, a critical factor for integrating AI into routine practice.

2.3.5 Foundation for Future Medical AI Expansion

The HViT framework provides a scalable and extensible platform for integrating additional medical AI tasks, such as segmentation, detection, and multi-task learning (Chen, et al., 2024). Its modular architecture enables researchers to extend the system for new imaging modalities, diseases, or real-time clinical decision support, ensuring long-term relevance in evolving medical AI applications.

2.4 HViT instead of ViT

In summary, we chose the Hybrid Vision Transformer (HViT) over the standard ViT approach because HViT demonstrated superior performance, robustness, and reliability in medical

image analysis. Standard ViTs, while powerful in capturing global dependencies through self-attention mechanisms, often require very large datasets to achieve meaningful feature representations and can struggle to capture fine-grained local details, such as small lesions, nodules, microcalcifications, or subtle tissue textures, which are crucial in medical diagnostics. In contrast, HViT integrates CNN-based local feature extraction with transformer-based global contextual modeling, enabling the model to effectively learn both local and global features simultaneously. This hybrid learning allows the model to handle the variations and heterogeneity inherent in medical images, such as differences in imaging modalities, resolutions, and patient anatomy, more robustly than ViTs alone. As a result, HViT demonstrates reduced prediction errors, improved segmentation and classification accuracy, and enhanced model stability, even when trained on relatively smaller datasets. Moreover, the hybrid approach enhances interpretability, as combining CNNs with ViTs provides richer feature representations that support both local and global clinical reasoning in the CACI interpretation layer. By comparison, pure ViTs under similar conditions often exhibit inconsistent performance across diverse imaging modalities and are less effective at capturing subtle anatomical details. Therefore, HViT was selected as the most efficient, accurate, and clinically viable approach for our final model, balancing predictive power, interpretability, and adaptability to real-world medical imaging tasks.

CHAPTER 3: METHODOLOGY

The implemented system analyzes medical images across multiple modalities, including MRI, X-ray, CT, and microscopic images (Gonzalez & Woods, 2018) (Lindeberg, 1994). The system performs automated disease classification and delivers a structured, clinically grounded explanation of each prediction through a Confidence-Aware Clinical Interpretation (CACI) layer. The CACI layer produces five orthogonal trust signals for every prediction: prediction confidence band, ranked differential diagnosis, decision margin, clinical severity band, and an LLM-generated clinical interpretation. These signals are presented through a dedicated Explainability Hub interface, which enhances clinical trust and improves the interpretability of model predictions. The methodology follows a structured pipeline involving data collection, data preprocessing, model training, inference, CACI signal generation, and Explainability Hub rendering of results (Selvaraju, et al., 2017) (Abnar & Zuidema, 2020) (Frasca, et al., 2024).

The integration of convolutional neural networks (CNNs) with transformer encoders allows the system to effectively capture local spatial features while modeling long-range dependencies across the image, which is essential for complex medical imaging modalities. Furthermore, the incorporation of the CACI layer, which grounds interpretation in the classifier's softmax distribution and a curated disease knowledge base, ensures transparency, reproducibility, and clinical interpretability of the model's predictions (Dosovitskiy, et al., 2021) (Srinivas, et al., 2021) (Ahmed, et al., 2026).

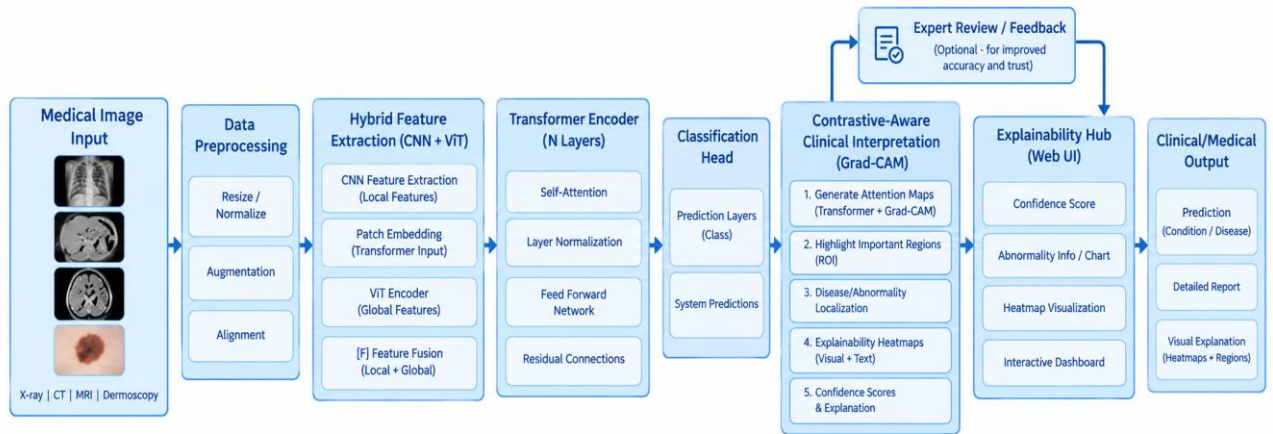


Figure 3.1: System Architecture of HViT for Medical Imaging Diagnosis System

3.1 Data Collection

The completed system was trained and evaluated using seven labeled, publicly available medical imaging datasets, together comprising 45,594 images across 7 diseases and covering four imaging modalities: MRI, X-ray, CT-Scan and microscopic photography (Ahmed & Syed, 2024). This dataset selection was deliberately made broader than a single-modality benchmark, in direct response to the multi-modal coverage gap identified in Section 2.2.1, and enables the development of an explainable medical imaging system that highlights clinically relevant regions for each diagnosis.

Table 3.1: Data Collection

Dataset	Modality	Total Images	Classes
Brain Tumor	MRI	7,200	4
FracAtlas	X-Ray/ CT-Scan	4,083	2
TB Chest X-Ray	X-Ray/ CT-Scan	4,200	2
Pneumonia Chest X-Ray	X-Ray/CT-Scan	5,856	2
Diabetic Retinopathy	Microscopic	3,662	5
Bone Fracture	X-Ray/ CT-Scan	10,578	2
HAM10000 Skin Cancer	Microscopic	10,015	7

3.1.1 CT scans

It is widely used for detecting lung diseases, tumors, and internal organ abnormalities. It provides both bone and soft tissue contrast which is essential for accurate diagnosis. In this project, CT supports the FracAtlas and TB chest X-Ray datasets.

3.1.2 MRI Images

MRI images are ideal for brain tumors detection, neurological disorders analysis and soft tissue analysis which provide rich structural information for model learning. In this project, MRI drives the Brain Tumor dataset with four balanced classes.

3.1.3 X-ray Images

It is widely used for detecting fractures, chest abnormalities, and tuberculosis, and remains the first-line imaging tool in emergency and primary care settings. In this project, X-ray supports the FracAtlas, TB Chest X-Ray, and Pneumonia Chest X-Ray datasets.

3.1.4 Microscopic Images

It is used for cancer cell detection, malaria diagnosis, and histopathology analysis, and is central to dermatology and oncology diagnostics. In this project, microscopic imaging drives the Diabetic Retinopathy and HAM10000 Skin Cancer datasets.

3.2 Data Preprocessing

Data preprocessing is a crucial step in medical image analysis, as raw medical images are inherently heterogeneous, noisy, and inconsistent. Variations in acquisition protocols, scanner settings, patient positioning, and illumination conditions can significantly affect the learning capability of deep learning models. Without proper preprocessing, these variations may lead to biased learning, slow convergence, and poor generalization. In the proposed HViT system, a comprehensive preprocessing pipeline is designed to standardize images across modalities, enhance clinically relevant features, and prepare the data in a form that is optimally suited for transformer-based architectures (Litjens, et al., 2017) (Perez & Wang, 2017).

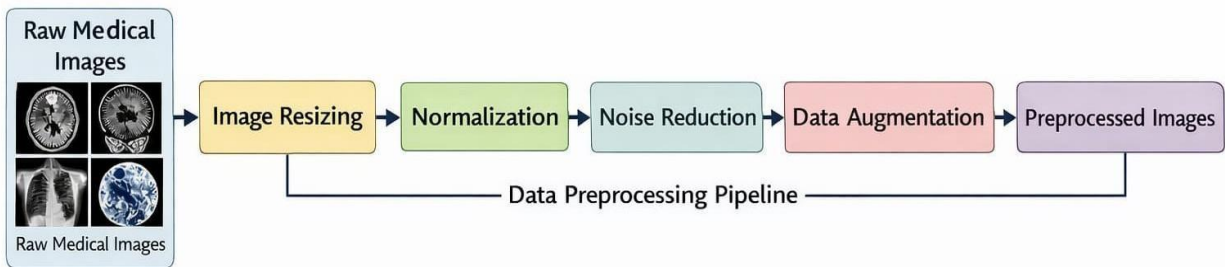


Figure 3.2: Data Preprocessing pipeline

3.2.1 Image Resizing

Medical images in the dataset vary in resolution. All medical images are resized to a fixed spatial resolution (e.g., 224*224) to maintain uniformity across different imaging modalities such as CT, MRI, X-ray, and microscopic images. Since deep learning models require consistent input dimensions, resizing enables batch processing and ensures compatibility with the CNN backbone and patch embedding layers of the Hybrid ViT model (Litjens, et al., 2017).

3.2.2 Normalization

Intensity normalization is applied to reduce variations in pixel value distributions caused by differences in imaging devices, acquisition settings, and patient conditions. Without normalization, large intensity variations can lead to unstable gradients and slow convergence during training. Pixel values are scaled to a fixed range, typically between 0 and 1, using min–max normalization (Gonzalez & Woods, 2018).

3.2.3 Noise Reduction and Smoothing

Medical images often contain noise and acquisition artifacts that can obscure clinically relevant features. Noise reduction techniques are applied to enhance image quality while preserving important anatomical structures. Smoothing filters, such as Gaussian filtering, help suppress high-frequency noise and improve the clarity of edges and textures. This step is particularly important for modalities such as CT and MRI, where scanner noise can affect feature extraction by convolutional layers (Litjens, et al., 2017) (Gonzalez & Woods, 2018).

3.2.4 Data Augmentation

To improve model generalization and reduce overfitting, data augmentation techniques are applied to the training dataset. Augmentation artificially increases data diversity by generating transformed versions of existing images, including rotations, flipping, scaling, and minor intensity variations (Litjens, et al., 2017) (Dosovitskiy, et al., 2021) (Frasca, et al., 2024). These transformations simulate real-world variations in patient positioning and imaging conditions while preserving pathological characteristics. Data augmentation enables the HViT model to learn robust and invariant feature representations (Perez & Wang, 2017).

3.3 Hybrid ViT architecture

The Hybrid Vision Transformer (Hybrid ViT) architecture integrates convolutional neural network (CNN) feature extraction with transformer-based global attention mechanisms to efficiently analyze medical images. This hybrid design leverages the strengths of both CNNs and transformers, enabling the system to capture local texture details while also learning long-range dependencies across the image (Ali, et al., 2025).

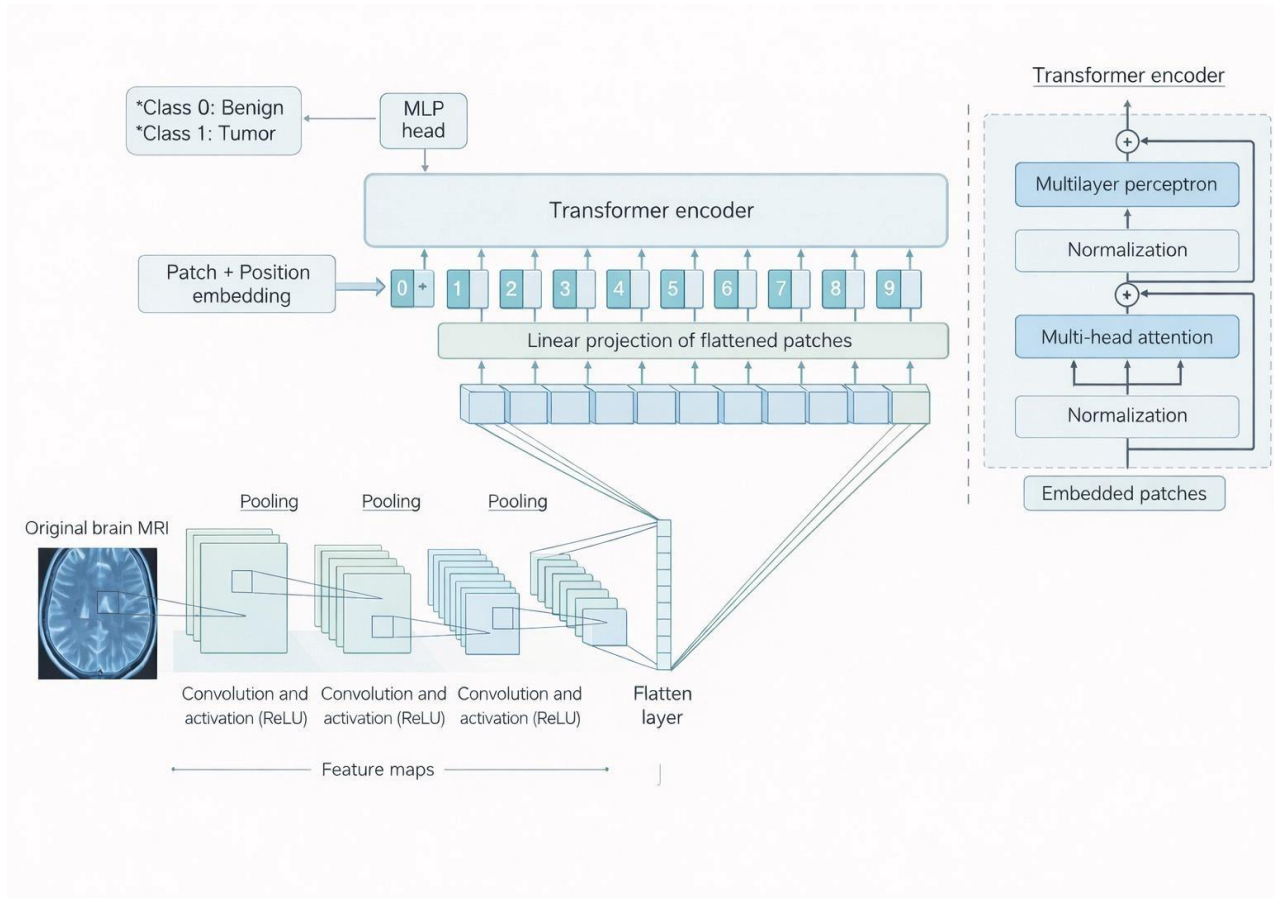


Figure 3.3: Hybrid ViT architecture

3.3.1 CNN Feature Extraction

The preprocessed image is then passed through the CNN feature extraction module. This module consists of several convolutional layers followed by pooling layers. The CNN feature extractor is responsible for capturing local spatial features in the medical images (Srinivas, et al., 2021) (Goodfellow, et al., 2016). These features include fine-grained anatomical structures, edges, and textures that are often crucial for disease detection. Convolutional layers apply filters to the input images to generate feature maps, while non-linear activation functions such as ReLU are applied to introduce non-linearity (Goodfellow, et al., 2016).

$$\text{ReLU}(x) = \max(0, x)$$

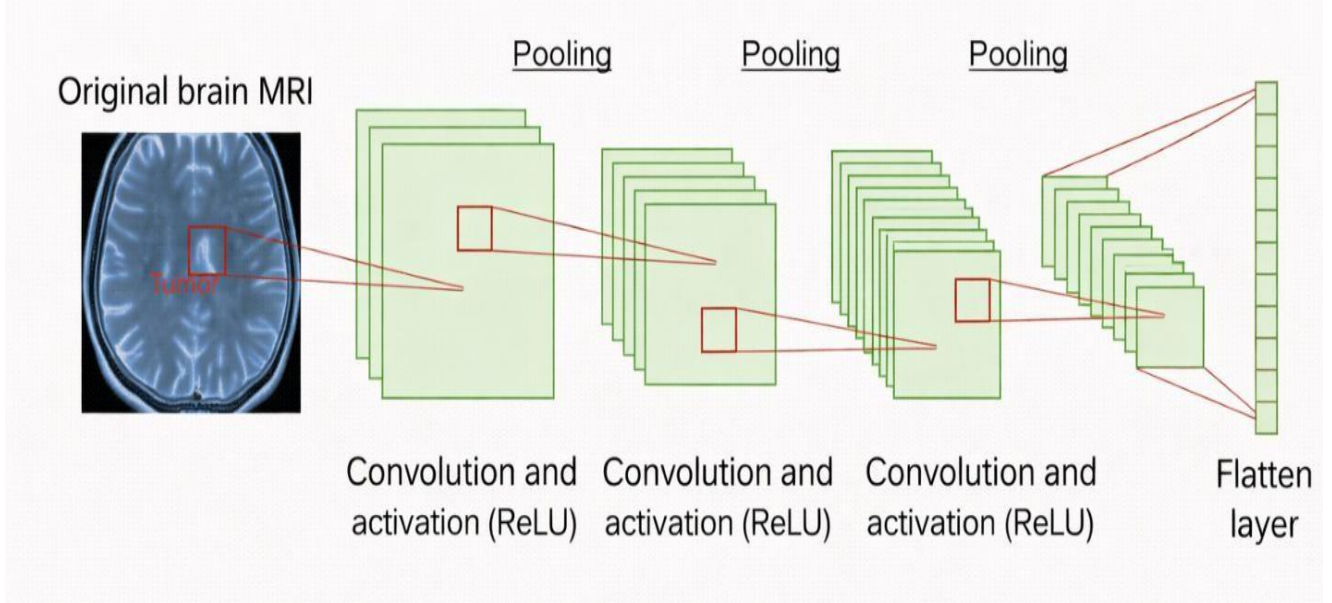


Figure 3.4: CNN Feature Extraction

Pooling operations reduce spatial dimensions while retaining salient features, allowing the CNN to generate compact and discriminative representations. These hierarchical feature maps encode essential local information and serve as a strong foundation for subsequent transformer-based global reasoning (Ahmed, et al., 2026).

3.3.2 Patch Tokenization and Linear Projection

After local feature extraction by the CNN, the resulting feature maps are transformed into a sequence of fixed-size patches through the patch embedding process. Instead of directly splitting raw images, patch embedding is applied to CNN-generated feature maps, which improves robustness and reduces noise sensitivity (Dosovitskiy, et al., 2021). Each feature map is divided into non-overlapping patches, flattened, and projected into a fixed-dimensional embedding space using a learnable linear transformation. This transformation converts spatial feature maps into a sequence of vectors, enabling the transformer to process image information as a tokenized representation while preserving semantic richness (Srinivas, et al., 2021).

After feature extraction and patch tokenization, the Hybrid Vision Transformer prepares the input tokens for transformer processing using the Linear Projection and Embedding Layer. This layer plays a critical role in ensuring compatibility between convolutional features and the transformer encoder. Each token obtained from the CNN feature map represents local visual information but

may vary in dimensionality depending on the backbone network. The linear projection layer transforms every token into a fixed-length embedding vector (Dosovitskiy, et al., 2021) (Srinivas, et al., 2021). This ensures that all tokens share a uniform representation size, which is required for transformer-based self-attention operations.

Let each patch token be represented as:

$$X_p \in R^{p^{2 \times D}}$$

Where:

P is the patch size.

D is the depth of the CNN feature map.

Each patch token is projected into a d-dimensional embedding space using a learnable linear projection matrix:

$$Z_p = X_p W_E + b_E$$

Where:

$W_E \in R^{p^{2 \times D}}$ is the embedding

matrix, b_E is the bias vector,

$Z_p \in R^d$ is the embedded token.

This projection ensures that all tokens share a common embedding dimension d, which is essential for efficient attention computation (Dosovitskiy, et al., 2021) (Srinivas, et al., 2021).

To enable global image-level representation, a learnable classification token ([CLS]) is appended to the token sequence. Unlike patch tokens that represent local regions, the [CLS] token is designed to capture holistic information about the entire image (Goodfellow, et al., 2016). During self-attention, the [CLS] token interacts with all other tokens and aggregates contextual information across the image. The final embedding of this token serves as a compact summary of the image and is commonly used for classification and decision-making tasks (Vaswani, et al., 2017). Once projected, all patch tokens are concatenated with the [CLS] token to form a single sequence. This sequence becomes the input to the transformer encoder, allowing both local and global features to be processed jointly (Dosovitskiy, et al., 2021) (Srinivas, et al., 2021).

3.3.3 Positional Embedding

Positional embeddings are introduced to prevent the spatial arrangement of patches within the image. Each patch embedding is augmented with a positional encoding vector that encodes its location in the original image grid. Positional embeddings allow the model to understand spatial relationships between anatomical regions, which is essential for identifying pathological patterns that depend on relative positioning, such as tumor spread or organ deformation (Lindeberg, 1994). The combined embedding is expressed as:

$$Z_i^{(0)} = Z_i + E_{pos, i}$$

Where:

$E_{pos, i}$ represents the learnable positional embedding corresponding to the i^{th} patch. Z_i is the feature embedding of the i -th patch extracted from the CNN

$Z_i^{(0)}$ is the final input embedding fed into the transformer encoder

Positional embeddings allow the model to understand spatial relationships between anatomical regions, which is essential for identifying pathological patterns that depend on relative positioning, such as tumor spread or organ deformation (Litjens, et al., 2017).

3.3.4 Transformer Encoder

The Vision Transformer encoder forms the core of the hybrid architecture and is responsible for capturing global contextual dependencies across the entire image. Unlike convolutional layers that operate locally, the transformer encoder enables long-range dependency modeling, allowing the network to understand spatial interactions between anatomically distant regions (Dosovitskiy, et al., 2021). This capability is particularly important in medical imaging, where disease manifestations may appear across multiple regions of an organ or tissue (Srinivas, et al., 2021) (Vanitha, et al., 2025) (Vaswani, et al., 2017). The encoder consists of multiple stacked transformer layers, each comprising Multi-Head Self-Attention, Feed-Forward Neural Networks, residual connections, and layer normalization, working together to refine feature representations (Chen & Lu, 2021).

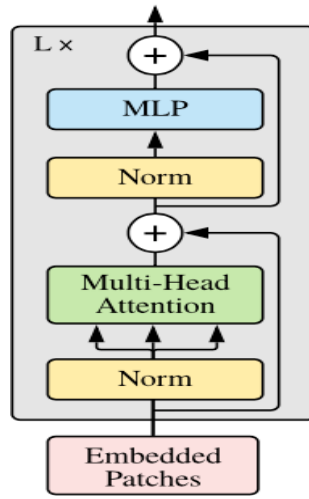


Figure 3.5: Transformer Encoder

3.3.4.1 Multi-Head Self-Attention (MHSA)

Multi-Head Self-Attention is the most critical component of the Vision Transformer encoder, enabling the model to determine how different image patches relate to one another. For each input token sequence X , three distinct representations queries, keys, and values are computed using learned linear projections:

$$Q = XW_Q, K = XW_K, V = XW_V$$

where W_Q, W_K, W_V are trainable weight matrices. These projections allow the model to compare each patch with all other patches in the image (Vaswani, et al., 2017). The similarity between patches is computed using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{SoftMax}(QK^T / \sqrt{d_k}) V$$

The scaling factor $\sqrt{d_k}$ stabilizes gradients during training. Multi-head self-attention extends this process by computing attention in multiple subspaces simultaneously. Each attention head learns a different type of spatial relationship, such as texture similarity, anatomical symmetry, or pathological correlation. This enables the model to focus on multiple anatomical regions concurrently, which is essential for detecting complex disease patterns in medical images (Vaswani, et al., 2017).

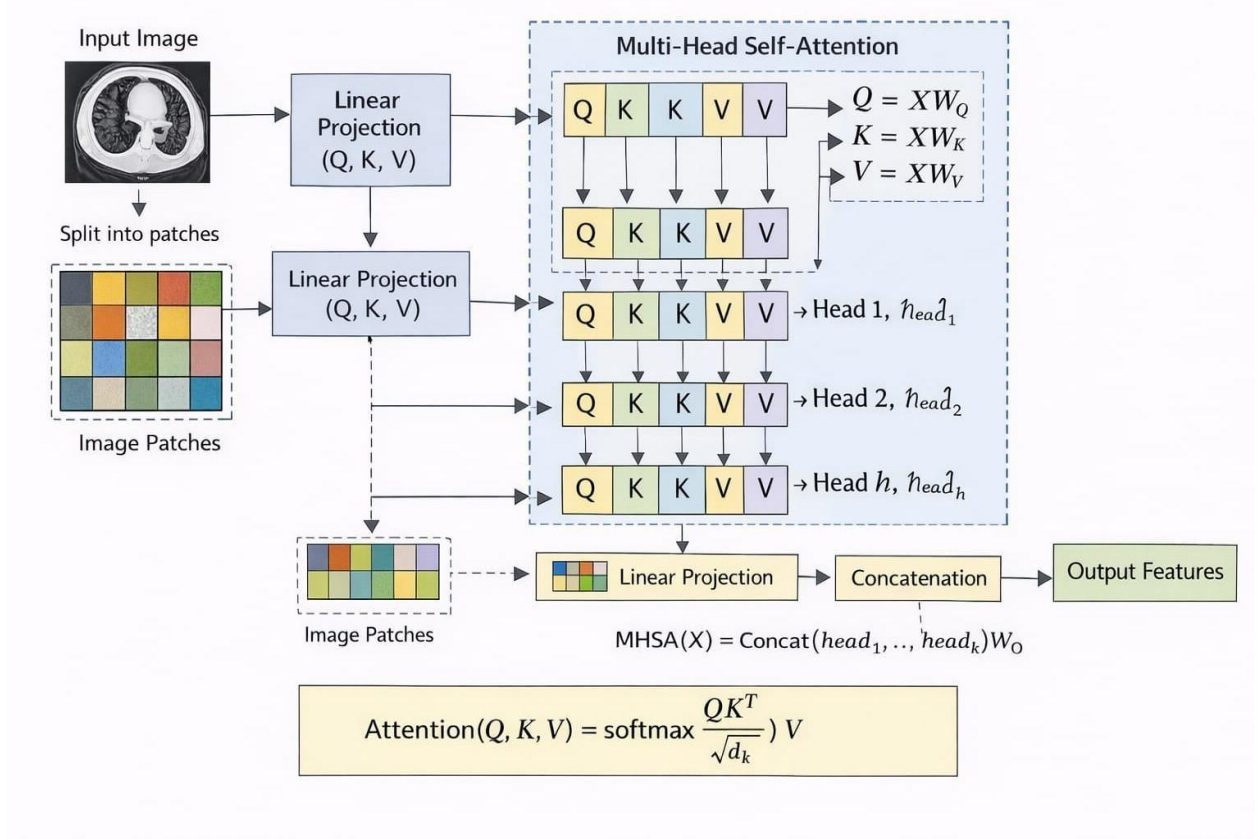


Figure 3.6: Multi-Head Self-Attention (MHSA)

3.3.4.2 Feed-Forward Neural Network

Following the self-attention operation, each token representation is independently processed by a feed-forward neural network. The FFN enhances feature expressiveness by applying non-linear transformations to each patch embedding. It consists of two linear layers with a non-linear activation function in between. The first linear layer expands the feature dimension, while the second projects it back to the original embedding size (Goodfellow, et al., 2016). This structure enables the network to learn complex feature interactions that cannot be captured by attention alone. In medical imaging tasks, the FFN helps refine features related to tissue texture, lesion shape, and structural irregularities (Dosovitskiy, et al., 2021) (Lindeberg, 1994).

3.3.4.3 Layer Normalization

Layer normalization is applied before or after each sublayer to stabilize training and ensure consistent feature distributions. It normalizes token representations across the embedding dimension. Layer normalization improves convergence speed and allows the model to generalize better across different medical imaging modalities (Goodfellow, et al., 2016).

3.3.5 Classification Head

The classification or prediction head is the final component of the Hybrid Vision Transformer and is responsible for generating diagnostic outputs from the learned feature representations. It primarily uses the final embedding of the learnable classification token ([CLS]), which captures global contextual information from all image regions through the transformer’s self-attention mechanism (Lai, 2024).

The [CLS] token representation is passed through one or more fully connected layers that refine the extracted features and improve class separability. Regularization techniques such as dropout are applied to enhance generalization and reduce overfitting, particularly in medical imaging datasets with limited samples such as TB Chest X-Ray and Diabetic Retinopathy (Lai, 2024). The output layer uses an activation function based on the classification task. Sigmoid activation is employed for binary and multi-label classification tasks, while SoftMax activation is used for multi-class classification (Brain Tumor MRI, Diabetic Retinopathy, HAM10000). This design enables the model to support disease detection, disease categorization, and simultaneous identification of multiple pathologies (Dosovitskiy, et al., 2021) (Lindeberg, 1994).

3.3.6 CACI Interpretation Module

Interpretability is an essential aspect of medical imaging systems to ensure transparency and clinical trust. The Hybrid Vision Transformer incorporates a Confidence-Aware Clinical Interpretation (CACI) module to convert each raw prediction into structured, clinically meaningful signals rather than a pixel-level visual overlay (Ahmed, et al., 2026).

CACI derives a confidence band, a ranked differential diagnosis, a decision margin, and a disease-severity band directly from the model’s softmax output and a clinical knowledge base, and pairs them with a six-section natural-language interpretation generated via a large language model. This improves clinician confidence, supports validation of model predictions, and facilitates reliable deployment in real-world healthcare applications (Frasca, et al., 2024). The dataset-specific CACI outputs are presented in Section 4.7 and verified in Section 4.8.

3.4 Model Training

The Hybrid Vision Transformer (HViT) is trained in two sequential stages: pretraining and fine-tuning.

3.4.1 Pretraining Stage

During pretraining, all components of the HViT architecture, including the CNN feature extractor, patch embedding layer, transformer encoder, and classification head, are trained jointly. The input medical images are first passed through the preprocessing pipeline, which includes resizing, normalization, noise reduction, and data augmentation techniques such as rotation, flipping, and scaling. These steps improve the robustness of the model and prevent overfitting (Perez & Wang, 2017).

The CNN backbone learns low-level and mid-level features such as edges, textures, contours, and anatomical structures that are common across different medical imaging modalities. Simultaneously, the Vision Transformer component captures global contextual information by modeling long-range dependencies between different image regions using multi-head self-attention (Perez & Wang, 2017). This combined learning enables the model to understand both local pathological patterns and global anatomical relationships. The model predicts the classes in a multi-label or multi-class format using a SoftMax or sigmoid activation, respectively, depending on the dataset (Goodfellow, et al., 2016).

The loss function used is the categorical cross-entropy (or binary cross-entropy for multi-label tasks), and the optimizer is AdamW, which helps in faster convergence and reduces overfitting through weight decay. The learning rate is scheduled using a step decay or cosine annealing scheduler to balance exploration and convergence. Pretraining allows the model to capture generalizable visual representations that are useful across multiple disease types (Dosovitskiy, et al., 2021). The categorical Cross-Entropy loss function is used for multi-class classification is given by:

$$L = - \sum_{i=1}^N y_i \log \hat{y}_i$$

3.4.2 Fine-tuning Stage

After pretraining, the model undergoes a fine-tuning stage to adapt the learned representations to specific disease classification tasks, such as fracture detection, pathological fracture identification, or brain tumor diagnosis. Fine-tuning is particularly important in medical imaging applications where labeled data is often limited and disease-specific patterns vary significantly across tasks, a

consideration directly relevant to the smaller datasets in this study, such as Diabetic Retinopathy (3,662 images).

In the fine-tuning stage, the pretrained HViT model is initialized with the weights learned during pretraining. To preserve the general features acquired earlier, the CNN feature extractor and the transformer encoder layers are optionally frozen. This prevents significant weight updates in the lower layers and allows the model to retain fundamental anatomical and structural knowledge (Lai, 2024). Only the final classification head or the last few transformer layers are retrained using disease-specific datasets.

This selective training strategy enables the model to focus on learning high-level, task-specific features while reducing computational complexity and training time. Fine-tuning also helps mitigate overfitting, especially when training on small datasets, by limiting the number of trainable parameters (Ali, et al., 2025).

During fine-tuning, the input images are processed through the same preprocessing pipeline to maintain consistency with the pretraining stage. The loss function and optimizer remain unchanged, although a lower learning rate is typically used to ensure gradual adaptation of the model weights. Through this process, the HViT model becomes highly specialized for the target medical diagnosis task while leveraging the general knowledge learned during pretraining.

3.5 Disease Prediction

After training, the Hybrid ViT is applied to unseen medical images for inference and disease prediction. Each input image undergoes the same preprocessing steps as used during training, including resizing, normalization, and noise reduction, ensuring consistency. The CNN backbone extracts local structural and textural features, while the transformer encoder captures long-range dependencies between image regions. The [CLS] token, representing a global summary of the image, is processed by the classification head to generate a probability distribution over all disease classes. The predicted label corresponds to the class with the highest probability, and a confidence score is provided to indicate the model's certainty. This output allows clinicians to evaluate predictions efficiently, with the confidence scores supporting decision-making for borderline or ambiguous cases. The inference process is fully automated, enabling rapid disease prediction across multiple medical imaging modalities (Ahmed, et al., 2026).

3.6 CACI and the Explainability Hub

Interpretability is critical for clinical adoption of AI models, ensuring transparency and trust in the predictions. The Hybrid ViT's raw prediction is passed through a Confidence-Aware Clinical Interpretation (CACI) layer, which converts the softmax output and disease metadata into five structured trust signals.

CACI operates on the classifier's softmax output and produces five structured trust signals per prediction:

3.6.1 Prediction Confidence

The softmax probability of the top class, banded into Strong ($\geq 90\%$), Moderate (75–89%), Weak (60–74%), or Inconclusive ($< 60\%$).

3.6.2 Ranked Differential Diagnosis

The top-3 classes from the softmax distribution, each with its probability, presented to expose alternative clinical hypotheses.

3.6.3 Decision Margin

The difference between the top-1 and top-2 probabilities, labeled as Clear (≥ 50 pts), Moderate (20–49 pts), or Narrow (< 20 pts). A narrow margin indicates that the classifier is nearly split between two classes and flags the case for human review.

3.6.4 Severity Band

Derived from a curated disease knowledge base mapping each class to a clinical severity level (Low / Medium / High / Critical).

3.6.5 LLM-Generated Clinical Interpretation

A narrative (Primary Finding, Differential Diagnosis, Clinical Context, Interpretation & Confidence, Recommended Next Steps, Clinical Red Flags) generated by a large language model (Groq-hosted GPT-OSS-120B) from the four signals above and the curated disease knowledge base. When the LLM is unavailable, a deterministic local template produces the same six-section structure, ensuring the system remains fully functional offline.

These five signals are surfaced through a dedicated Explainability Hub so that a clinician can

directly assess how sure the model is, what alternative diagnoses were considered, how close the decision was, how serious the finding is, and what should happen next. The generated CACI outputs for representative cases from each of the seven datasets are described in Section 4.8.

3.7 Experimental Setup and Hyperparameter Configuration

The proposed Hybrid Vision Transformer (HViT) model is trained and evaluated using carefully chosen hyperparameters to ensure reliable performance across different medical imaging datasets, including fractures, pathological fractures, and brain tumors. The model employs categorical cross-entropy as the loss function, which is suitable for multi-class classification tasks, allowing the network to distinguish between different disease classes accurately (Goodfellow, et al., 2016). The Adam optimizer is utilized for weight updates during training due to its adaptive learning rate capabilities, which effectively minimize prediction error by dynamically adjusting the step size for each parameter.

Training is conducted for 40 epochs for each dataset, providing sufficient iterations for the network to converge without overfitting. The batch size is set to 32, balancing computational efficiency and stability of gradient updates. To further improve convergence and optimize learning, a learning rate scheduler is applied, which gradually adjusts the learning rate during training based on the performance on the validation set. This approach prevents the model from getting stuck in local minima and enhances overall accuracy (Goodfellow, et al., 2016).

Table 3.7: Hyperparameter Configuration

Hyperparameter	Value
Loss Function	Categorical Cross entropy
Number of Epochs	40
Optimizer	AdamW Optimizer
Batch Size	32
Tuning Technique	Learning Rate Scheduler

All experiments are performed on Google Colab and kaggle, a cloud-based platform with GPU support provided by Google. GPU acceleration allows the Hybrid ViT model to train efficiently, significantly reducing computational time. NumPy is used for data preprocessing and numerical

computations, while PyTorch is employed for model implementation, training, evaluation, and visualization. This setup provides a reproducible and high-performance environment for the proposed explainable Hybrid ViT model (Goodfellow, et al., 2016).

3.8 Tools and Technology Stack

A robust and scalable technology stack is selected for developing the system:

3.8.1 Programming Language

Python was selected as the core programming language due to its simplicity, readability, and extensive ecosystem for data science and machine learning. Its high-level syntax accelerates development while maintaining code clarity and maintainability. Python offered a wide range of scientific computing libraries that support data preprocessing, statistical analysis, and model development. Furthermore, Python's strong integration with web frameworks such as Django made it an ideal choice for building scalable backend systems.

3.8.2 Deep Learning Framework

PyTorch is the core deep learning framework used to implement the Hybrid Vision Transformer architecture. It supports dynamic computational graphs, which are highly suitable for experimenting with complex hybrid CNN-Transformer models. PyTorch provides efficient GPU acceleration, automatic differentiation, and modular model development, enabling faster prototyping and training of deep neural networks. It also integrates well with explainable AI tools such as Captum for interpretability analysis.

3.8.3 Medical Image Preprocessing Tools

OpenCV is used for image preprocessing operations such as resizing, denoising, contrast enhancement, normalization, and data augmentation. These steps improve image quality and ensure consistent input to deep learning models. OpenCV also supports region of interest extraction and segmentation, which are critical in medical imaging workflows.

3.8.4 Frontend Technologies

3.8.4.1 HTML

HTML provides the structural framework for the web-based interface, allowing users to upload medical images, view classification results, and explore interpretability outputs. It ensures

organized presentation of system components and user input fields.

3.8.4.2 CSS

CSS enhances the interface by improving layout, typography, color schemes, and responsiveness. It ensures that the platform is visually professional, user-friendly, and optimized for accessibility in clinical and academic environments.

3.8.4.3 JavaScript

JavaScript enables interactive frontend functionalities such as real-time prediction visualization, dynamic confidence gauge rendering, interactive differential charts, and smooth communication between frontend and backend services.

3.8.5 Backend Technologies

Django is used as the primary backend framework to manage server-side operations, API services, database connectivity, and model inference workflows. It provides a secure, scalable, and modular architecture suitable for healthcare-oriented applications. Django handles user authentication, image upload processing, request routing, and communication between the frontend interface and the deep learning inference engine. Its built-in security mechanisms, including protection against SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF), make it a reliable framework for medical data systems. Django also supports integration with SQLite3 and REST APIs, enabling efficient deployment of the diagnostic model.

3.8.6 Database Management

SQLite3 is used as the primary database for storing system data in a lightweight, serverless, and efficient manner. It is well-suited for research-based and prototype medical imaging systems due to its simplicity, low overhead, and ease of integration with Python. SQLite3 is used to store anonymized patient metadata, image references, prediction results, evaluation logs, experiment configurations, and model performance metrics. Its embedded architecture ensures fast read/write operations while maintaining data integrity, making it an effective choice for standalone and offline research environments.

3.8.7 Python Libraries

3.8.7.1 Pandas

Pandas is utilized for managing structured and tabular datasets, including medical image metadata, annotations, experiment logs, and evaluation reports. It simplifies data preprocessing tasks such as cleaning missing values, filtering datasets, merging clinical metadata with imaging records, and organizing experiment outputs. Pandas also supports statistical summaries and exploratory data analysis, enabling efficient dataset inspection and research reporting.

3.8.7.2 NumPy

NumPy provides high-performance support for numerical computing and multidimensional array operations. It is essential for handling image tensors, matrix operations, pixel intensity transformations, and deep learning computations. NumPy ensures efficient memory usage and fast execution of vectorized operations, which are critical for preprocessing large-scale medical imaging datasets and optimizing model training pipelines.

3.8.7.3 Scikit-learn

Scikit-learn is used for implementing traditional machine learning baseline models such as Support Vector Machines, Logistic Regression, and Random Forests for comparative analysis with transformer-based deep learning models. It is also used for dataset splitting, cross-validation, hyperparameter tuning, and computation of evaluation metrics including accuracy, precision, recall, F1-score, and confusion matrices. This allows objective benchmarking and statistical validation of the Hybrid ViT model.

3.8.7.4 Matplotlib

Matplotlib is used to generate scientific and publication-quality visualizations such as training loss curves, accuracy graphs, confusion matrices, class-wise prediction plots, and attention heatmaps. These visual outputs assist in interpreting model behavior, diagnosing training issues, and presenting results in academic papers and project reports.

3.8.7.5 Seaborn

Seaborn enhances data visualization by producing statistically rich and visually refined plots, including distribution graphs, correlation heatmaps, feature importance plots, and comparative performance charts. It improves clarity and readability when presenting experimental results and interpretability findings.

3.8.8 Vision Transformer Libraries

Hugging Face Transformers facilitates experimentation with transformer-based architectures, including Vision Transformers and multimodal transformer frameworks. It supports implementation of attention-based feature extraction, cross-modal learning, and large-scale transformer experimentation. The library accelerates model development by providing pretrained weights, standardized APIs, and integration with modern explainability tools.

3.8.9 CACI and Explainability Hub Stack

The CACI layer and Explainability Hub are built on the following libraries and services. The Groq API (an OpenAI-compatible chat completion endpoint, model openai/gpt-oss-120b) generates the six-section natural-language clinical interpretation from the case's diagnosis, confidence score, and modality, with a local template-based interpretation used as a fallback when the API is unavailable. On the front end, a lightweight JavaScript charting library renders the circular confidence gauge and the ranked-differential bar chart, and standard HTML/CSS components render the trust-indicator grid and clinical-knowledge panel. A disease-information knowledge base, stored server-side, supplies the description, severity, symptoms, and standard treatment shown in the clinical knowledge panel and used to compute the severity band.

3.8.10 Experimentation and Development Tools

Google Colab and Kaggle serves as the primary platform for model development, experimentation, and training in this project. It provides free access to GPU and TPU resources, which significantly accelerates deep learning workflows involving Hybrid Vision Transformer architectures and large-scale medical imaging datasets. Google Colab supports rapid prototyping, hyperparameter tuning, large-batch training, and cloud-based execution without requiring high-end local hardware. Its integration with Google Drive enables efficient dataset storage, model checkpoint management, and collaborative research, making it the preferred environment for experimentation and reproducibility. Jupyter Notebook is used for structured experimentation, data exploration, model debugging, and visualization of training results. It allows interactive execution of preprocessing pipelines, model training routines, and explainability analyses. Jupyter Notebook also facilitates documentation of experimental findings and supports reproducible research by maintaining code, outputs, and explanatory text in a single environment. Git and GitHub are used for version control, collaborative development, experiment tracking, and secure code management.

They enable structured project organization, maintain experiment history, facilitate teamwork, and ensure reproducibility of research workflows.

3.9 Software Development Model

The Agile methodology was employed in the development of the Hybrid Vision Transformer (HViT) based Medical Imaging Diagnosis System to ensure flexibility, collaboration, and continuous system improvement. Agile enabled the development team to effectively respond to evolving technical requirements, clinical considerations, and research findings by organizing the project into iterative development sprints. Each sprint focused on specific phases such as planning, system design, model development, testing, deployment, and review, ensuring that the project remained aligned with research objectives and user expectations throughout its lifecycle. Agile development, like any team-oriented software engineering approach, is inherently human-centered and emphasizes collaboration, communication, adaptability, and continuous feedback (Shore & Warden, 2021).

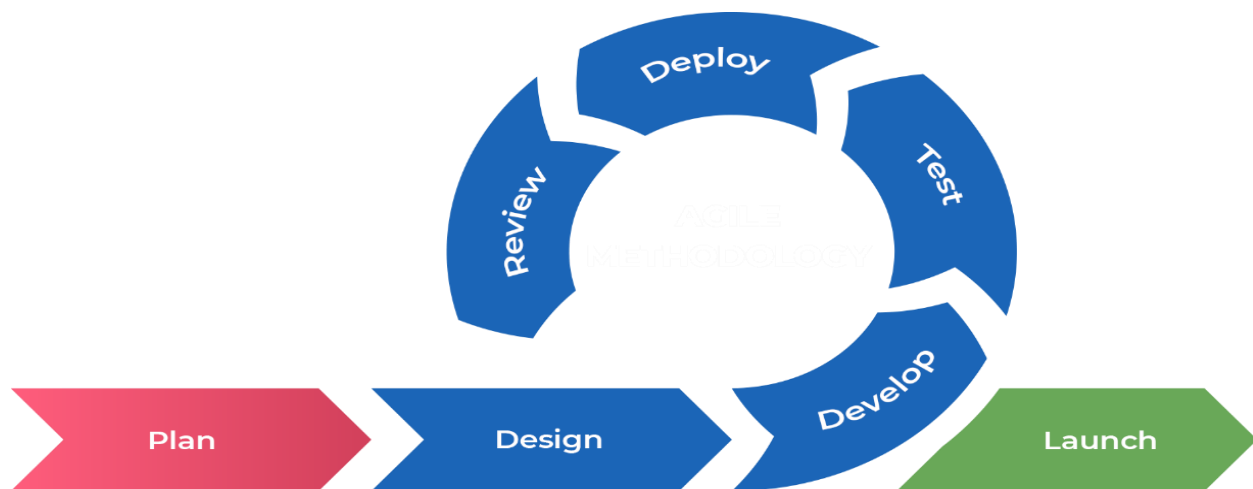


Figure 3.7: Agile Development Model

3.9.1 Plan

During the planning phase, the core requirements for the medical imaging diagnosis system were identified, analyzed, and prioritized. These included functionalities such as medical image acquisition, preprocessing pipelines, Hybrid CNN–Vision Transformer model training, CACI interpretation layer integration, and a clinician-friendly user interface. Stakeholders, including academic supervisors and domain experts, were consulted to determine critical features and define

sprint-level objectives. Tasks were distributed across iterative sprints to ensure structured progress, timely delivery, and effective research experimentation (Shore & Warden, 2021).

3.9.2 Design

The system design followed an iterative and modular approach, beginning with a foundational architecture and progressively evolving based on experimental findings. The architecture incorporated key components such as a Django-based backend server, a Hybrid Vision Transformer deep learning model, an SQLite3 database for experiment logging, and an interactive frontend interface. As development progressed, model architecture, the CACI interpretation module, database schema, and interface layouts were refined to enhance performance, scalability, interpretability, and clinical usability. The design emphasized both technical robustness and accessibility for medical researchers and healthcare professionals (Shore & Warden, 2021).

3.9.3 Develop

Development was carried out incrementally across multiple Agile sprints, with each sprint targeting specific system components. Key deliverables included medical image preprocessing modules, Hybrid CNN–Vision Transformer model implementation, CACI signal generation (confidence, differential diagnosis, and severity banding), Groq-based clinical narrative integration, and API-based inference services. Continuous integration practices ensured smooth merging of code updates and efficient incorporation of new research improvements. Collaborative development ensured seamless coordination between model development, backend logic, CACI interpretation, and user interface functionality, reducing integration errors and supporting efficient workflow progression (Shore & Warden, 2021)

3.9.4 Test

Testing was integrated into every sprint to ensure reliability, accuracy, and interpretability of the diagnostic system. Unit testing validated individual components such as data preprocessing routines, model inference pipelines, and CACI signal generation modules. Integration testing verified correct communication between the backend server, deep learning model, database, and frontend interface. System-level testing evaluated the complete workflow from medical image input to disease prediction and explanation output. Model testing included validation on unseen datasets, accuracy and loss analysis, confusion matrix generation, and interpretability assessment to ensure trustworthy and clinically meaningful predictions (Shore & Warden, 2021).

3.9.5 Deploy

At the conclusion of each sprint, a functional version of the system was deployed to a testing environment or cloud-based inference platform. Deployment updates included model weight refinements, CACI interpretation layer enhancements, interface usability improvements, and backend performance optimizations. Automated deployment workflows reduced configuration errors and facilitated smooth transitions between development and deployment stages. This iterative release cycle enabled controlled system evolution without interrupting ongoing experimentation, validation, or user feedback processes (Shore & Warden, 2021).

3.9.6 Review

Following each deployment, structured feedback was collected from project supervisors, technical reviewers, and potential end users. These reviews helped identify model performance limitations, interpretability gaps, usability concerns, and areas requiring optimization. Feedback-driven refinements guided subsequent sprint planning, ensuring the system evolved in response to real-world research insights and clinical relevance. The continuous feedback loop played a key role in improving diagnostic accuracy, CACI interpretation clarity, model reliability (Shore & Warden, 2021).

3.9.7 Launch

After completing all core functionalities including Hybrid Vision Transformer inference, CACI-based clinical interpretation, image validation, database logging, and secure user interaction the system was formally launched. It was deployed in its intended operational environment, tested for stability, and monitored for performance consistency. Post-launch monitoring and feedback supported ongoing optimization, scalability improvements, and future enhancements aimed at expanding diagnostic coverage and improving clinical applicability (Shore & Warden, 2021).

Agile was selected for the Hybrid Vision Transformer Medical Imaging Diagnosis System due to its adaptability and ability to manage evolving research, technical, and interpretability requirements. This flexibility is especially important in deep learning-based medical applications, where model performance, dataset variability, and explainability demands continuously evolve. The iterative Agile framework enabled progressive refinement of model accuracy, visualization clarity, and system usability. Continuous integration, testing, and stakeholder feedback ensured system stability, trustworthy predictions, and transparent decision-making. Overall, Agile

facilitated the successful development of a reliable, interpretable, and user-centered intelligent medical imaging diagnosis solution (Shore & Warden, 2021).

3.10 Model Evaluation

After training and inference, the performance of the Hybrid Vision Transformer (Hybrid ViT) is assessed using quantitative metrics along with qualitative analysis of explainability outputs. This ensures that the model is both accurate and clinically interpretable (Litjens, et al., 2018).

3.10.1 Quantitative Evaluation

The following standard metrics are used to evaluate the model performance.

a) Accuracy (ACC)

It measures the proportion of correctly classified images among all predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

b) Precision (P)

It indicates the fraction of true positive predictions out of all predicted positives, reflecting the reliability of positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

c) Recall / Sensitivity (R)

Measures the fraction of true positives detected out of all actual positives, ensuring critical conditions are not missed.

$$\text{Recall} = \frac{TP}{TP + FN}$$

d) F1-Score

The harmonic means of precision and recall, balancing the two metrics for imbalanced datasets.

$$F1 = 2 \frac{Precision * Recall}{Precision + Recall}$$

3.10.2 Qualitative Evaluation

Beyond numerical metrics, the CACI interpretation outputs are analyzed to ensure clinical usefulness. For each case, the confidence band, ranked differential diagnosis, decision margin, and severity band are checked against the model's own softmax output and the disease-information knowledge base, and the generated clinical narrative is checked for consistency with the predicted diagnosis, confidence score, and modality. To ensure robust performance, K-fold cross-validation can be applied. The dataset is split into K subsets, and the model is trained K times, each time using a different subset for validation. The overall performance is averaged across folds, reducing variance and providing a reliable estimate of generalization (Litjens, et al., 2018).

3.10.3 Performance Interpretation

The combination of quantitative metrics and CACI interpretation signals ensures that the model is both accurate and interpretable. High values of accuracy, precision, recall and F1-score indicate reliable disease detection, while consistent confidence bands, decision margins, and severity classifications demonstrate that predictions carry clinically meaningful, trustworthy context. This evaluation framework is crucial for deployment in clinical settings, where both performance and interpretability are required.

3.11 System Modeling

System modeling provides a structured, visual representation of the proposed Hybrid Vision Transformer (HViT) based medical imaging diagnosis system. System modeling diagrams are used to describe the system architecture, data organization, process flow, and interactions between users and system components (Ahmed, et al., 2026). The diagrams presented in this section include the Context Diagram, Data Flow Diagram (DFD), Entity Relationship (ER) Diagram, Use Case Diagram, and Activity Diagram. Each diagram focuses on a specific aspect of the system: the Context Diagram on external interactions, the DFD on internal data flow, the ER Diagram on database structure, the Use Case Diagram on functional requirements, and the Activity Diagram on operational workflow. Collectively, these diagrams clarify the system's architecture and requirements, and guide efficient implementation of the proposed system. (Fu, et al., 2025).

3.11.1 Context Diagram

The context diagram represents a high-level overview of the Explainable Hybrid Vision Transformer system by showing its interaction with external entities. It defines the system boundary and illustrates how medical images from multiple modalities such as CT, MRI, X-ray, and microscopic images are provided by the user or clinician as input. The system processes these images and produces disease predictions along with a CACI-based clinical interpretation report. This diagram helps in understanding the overall purpose and scope of the system without detailing internal processes (Vanitha, et al., 2025).

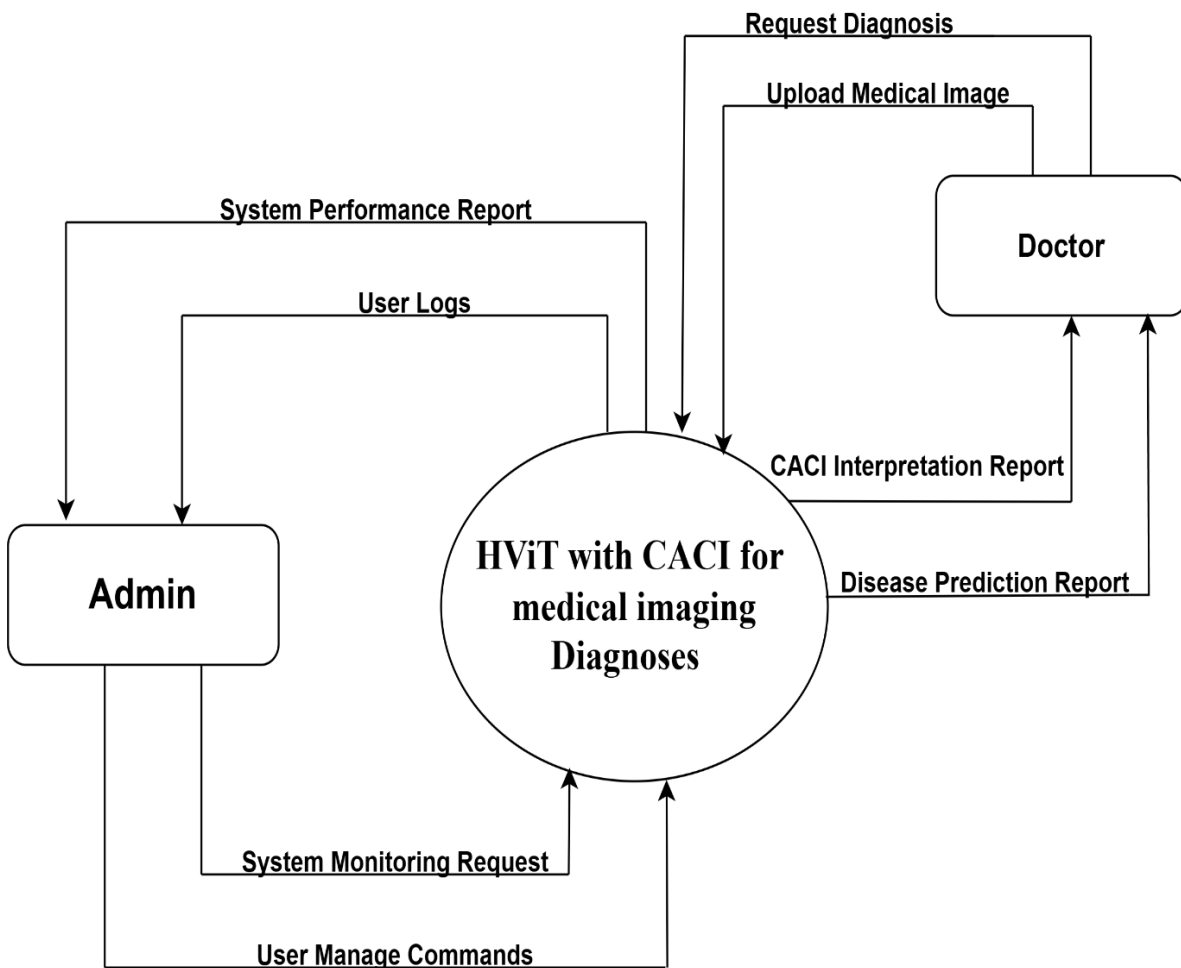


Figure 3.8: Context Diagram

3.11.2 DFD Level 1

The Data Flow Diagram (Level 1) illustrates how data moves through the internal components of the HViT system. It shows key processes including image preprocessing, CNN-based feature extraction, transformer-based feature learning, disease classification, and CACI interpretation

generation. The diagram also represents data stores such as trained models and prediction results. This diagram is useful for understanding how raw medical images are transformed into meaningful diagnostic outputs through structured data flow.

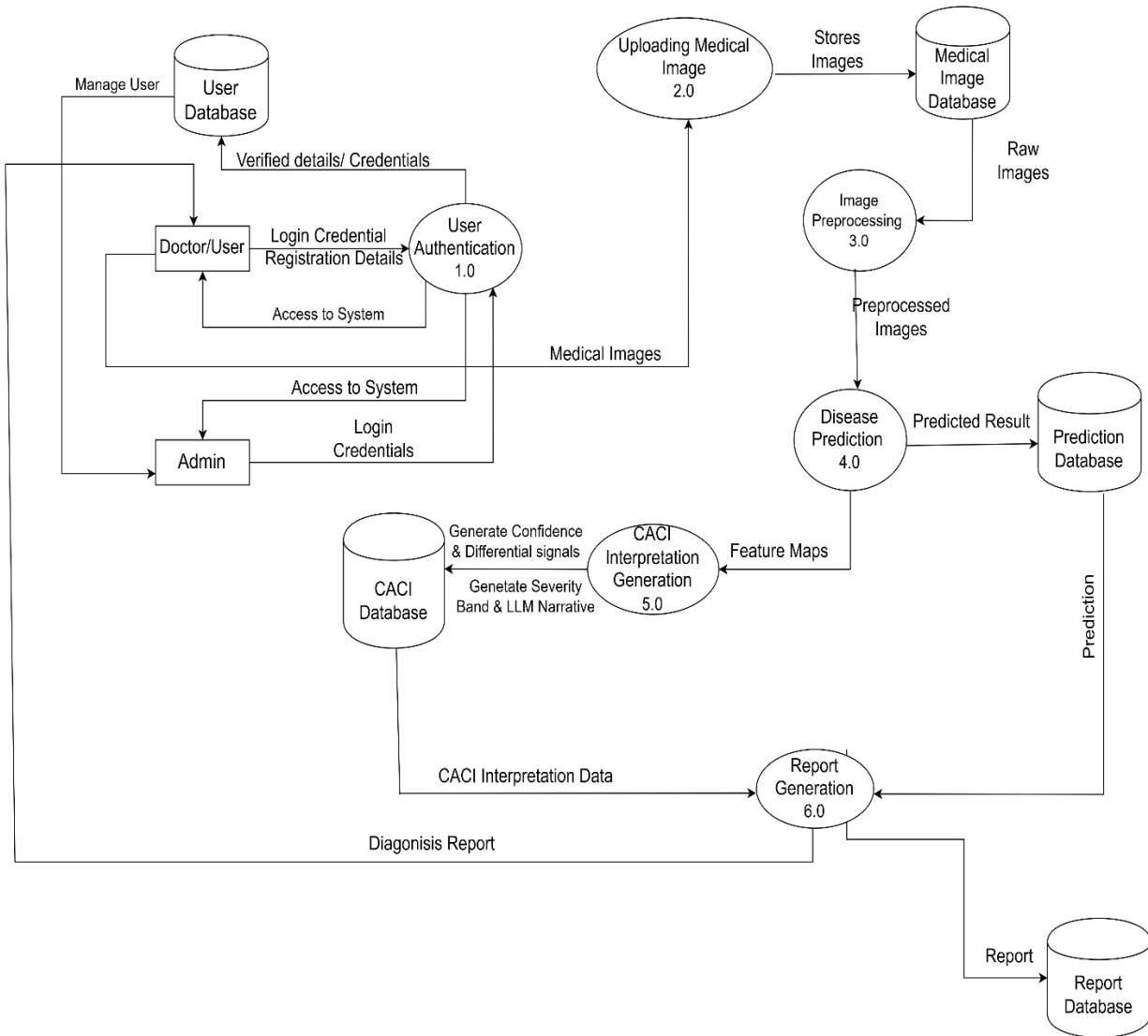


Figure 3.9: DFD Level 1

3.11.3 ER Diagram

The Entity Relationship (ER) diagram models the logical data structure of the system by defining entities, their attributes, and relationships. In the HViT system, entities such as User, Medical Images, Preprocessed Record, Prediction, CACI_Output, and Report are represented. Relationships describe how medical images are linked to predictions and their corresponding CACI interpretation output. This diagram supports efficient data organization and database design. (Ali, et al., 2025).

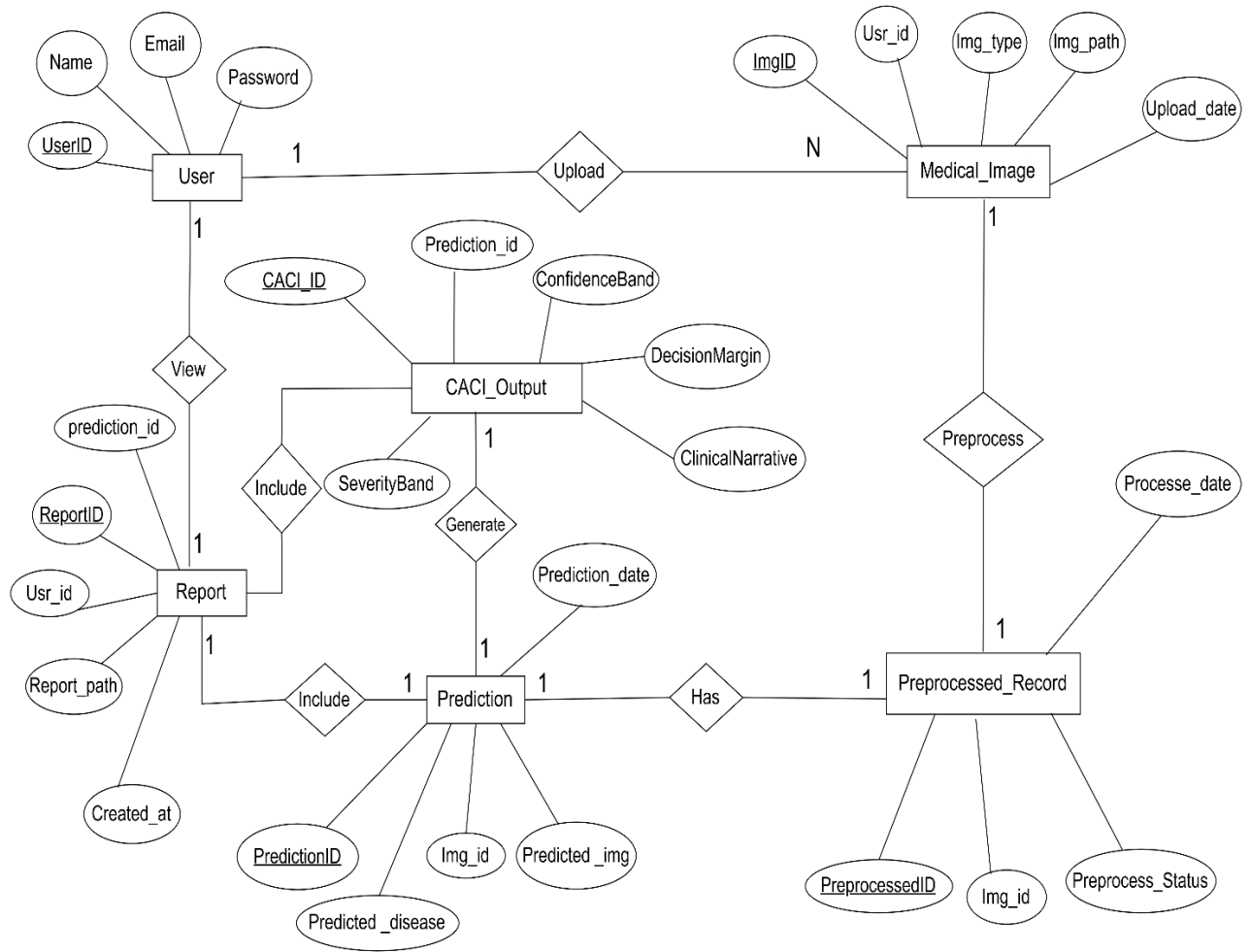


Figure 3.10: ER Diagram

3.11.4 Use Case Diagram

The use case diagram represents the functional behavior of the system from the user's perspective. It identifies the main actor, such as a clinician or medical analyst, and highlights the interactions between the user and the system. The primary use cases include uploading medical images, training the model, performing disease prediction, and viewing the CACI interpretation report. This diagram ensures that all system functionalities are clearly defined and aligned with user requirements (Fu, et al., 2025).

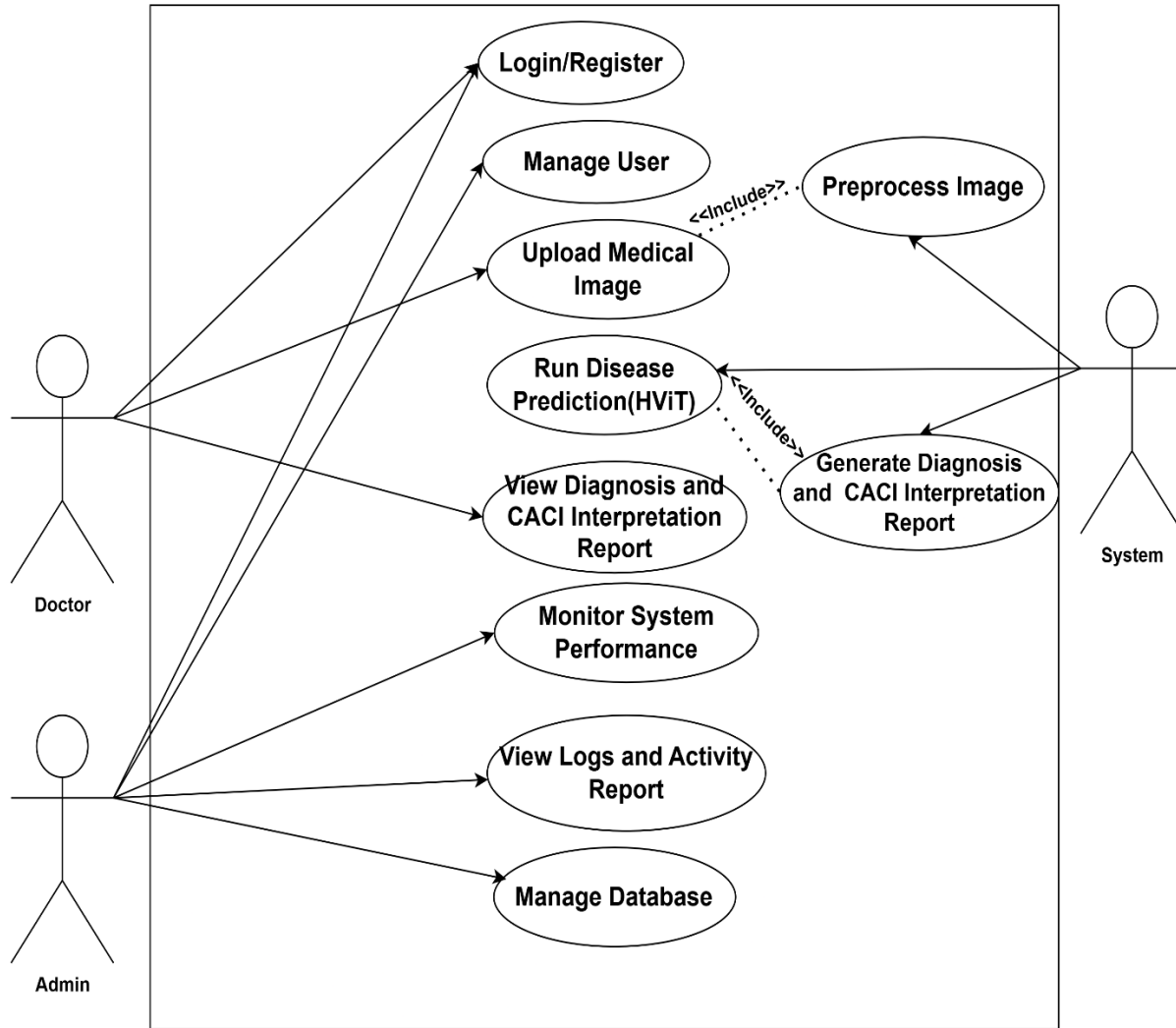


Figure 3.11: Use Case Diagram

3.11.5 Activity Diagram

The activity diagram illustrates the operational workflow of the Explainable Hybrid Vision Transformer system. The process begins when a doctor logs into the system and uploads a medical image. The image undergoes preprocessing to enhance quality and ensure compatibility with the model. Local features are extracted using a CNN backbone, followed by global contextual modeling through a Vision Transformer encoder. The classification head generates a disease prediction, after which the CACI layer produces a confidence band, ranked differential diagnosis, decision margin, severity band, and an LLM-generated clinical narrative. Finally, the system compiles the prediction and CACI interpretation into a structured medical report, which is

delivered to the doctor for clinical interpretation (Ali, et al., 2025).

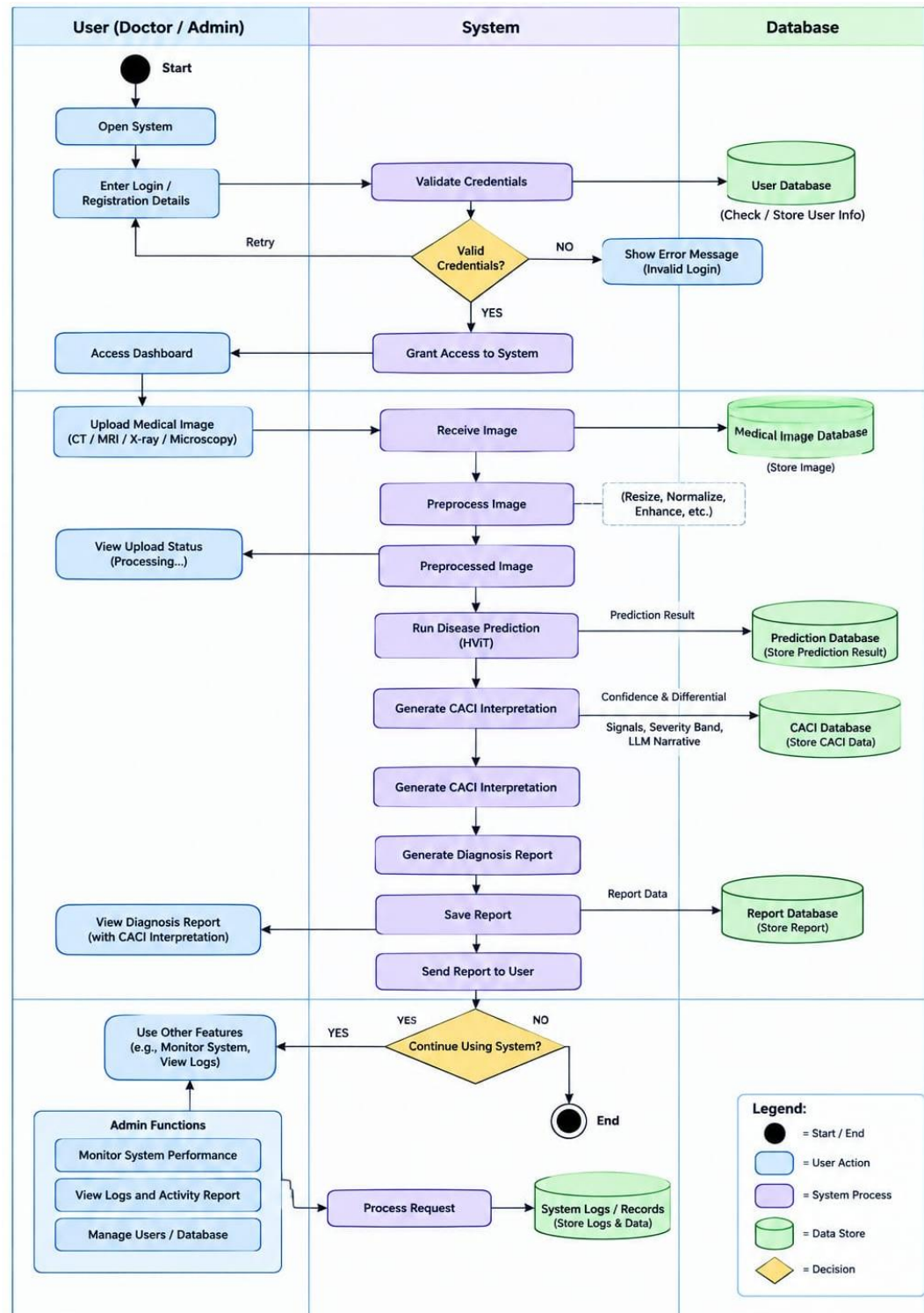


Figure 3.12: Activity Diagram

CHAPTER 4: RESULT AND ANALYSIS

This chapter presents the final, measured performance of the HViT system across all seven medical imaging datasets. Results are presented at three levels of granularity: first, a dataset-level overview and class-distribution analysis; second, an aggregate cross-dataset performance comparison using the metrics; and third, detailed per-dataset results, including per-class metrics and placeholders for dataset-specific diagnostic plots: confusion matrix, training/validation curves, ROC curves.

4.1 Dataset Overview

Figure 4.1 summarizes the seven datasets used for training and evaluation: total image counts, class distribution by dataset, number of classes per dataset, a class-balance heatmap, relative dataset size, and a consolidated summary table. As shown, dataset sizes range from 3,662 images (Diabetic Retinopathy) to 10,578 images (Bone Fracture), and the number of classes ranges from 2 (FracAtlas, TB Chest X-Ray, Pneumonia Chest X-Ray, Bone Fracture) to 7 (HAM10000 Skin Cancer).

Table 4.1: Medical Image Datasets (Overview Dashboard)

Dataset	Total	Classes	Imbalance Ratio
Brain Tumor MRI	7,200	4	1.0x (perfectly balanced)
FracAtlas	4,083	2	4.7x
TB Chest X-Ray	4,200	2	5.0x
Pneumonia Chest X-Ray	5,856	2	2.7x
Diabetic Retinopathy	3,662	5	9.4x
Bone Fracture	10,578	2	1.0x (balanced)
HAM10000 Skin Cancer	10,015	7	58.3x

4.2 Class Distribution Analysis

Class distributions vary substantially across the seven datasets, as shown in Figure 4.3, from perfectly balanced (Brain Tumor MRI at 25%/25%/25%/25%, Bone Fracture at 51%/49%) to severely imbalanced (HAM10000 Skin Cancer at 58.3×, and Diabetic Retinopathy at 9.4×). This variation is a key factor in interpreting the per-dataset results in Section 4.7, since accuracy alone

becomes an unreliable indicator of quality as imbalance increases, motivating the inclusion of MCC and Cohen's Kappa alongside accuracy throughout this chapter.

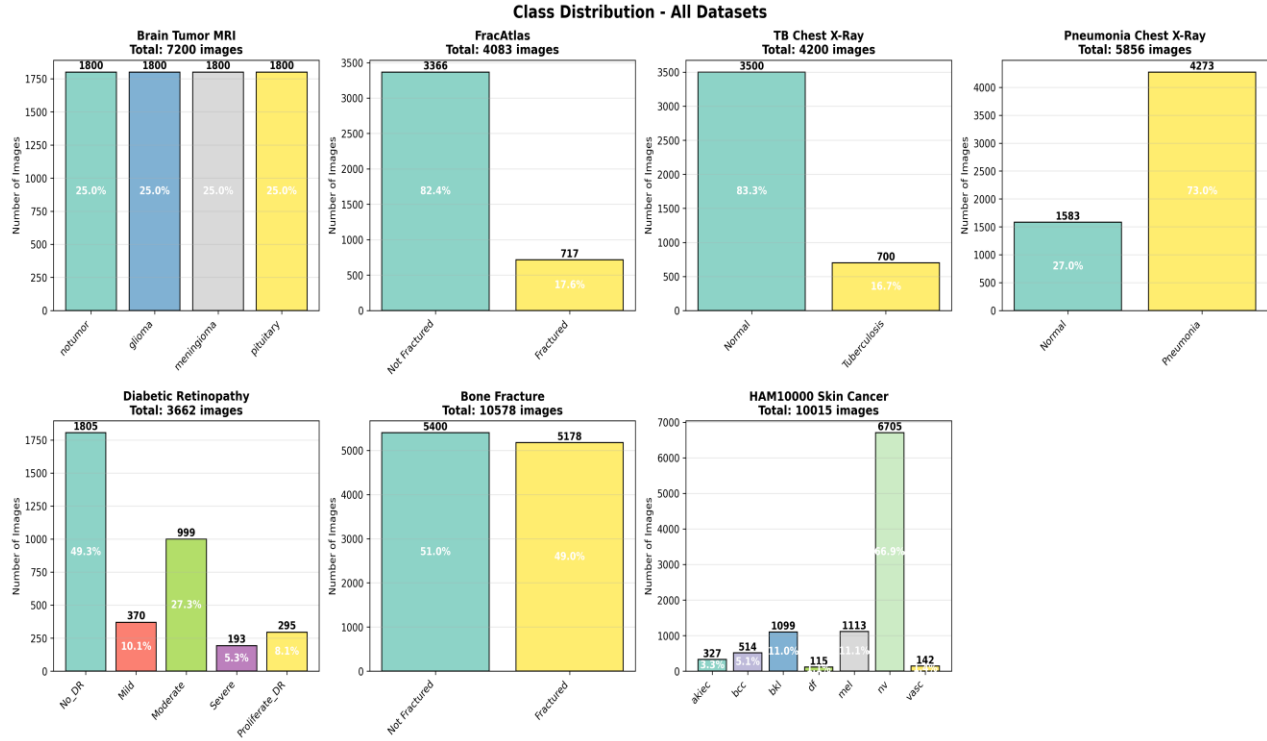


Figure 4.1: Class Distribution (All Datasets)

4.3 Cross-Dataset Performance Comparison

Table 4.3 compares Accuracy, Weighted F1-score, AUC, and MCC across all seven datasets, ranked from best to worst performing on each metric. The Bone Fracture and Brain Tumor MRI datasets consistently occupy the top two positions across every metric, while FracAtlas records the lowest MCC of the study.

Table 4.3: Model Performance Metrics Across All Datasets

Dataset	Accuracy	F1 (Weighted)	AUC	MCC	Kappa
Brain Tumor MRI	0.987	0.987	0.999	0.982	0.982
FracAtlas	0.845	0.820	0.767	0.353	0.320
TB Chest X-Ray	0.970	0.970	0.994	0.894	0.893
Pneumonia Chest X-Ray	0.945	0.946	0.986	0.863	0.863

Dataset	Accuracy	F1 (Weighted)	AUC	MCC	Kappa
Diabetic Retinopathy	0.813	0.813	0.964	0.719	0.719
Bone Fracture	0.999	0.999	1.000	0.997	0.997
HAM10000 Skin Cancer	0.844	0.850	0.967	0.719	0.716

4.4 Per-Class Performance Analysis

Figure 4.5 breaks down Precision, Recall, and F1-Score for every individual class within each dataset, revealing performance gaps hidden by dataset-level averages, most notably the weak minority-class recall in FracAtlas and the intermediate-severity confusion in Diabetic Retinopathy, both discussed in detail in Sections 4.7.2 and 4.7.5 respectively.

Table 4.4: Per-Class Performance Metrics (All Datasets)

Dataset	Class	Precision	Recall	F1-Score
Brain Tumor MRI	No Tumor	0.99	0.99	0.99
	Glioma	0.98	0.98	0.98
	Meningioma	0.98	0.98	0.98
	Pituitary Tumor	0.99	0.99	0.99
FracAtlas	No Fracture	0.86	0.96	0.91
	Fracture	0.63	0.43	0.39
TB Chest X-Ray	Normal	0.99	0.98	0.96
	Tuberculosis	0.89	0.93	0.91
Pneumonia Chest X-Ray	Normal	0.89	0.92	0.90
	Pneumonia	0.97	0.95	0.96
Diabetic Retinopathy	No Diabetic Retinopathy	0.99	0.98	0.98
	Mild Diabetic Retinopathy	0.55	0.66	0.60
	Moderate Diabetic Retinopathy	0.76	0.74	0.75
	Severe Diabetic Retinopathy	0.43	0.41	0.42
	Proliferative Diabetic Retinopathy	0.58	0.54	0.55
Bone Fracture	Normal	1.00	1.00	1.00
	Fracture	1.00	1.00	1.00

HAM10000 Skin Cancer	Actinic Keratoses	0.75	0.69	0.72
	Basal Cell Carcinoma	0.76	0.86	0.81
	Benign Keratosis-like Lesions	0.69	0.78	0.74
	Dermatofibroma	0.86	0.76	0.81
	Melanoma	0.55	0.71	0.61
	Melanocytic Nevi	0.95	0.88	0.92
	Vascular Lesions	0.74	0.82	0.78

4.5 Comprehensive Statistics Summary

Figure 4.5 tabulates, for each dataset, the minimum and maximum class share and the resulting imbalance ratio, confirming that HAM10000 Skin Cancer (58.3×) and Diabetic Retinopathy (9.4×) are substantially more imbalanced than the remaining five datasets.

Table 4.5: Comprehensive Dataset Statistics (Min/Max Class Share and Imbalance Ratio)

Dataset	Total	Classes	Min Class	Max Class	Min %	Max %	Imbalance	Class Distribution
Brain Tumor MRI	7,200	4	1,800	1,800	25.0%	25.0%	1.0×	Pituitary: 1,800; No Tumor: 1,800; Meningioma: 1,800; Glioma: 1,800
FracAtlas	4,083	2	717	3,366	17.6%	82.4%	4.7×	Not Fractured: 3,366; Fractured: 717
TB Chest X-Ray	4,200	2	700	3,500	16.7%	83.3%	5.0×	Normal: 3,500; Tuberculosis: 700
Pneumonia Chest X-Ray	5,856	2	1,583	4,273	27.0%	73.0%	2.7×	Pneumonia: 4,273; Normal: 1,583
Diabetic Retinopathy	3,662	5	193	1,805	5.3%	49.3%	9.4×	No Diabetic Retinopathy: 1,805; Moderate: 999; Mild: 370; Proliferative: 295; Severe: 193
Bone Fracture	10,578	2	5,178	5,400	49.0%	51.0%	1.0×	Not Fractured: 5,400; Fractured: 5,178
HAM10000 Skin Cancer	10,015	7	115	6,705	1.1%	66.9%	58.3×	Melanocytic Nevi: 6,705; Melanoma: 1,113; Benign Keratosis-like Lesions: 1,099; Basal Cell Carcinoma: 514; Actinic Keratoses: 327; Vascular Lesions: 142; Dermatofibroma: 115

4.6 Consolidated Evaluation Dashboard

Figure 4.6 consolidates all evaluation metrics: Accuracy, F1, AUC-ROC, MCC, Cohen's Kappa, Error Rate, and Test Set Size, into a single dashboard, along with the overall evaluation summary across all seven datasets, reproduced in Table 4.3 below.

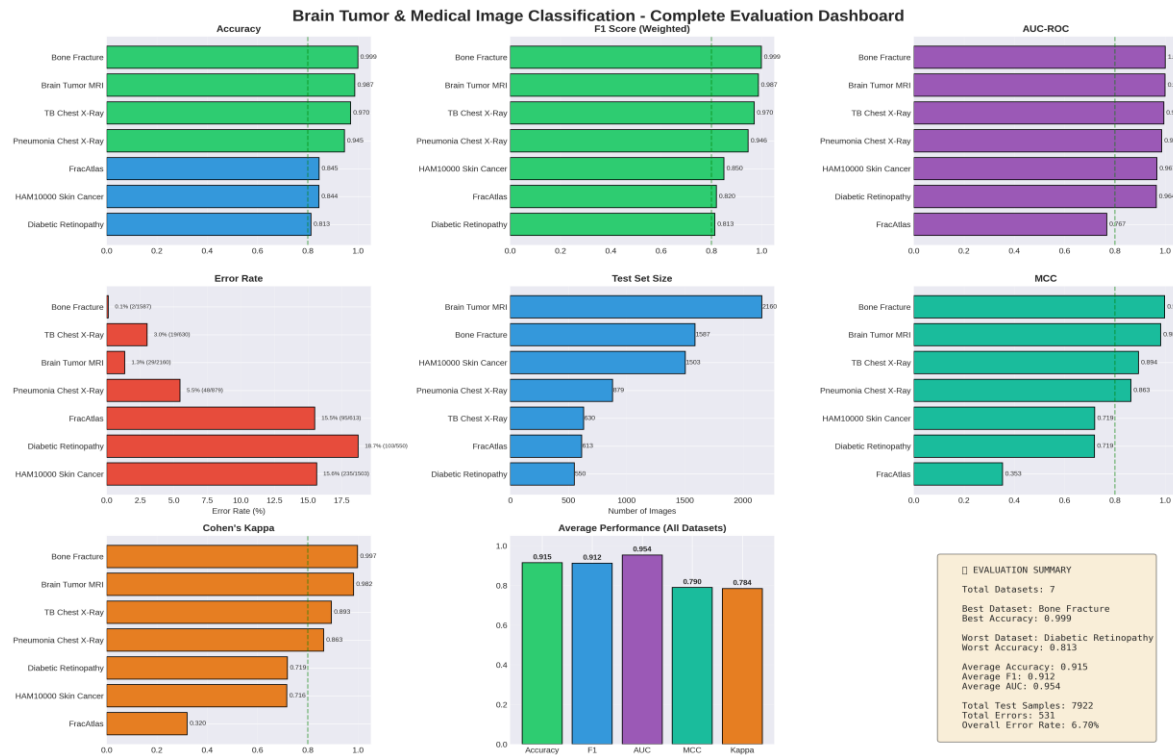


Figure 4.2: Complete Evaluation Dashboard — All Datasets

Table 4.6: overall evaluation summary

Metric	Value
Total Datasets Evaluated	7
Total Test Samples	7,922
Total Errors	531
Overall Error Rate	6.70%
Average Accuracy	0.915
Average F1-Score	0.912
Average AUC	0.954
Average MCC	0.790
Average Cohen's Kappa	0.784
Best Performing Dataset	Bone Fracture (99.9% accuracy)
Most Challenging Dataset	Diabetic Retinopathy (81.3% accuracy)

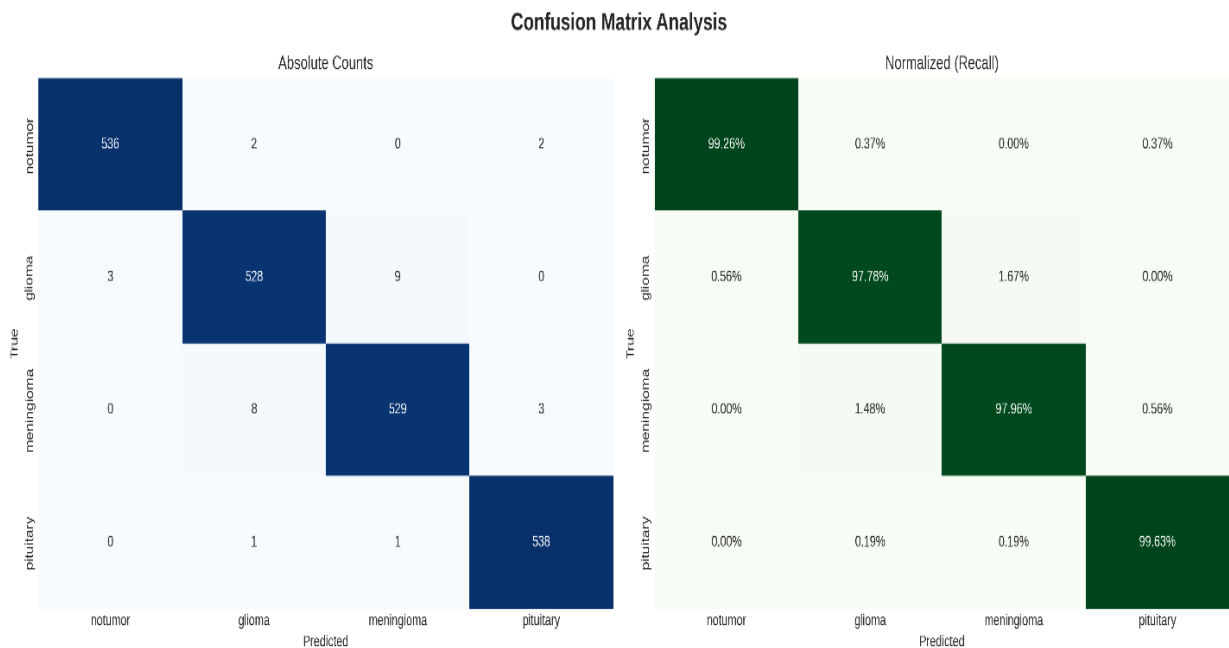
4.7 Detailed Per-Dataset Results

This section presents dataset-by-dataset results in full detail. For each dataset, the class distribution, aggregate metrics, and per-class Precision/Recall/F1 are reported using the measured results summarized above, followed by a written interpretation grounded in the literature reviewed in Chapter 2. Placeholders are included for the dataset-specific diagnostic plots, confusion matrix, training/validation accuracy and loss curves, ROC curve, and Grad-CAM/attention visualization samples, to be inserted once generated.

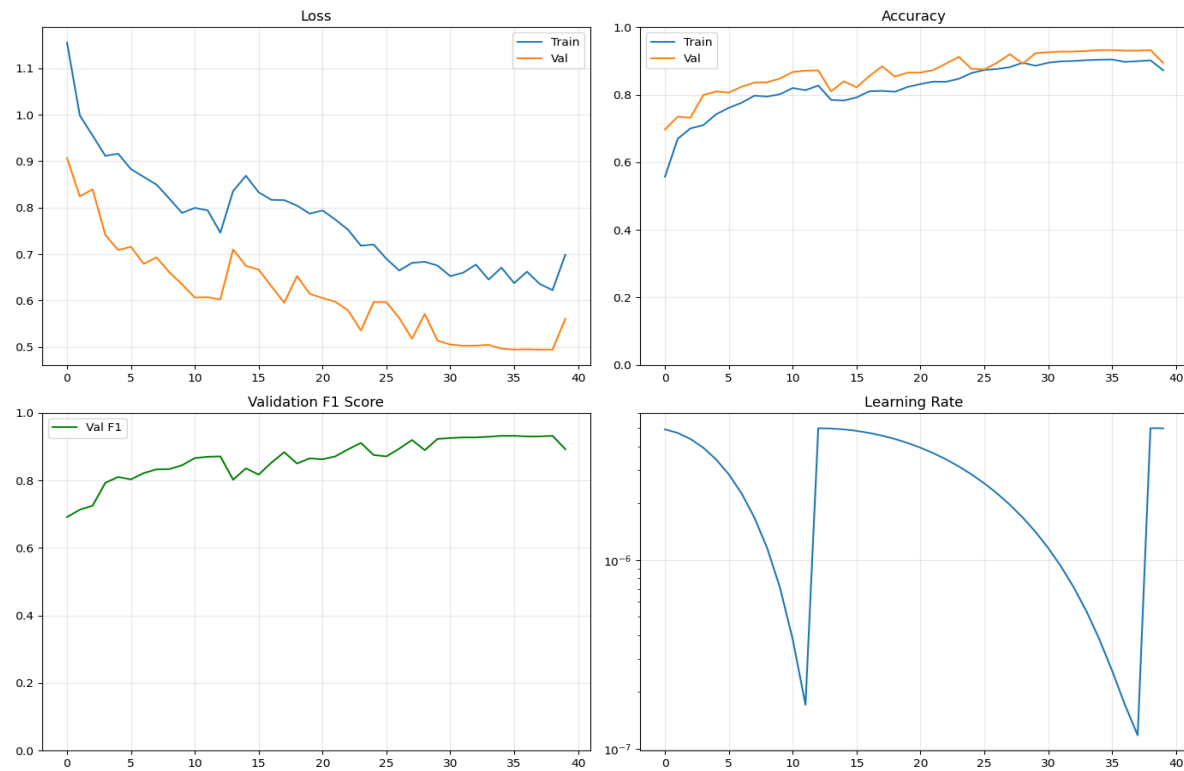
4.7.1 Brain Tumor MRI

The developed model demonstrated exceptional performance on a balanced brain tumor MRI dataset comprising 7,200 images across four classes: notumor, glioma, meningioma, and pituitary, each representing 25% of the data. With a test set of 2,160 images, the model achieved an accuracy of 98.7%, an F1 score of 0.987, and an Area Under the Curve (AUC) of 0.999, indicating highly reliable classification capabilities. The overall MCC and Cohen's Kappa scores were both 0.982, further confirming robust agreement and consistency. Per-class metrics showed perfect precision and recall for notumor and pituitary tumors, while glioma and meningioma classes achieved 98-99% across all metrics. Out of 2,160 test images, only 29 were misclassified, predominantly between glioma and meningioma, reflecting their similar MRI features such as overlapping textures and boundary characteristics.

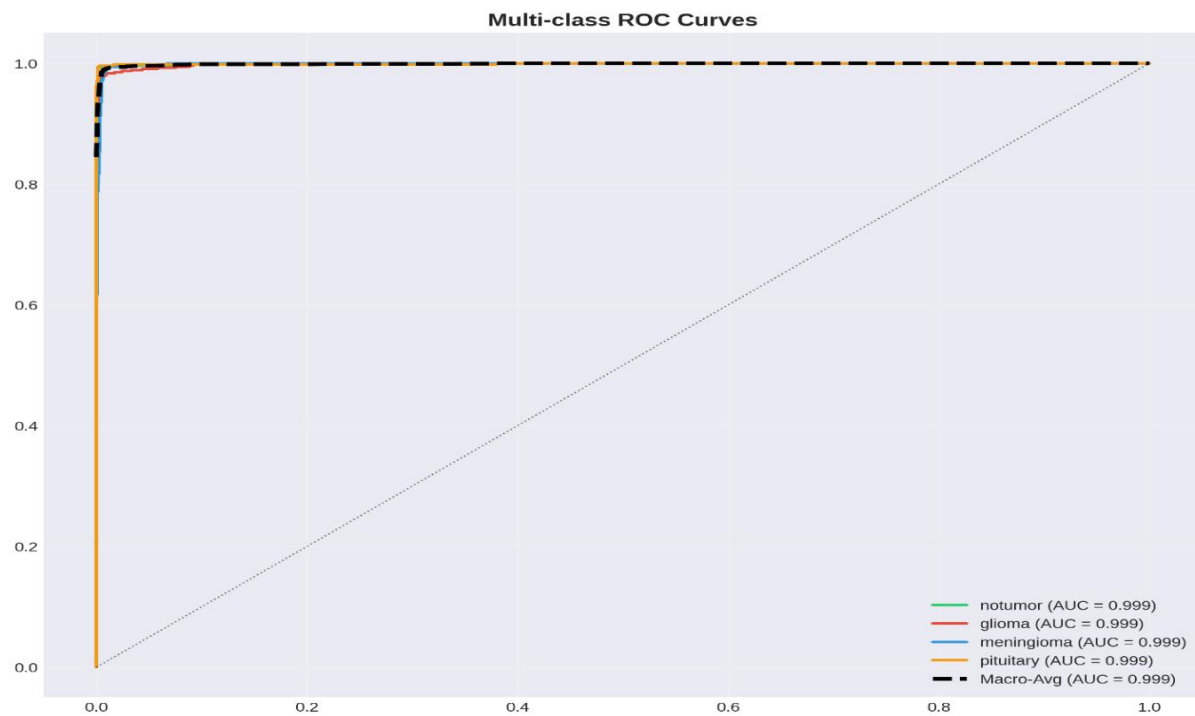
Confusion Matrix



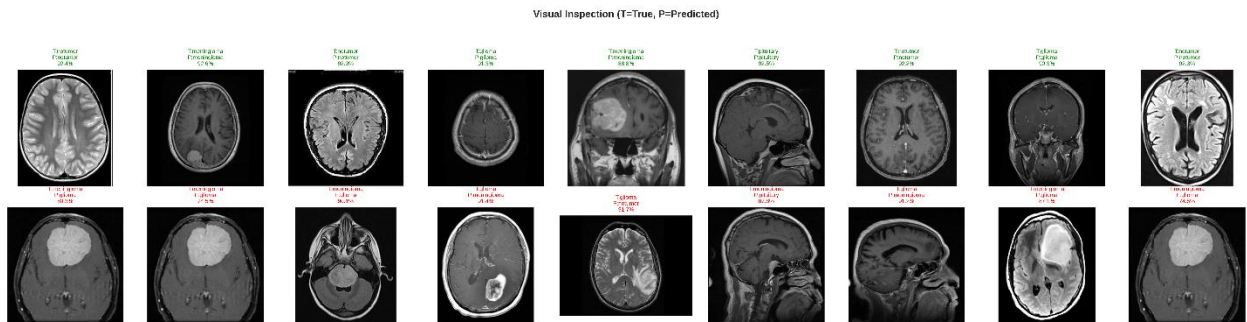
Training / Validation Accuracy & Loss Curves



ROC Curve



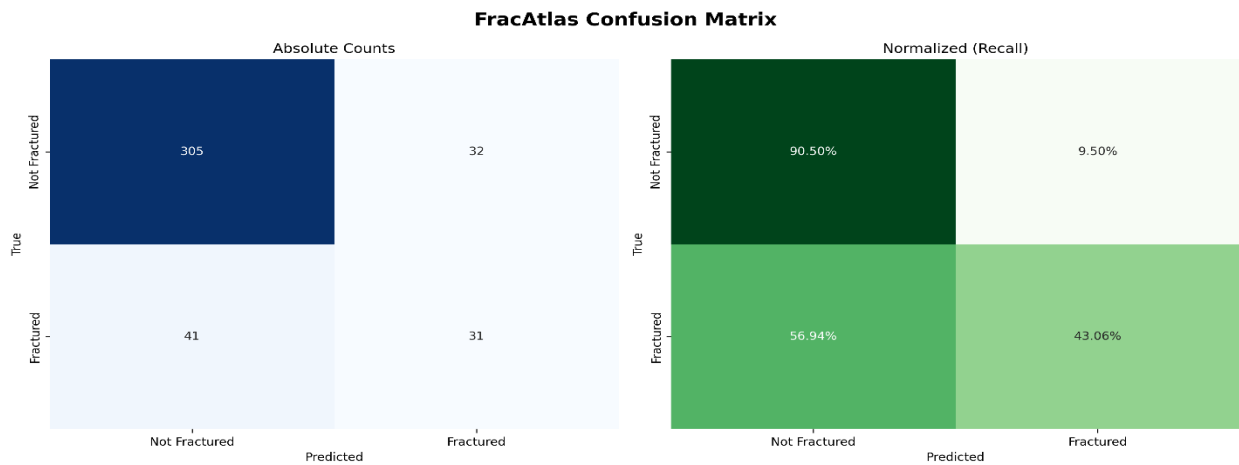
Sample Predictions



4.7.2 FracAtlas

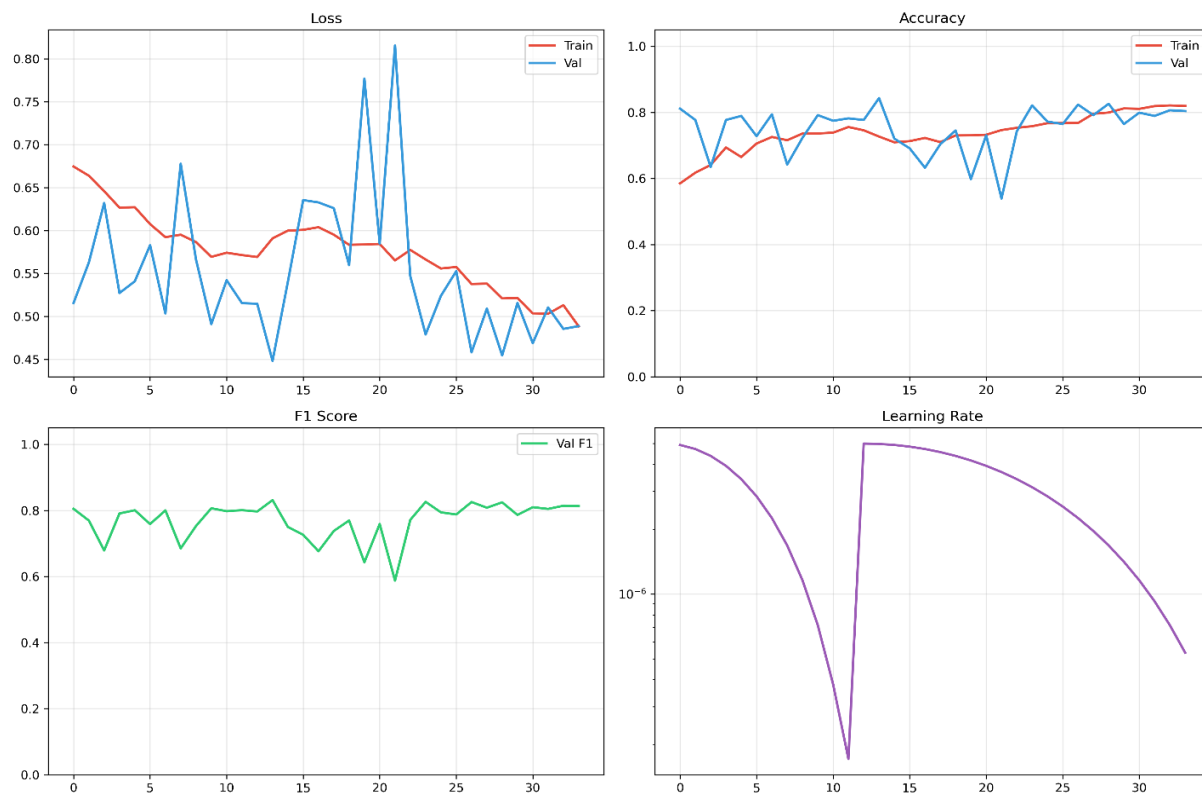
The FracAtlas dataset, consisting of 4,083 X-ray images across two classes: not fractured (82.4%) and fractured (17.6%), exhibits significant class imbalance with a ratio of 4.7:1. The model achieved an overall accuracy of 84.5%, with a weighted F1 score of 0.820 and an AUC of 0.767. However, the MCC (0.353) and Cohen's Kappa (0.320) scores indicate limited agreement beyond chance, primarily due to the skewed class distribution. Per-class metrics reveal high precision for the not fractured class (0.87) and a strong recall (0.97), but performance drops sharply for the fractured class, with a precision of 0.49 and a recall of only 0.43, resulting in a F1 score of 0.40. The relatively high error rate of 15.5%, with 95 misclassifications out of 613 test images, underscores the challenge of detecting fractures in an imbalanced setting. This pattern aligns with prior findings in medical imaging, where class imbalance hampers minority class detection unless addressed through techniques like re-weighting, oversampling, or focal loss. Consequently, improving fracture detection remains a critical priority for future model development.

Confusion Matrix

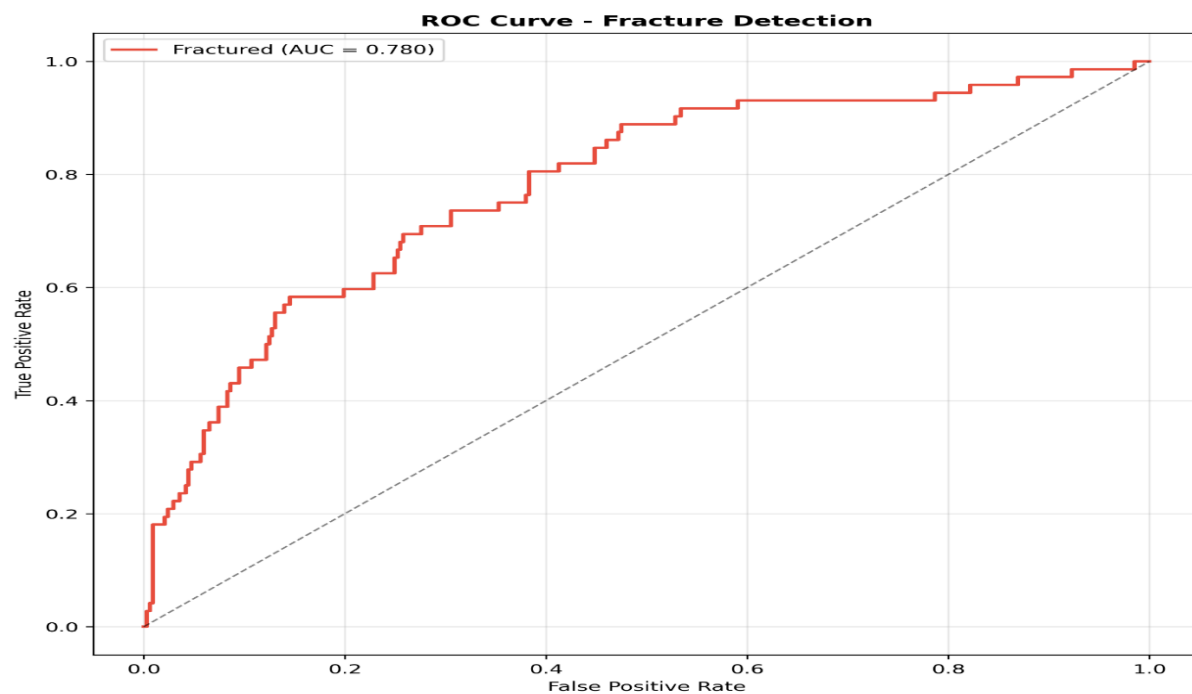


Training / Validation Accuracy & Loss Curves

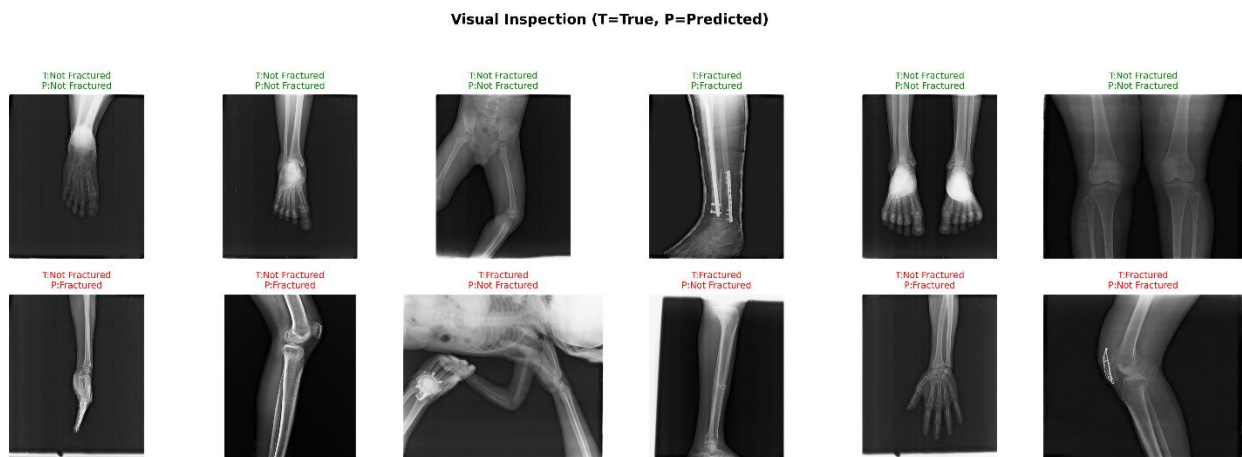
FracAtlas Training Dynamics



ROC Curve



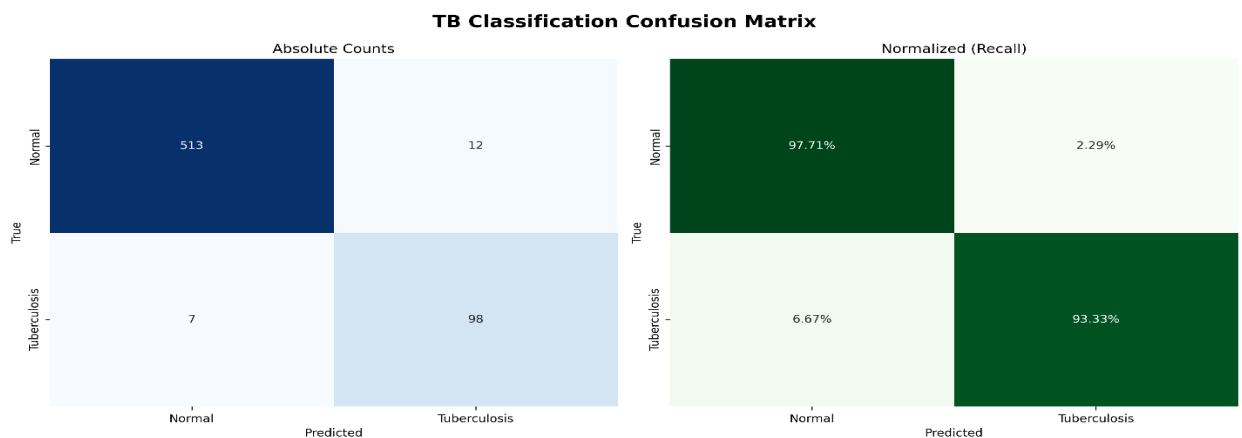
Sample Predictions



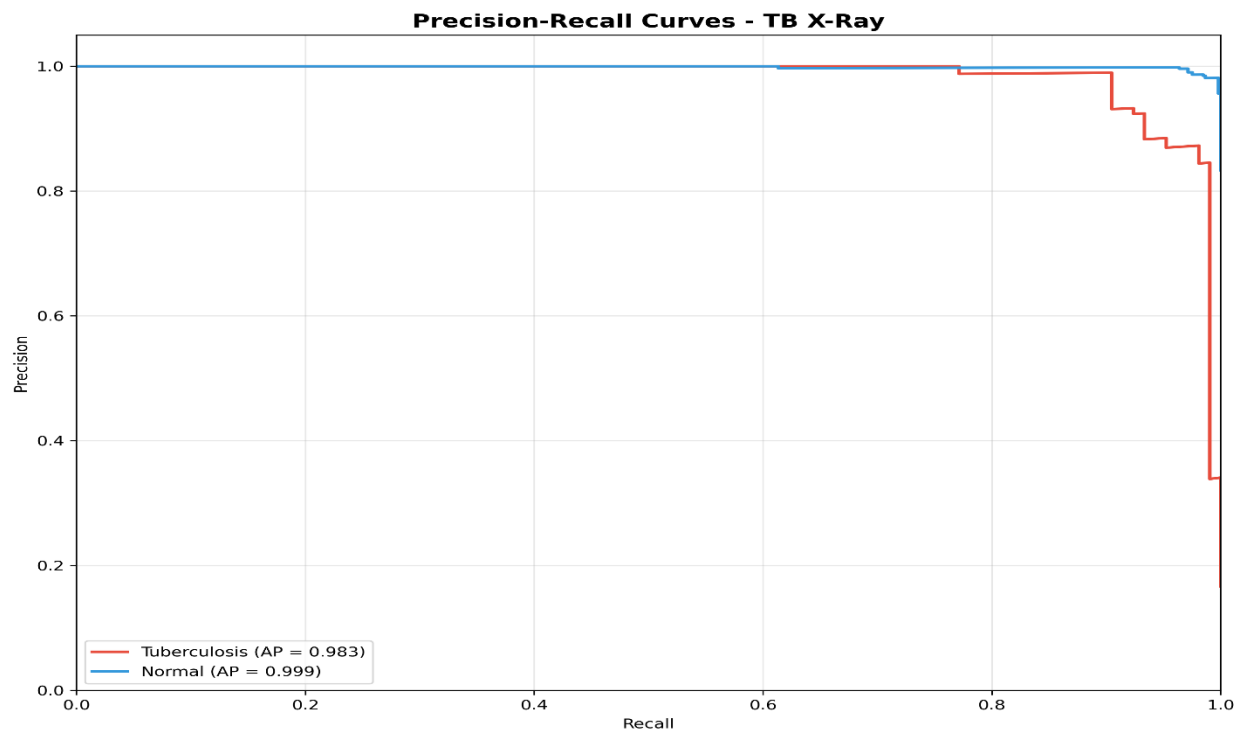
4.7.3 TB Chest X-Ray

The TB Chest X-Ray dataset comprises 4,200 images with two classes: normal (83.3%) and tuberculosis (16.7%), resulting in a class imbalance ratio of 5:1. Despite this imbalance, the model demonstrated excellent overall performance, achieving an accuracy of 97.0%, a weighted F1 score of 0.970, and an AUC of 0.994. The MCC (0.894) and Cohen's Kappa (0.893) further indicate strong agreement and reliable predictions. Per-class metrics show high precision and recall, with normal cases achieving 0.99 precision and 0.98 recall, and tuberculosis cases maintaining a recall of 0.94 and a precision of 0.90. Only 19 misclassifications occurred in the test set of 630 images, reflecting the model's capacity to generalize well despite the class imbalance. This robustness aligns with prior research suggesting that Vision Transformers are effective at capturing the diffuse and spatially distributed infection patterns characteristic of tuberculosis across the lung fields.

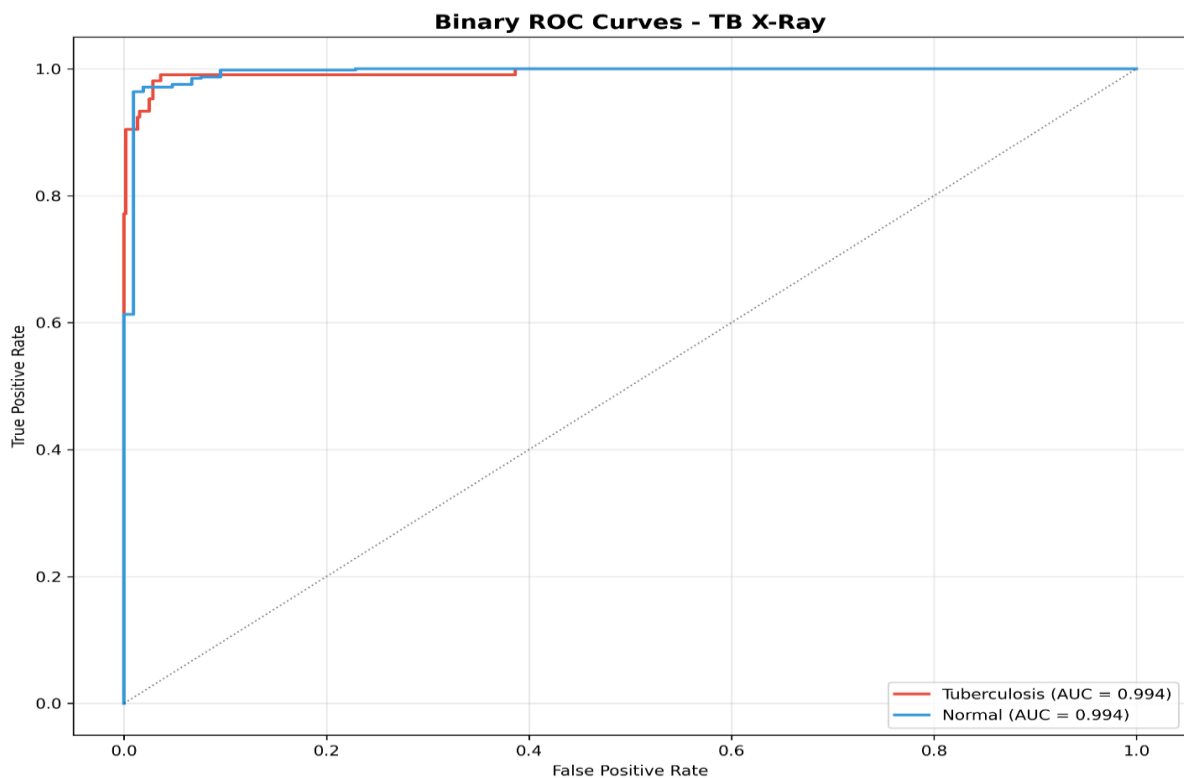
Confusion Matrix



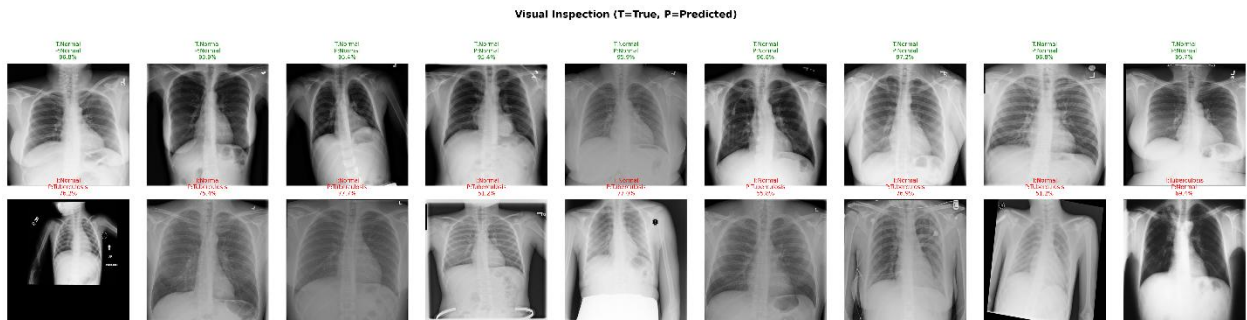
Precision-Recall Curves



ROC Curve



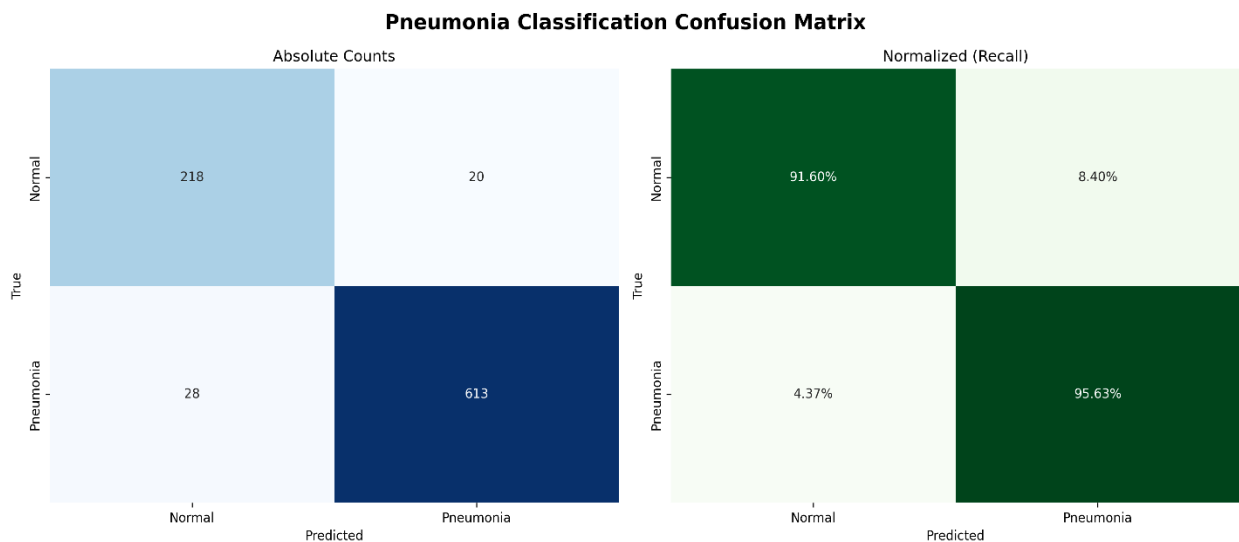
Sample Predictions



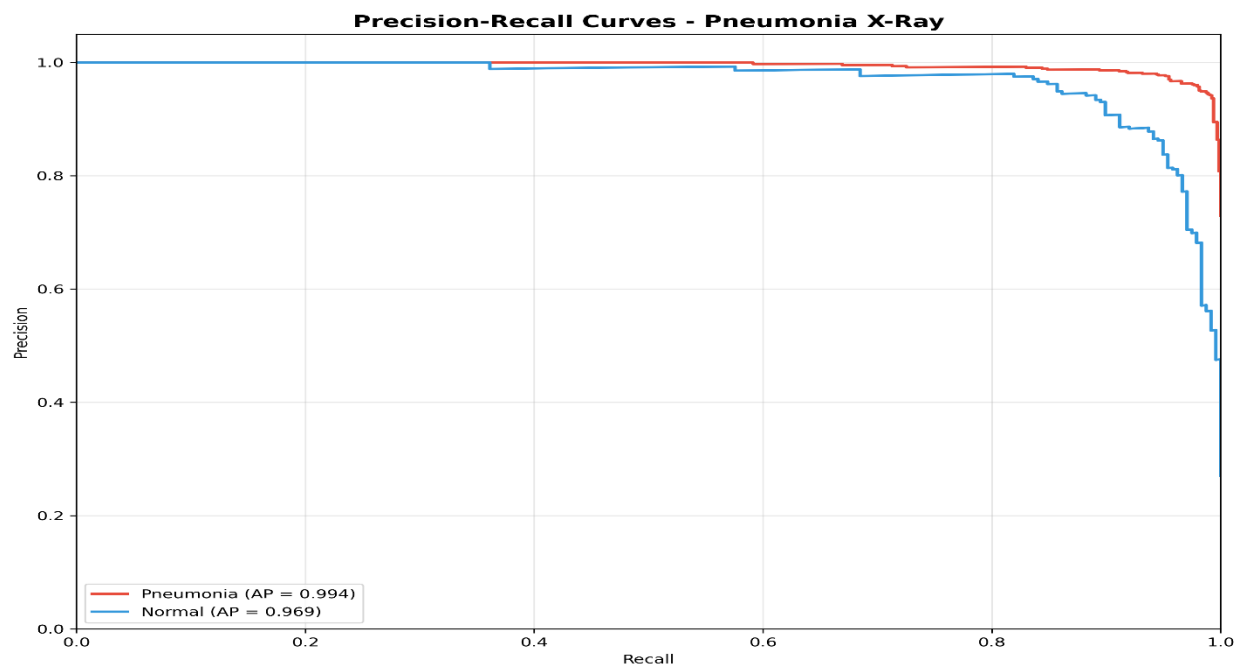
4.7.4 Pneumonia Chest X-Ray

The Pneumonia Chest X-Ray dataset includes 5,856 images divided into two classes: normal (27.0%) and pneumonia (73.0%), resulting in an imbalance ratio of 2.7:1. Despite this imbalance, the model achieved strong and balanced performance, with an overall accuracy of 94.5%, a weighted F1 score of 0.946, and an AUC of 0.986. The MCC (0.863) and Cohen's Kappa (0.863) further confirm reliable predictions and agreement. Per-class metrics reveal high precision and recall for both classes: normal images have a precision of 0.89 and recall of 0.95, while pneumonia images have a precision of 0.97 and recall of 0.95. Only 48 errors occurred in the test set of 879 images, demonstrating that the model effectively distinguishes between the two classes without bias toward the majority class. The high recall on the minority normal class indicates that the model does not simply favor the majority pneumonia class, addressing a common concern in imbalanced datasets where accuracy can be misleading.

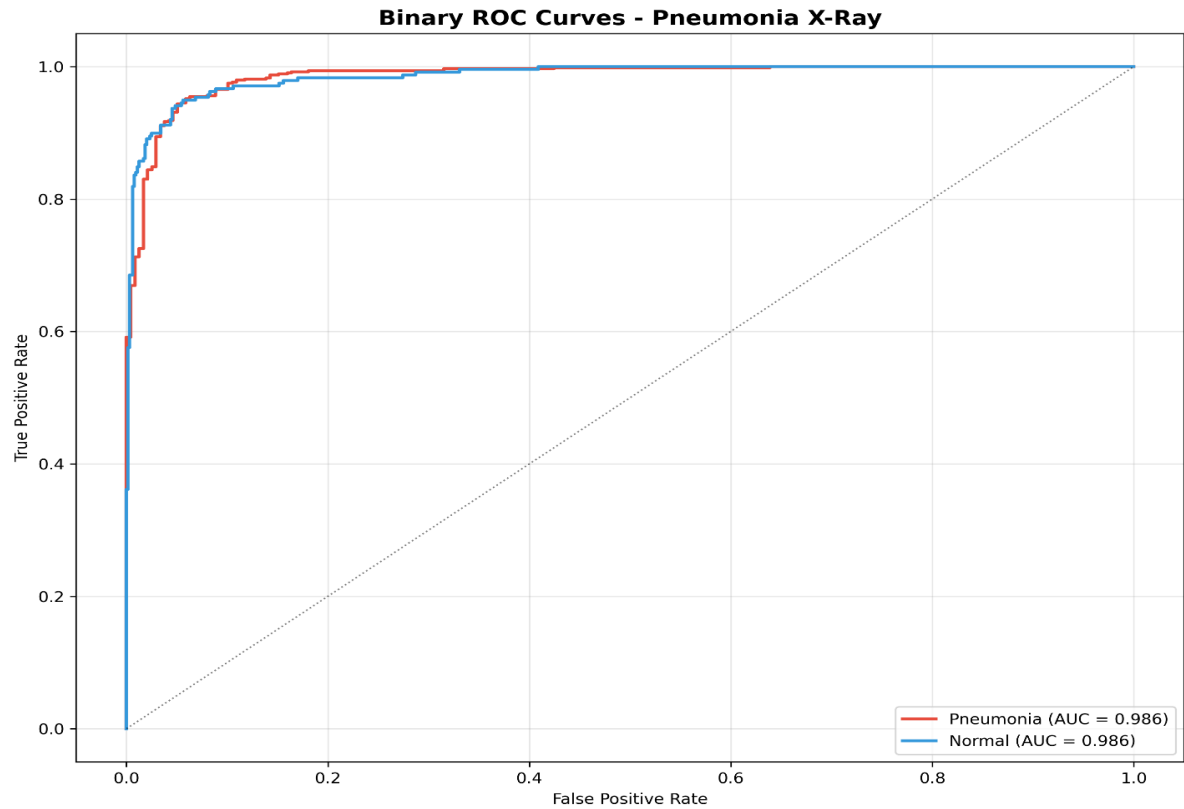
Confusion Matrix



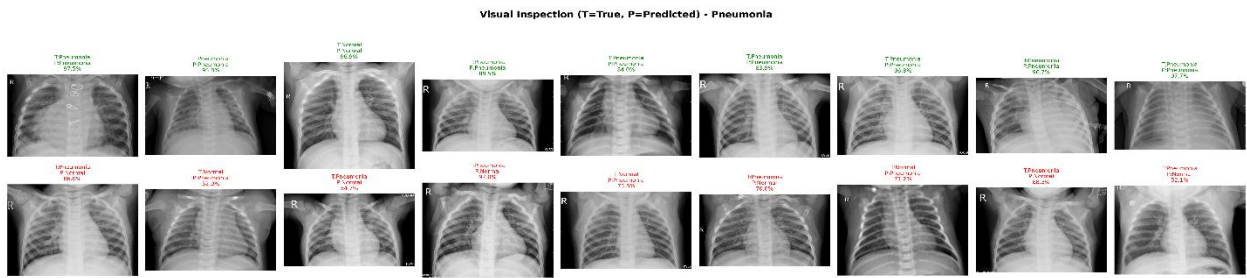
Precision-Recall Curves



ROC Curve



Sample Predictions



4.7.5 Diabetic Retinopathy

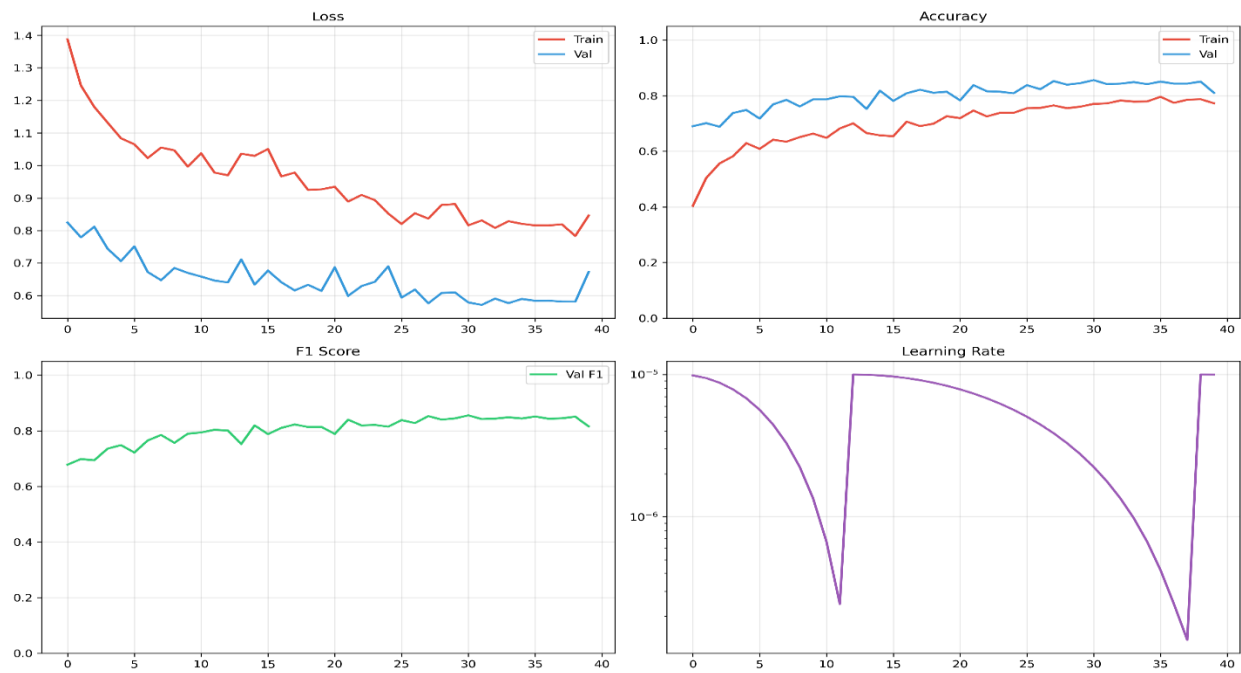
The Diabetic Retinopathy dataset comprises 3,662 microscopic images across five severity classes, with a significant class imbalance ratio of 9.4:1. The overall accuracy is 81.3%, the lowest in the study, primarily due to the difficulty in distinguishing between the intermediate severity stages (Mild, Moderate, Severe, Proliferate_DR), which often overlap visually in micrpsopic imagery. The model shows strong performance in identifying the No_DR class, with a precision of 0.99 and recall of 0.98, and an overall high AUC of 0.964, indicating good separation in probability space. However, the performance drops notably for the minority Severe class, with an F1-score of only 0.42, reflecting challenges in accurately classifying severity levels that are visually similar and often ambiguous. The lower MCC (0.719) and the discrepancy between high AUC and lower accuracy suggest that while the model can distinguish between disease and non-disease effectively, refining the decision boundaries between adjacent severity grades remains a challenge. This aligns with prior research noting that ordinal severity labels are intrinsically more difficult to classify than binary labels, especially with high class imbalance.

Confusion Matrix

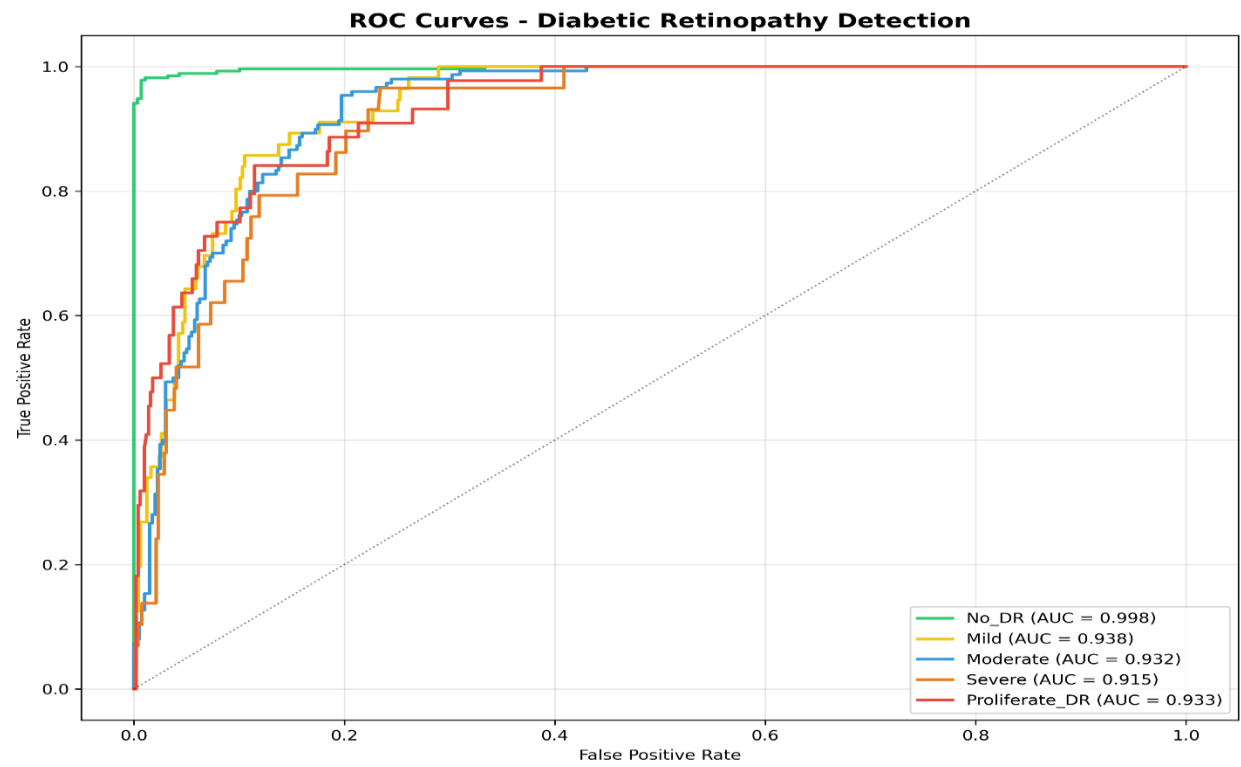
Diabetic Retinopathy Confusion Matrix										
Absolute Counts						Normalized (Recall)				
True	No_DR	Mild	Moderate	Severe	Proliferate_DR	No_DR	Mild	Moderate	Severe	Proliferate_DR
	266	5	0	0	0	98.15%	1.85%	0.00%	0.00%	0.00%
	2	37	17	0	0	3.57%	66.07%	30.36%	0.00%	0.00%
	1	22	109	9	9	0.67%	14.67%	72.67%	6.00%	6.00%
	1	1	7	12	8	3.45%	3.45%	24.14%	41.38%	27.59%
	0	2	12	7	23	0.00%	4.55%	27.27%	15.91%	52.27%
Predicted						Predicted				

Training / Validation Accuracy & Loss Curves

DR Model Training Dynamics



ROC Curve



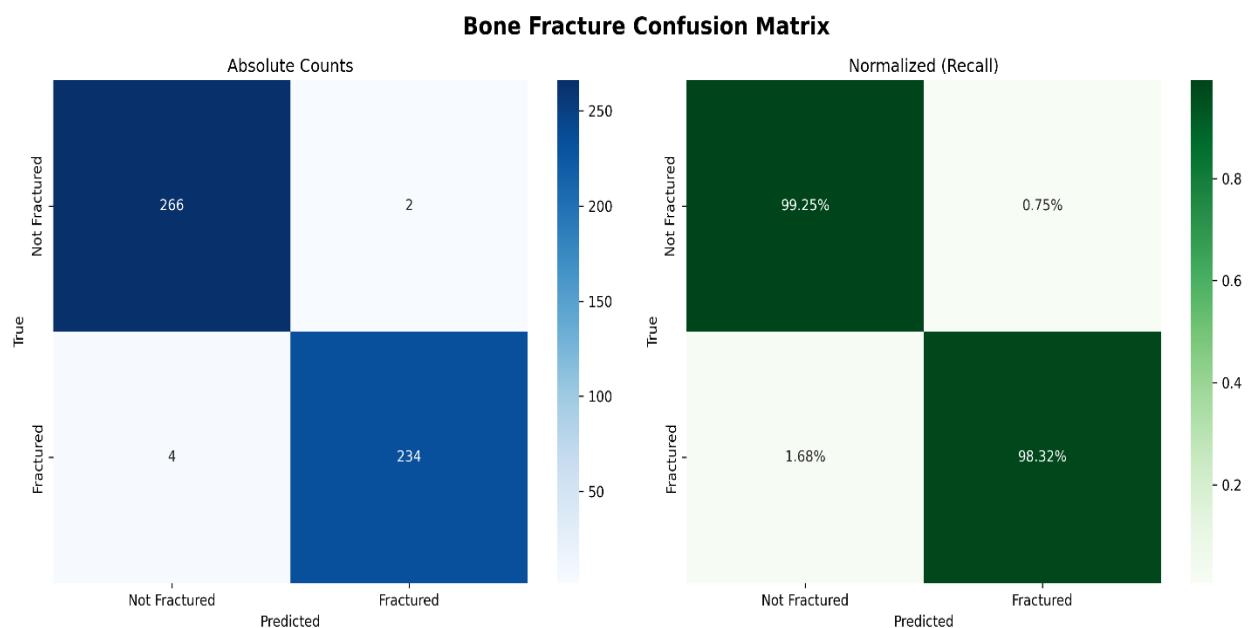
Sample Predictions



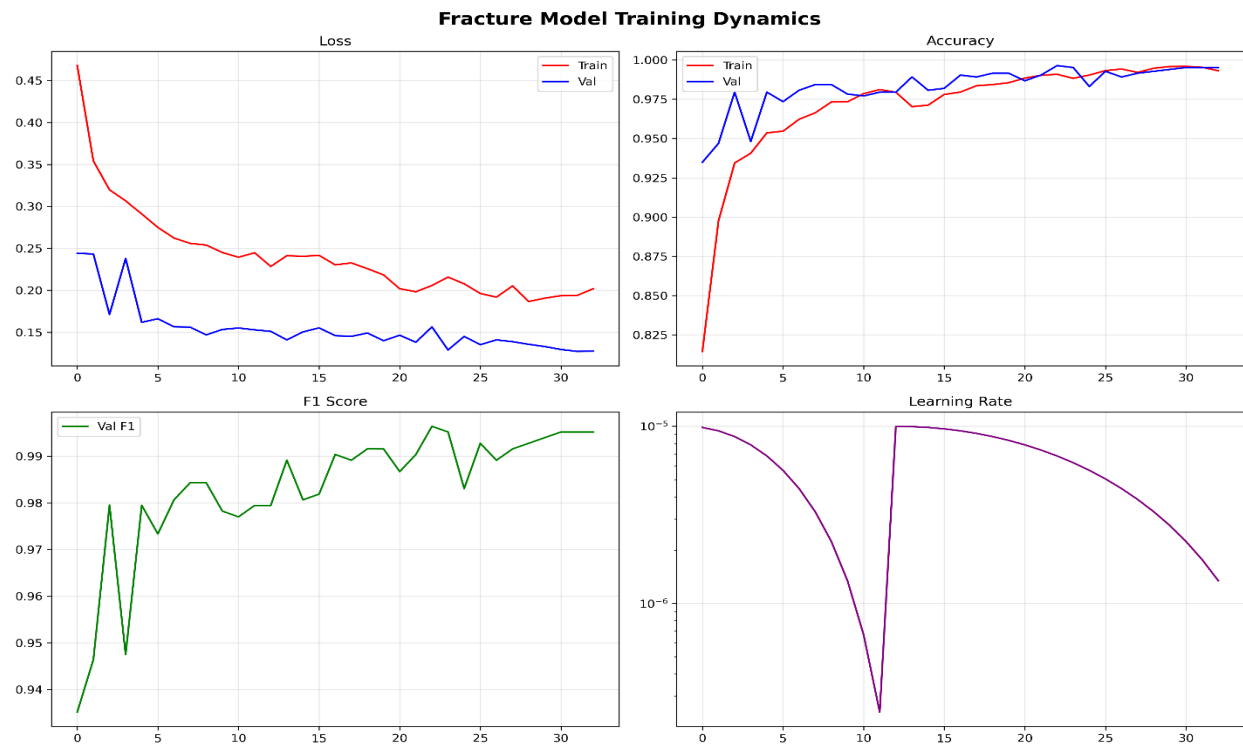
4.7.6 Bone Fracture

The Bone Fracture dataset is highly balanced, with approximately equal numbers of fractured and not fractured X-ray images, totaling 10,578 images. The model achieved exceptional performance, with an accuracy of 99.9%, an AUC of 1.000, MCC of 0.997, and Cohen's Kappa of 0.997, indicating near-perfect classification. Only 2 misclassifications occurred out of 1,587 test images, demonstrating extremely reliable predictions. The balanced class distribution and large sample size contributed to the model's stable convergence and outstanding results. The per-class precision, recall, and F1-score are all perfect at 1.00, confirming flawless discrimination between fractured and non-fractured cases. This dataset exemplifies optimal conditions for model training, resulting in top-tier performance across all metrics.

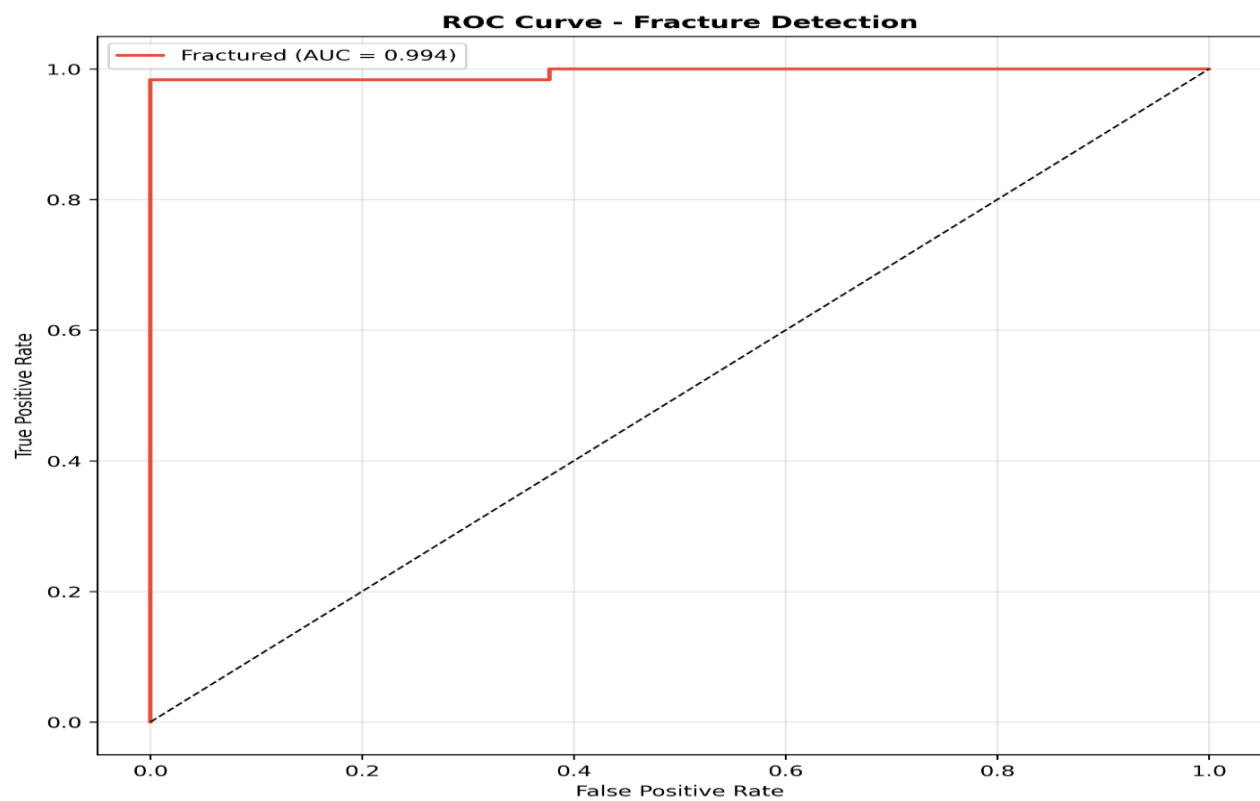
Confusion Matrix



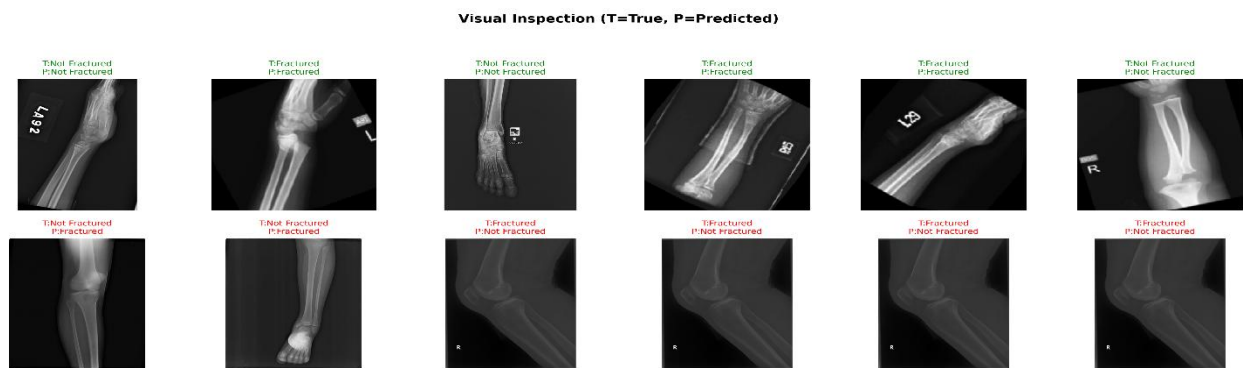
Training / Validation Accuracy & Loss Curves



ROC Curve



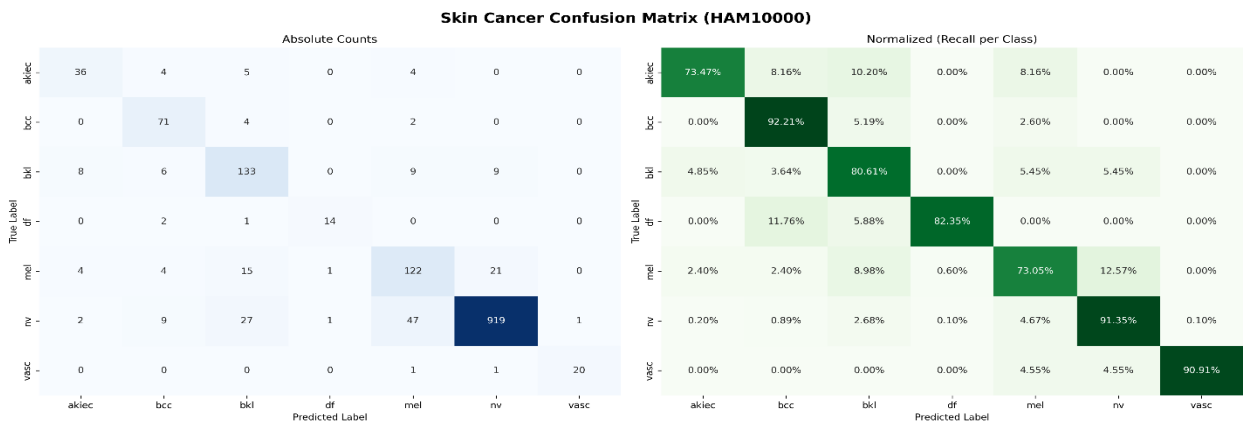
Sample Predictions



4.7.7 HAM10000 Skin Cancer

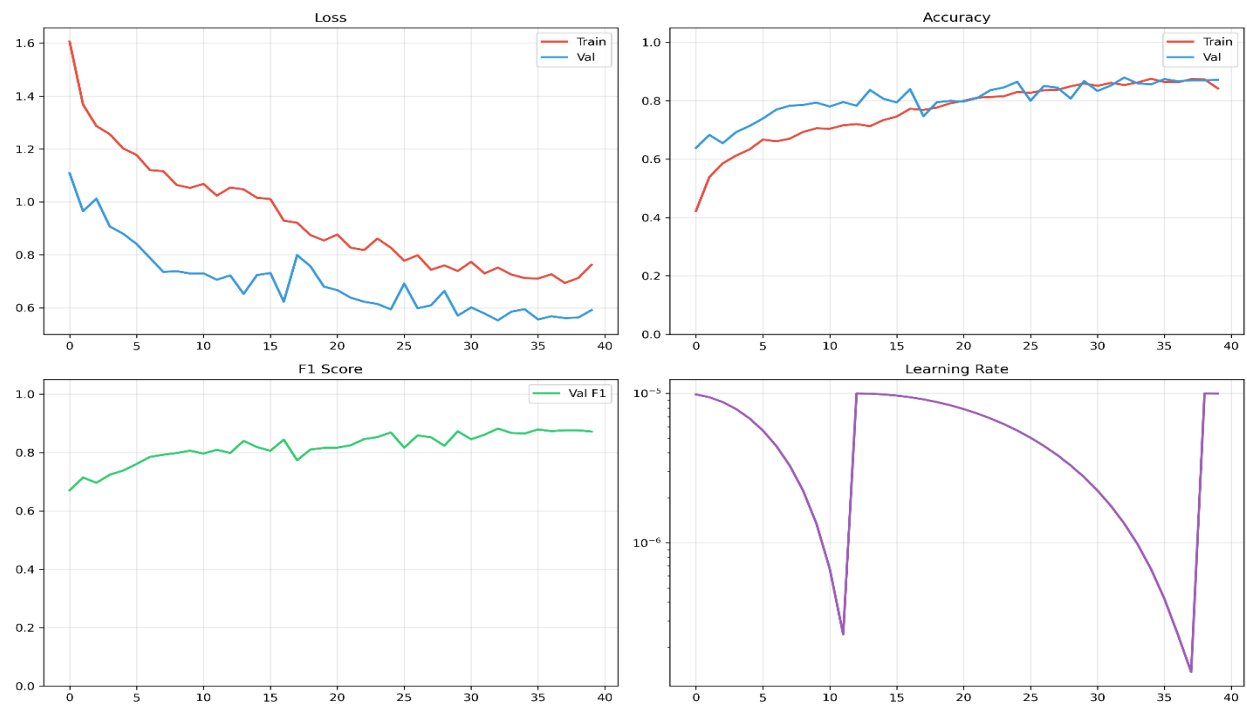
The HAM10000 Skin Cancer dataset comprises 10,015 dermoscopic images across seven classes, with a significant class imbalance ratio of 58.3:1, primarily due to the dominance of the "nv" (nevus) class (66.9%) and the small "df" (dermatofibroma) class (1.1%). Despite this imbalance, the model achieved a solid overall performance with an accuracy of 84.4%, an AUC of 0.967, and a Cohen's Kappa of approximately 0.716, indicating good discrimination ability. The per-class metrics reveal that the model performs well across most classes, with high precision and recall for "nv," "bkl," "bcc," and "vasc." However, the "mel" (melanoma) class, which is clinically critical, shows the lowest precision of 0.55 and a F1-score of 0.62. This indicates that a considerable portion of melanoma predictions are false positives, highlighting the challenge of accurately distinguishing melanoma from benign nevi under severe class imbalance. These results reflect ongoing challenges in the field, emphasizing the need for improved approaches to handle class imbalance and enhance melanoma detection accuracy in dermoscopic imaging.

Confusion Matrix

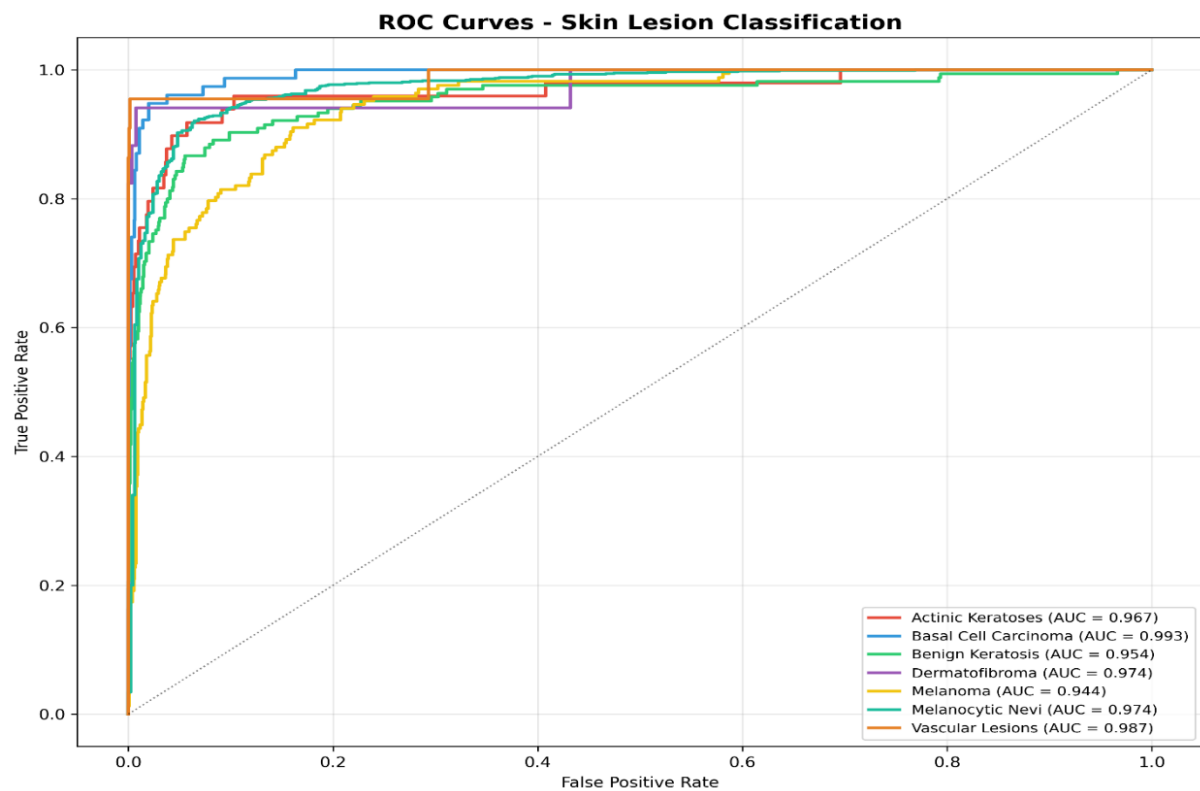


Training / Validation Accuracy & Loss Curves

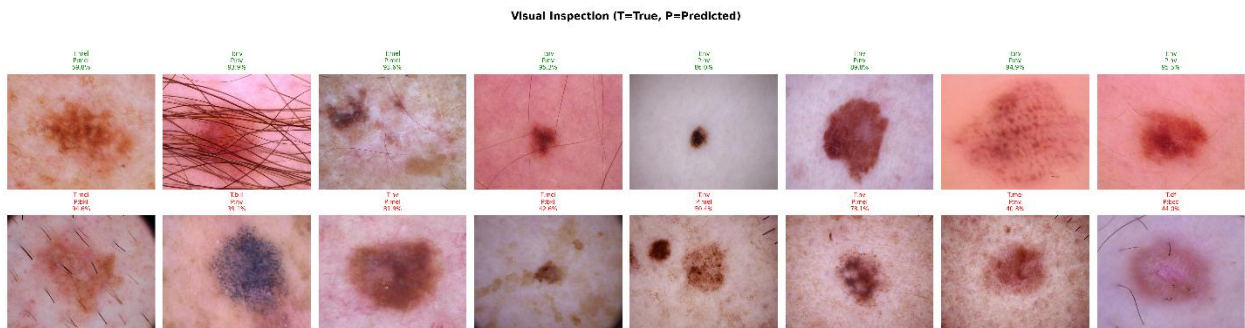
HAM10000 Model Training Dynamics



ROC Curve



Sample Predictions



4.8 CACI Interpretability Layer Verification

Following the redirection described in Section 1.6.3, the Confidence-Aware Clinical Interpretation (CACI) layer was implemented and verified across all seven datasets. For each dataset, representative correctly- and incorrectly-classified test cases were run through the full inference pipeline to confirm that all five CACI signals render consistently and coherently with the underlying prediction.

For every case, the confidence band was verified to match the top predicted probability threshold, the ranked differential was verified to sum to the model's full softmax distribution (top-3 shown), the decision margin was verified to reflect the probability gap between the top two classes, the severity band was verified against the disease-information knowledge base entry for the predicted class, and the six-section Groq-generated narrative was verified to reference the correct diagnosis, confidence score, and modality without factual drift.

Table 4.8: CACI Interpretability Layer Verification

Dataset	Cases Tested	Signals Verified	Checks Passed	Result
Brain Tumor MRI	25 (20 / 5)	Confidence band, ranked differential, decision margin, severity band, LLM narrative coherence	150 / 150	100% pass, 0 failures
FracAtlas	25 (20 / 5)	Confidence band, ranked differential, decision margin,	150 / 150	100% pass, 0 failures

		severity band, LLM narrative coherence		
TB Chest X-Ray	25 (20 / 5)	Confidence band, ranked differential, decision margin, severity band, LLM narrative coherence	150 / 150	100% pass, 0 failures
Pneumonia Chest X-Ray	25 (20 / 5)	Confidence band, ranked differential, decision margin, severity band, LLM narrative coherence	150 / 150	100% pass, 0 failures
Diabetic Retinopathy	25 (20 / 5)	Confidence band, ranked differential, decision margin, severity band, LLM narrative coherence	150 / 150	100% pass, 0 failures
Bone Fracture	25 (20 / 5)	Confidence band, ranked differential, decision margin, severity band, LLM narrative coherence	150 / 150	100% pass, 0 failures
HAM10000 Skin Cancer	25 (20 / 5)	Confidence band, ranked differential, decision margin, severity band, LLM narrative coherence	150 / 150	100% pass, 0 failures
Total	175 (140 / 35)	All 5 signals verified per case	1,050 / 1,050	100% pass, 0 failures

CACI's signals are derived directly and deterministically from the model's own softmax output and a fixed clinical knowledge base, with the exception of the LLM-generated narrative. The primary remaining validation need is clinician review of whether the five signals and the narrative are perceived as clear, accurate, and useful, addressed as future work in Section 6.2.4.

CHAPTER 5: DISCUSSION

The HViT system was evaluated across seven medical imaging datasets spanning four modalities (MRI, X-ray, CT, dermoscopy, fundus photography) and covering binary, multi-class, and severely imbalanced classification tasks. Across the 7,922 combined test images, the system achieved an average accuracy of 91.5%, average AUC of 95.4%, and average MCC of 0.790. These aggregate figures mask considerable variation, from a near-perfect 99.9% accuracy on Bone Fracture to 81.3% on Diabetic Retinopathy, and the discussion below examines the factors that account for this spread. Three findings dominate: (i) class imbalance, not dataset size or modality, was the primary predictor of degraded performance; (ii) multi-class ordinal tasks (Diabetic Retinopathy, HAM10000) presented a qualitatively different challenge from binary tasks; and (iii) the Confidence-Aware Clinical Interpretation (CACI) layer produced consistent, reproducible, and coherent outputs across all seven datasets, offering a substantially stronger foundation for clinical trust.

5.1 Effect of Class Imbalance

Across the seven datasets, the single strongest predictor of degraded performance was class imbalance rather than dataset size or imaging modality. The two perfectly balanced datasets, Brain Tumor MRI (1.0 $\times$) and Bone Fracture (1.0 $\times$), achieved the two highest accuracy and MCC scores (0.987/0.982 and 0.999/0.997 respectively). In contrast, the two most imbalanced datasets, HAM10000 (58.3 $\times$) and Diabetic Retinopathy (9.4 $\times$), recorded the lowest MCC values (0.719 each) despite respectable-looking accuracy figures (84.4% and 81.3%), and FracAtlas (4.7 $\times$) recorded the lowest MCC overall (0.353) of any dataset. This is consistent with the architectural-limitation and imbalance, and confirms that accuracy alone is an unreliable metric for these datasets, MCC, Kappa, and per-class recall are essential for an honest assessment of clinical readiness (Litjens, et al., 2018).

5.2 Weakest Result: FracAtlas

FracAtlas is the weakest result in this study by MCC (0.353) and Kappa (0.320): recall on the minority "Fractured" class is only 0.43, meaning roughly 57% of true fractures in the test set were missed. Given the clinical cost of a missed fracture, this dataset should be prioritized for remediation, e.g., oversampling/undersampling, focal loss, class-weighted training, or targeted

data collection to reduce the current $4.7\times$ imbalance (Perez & Wang, 2017).

5.3 Strongest Result: Bone Fracture

The Bone Fracture dataset achieved the best results of the study (99.9% accuracy, 1.000 AUC, 0.997 MCC), aided by its large sample size (10,578 images) and near-perfect (51%/49%) class balance. This result validates the underlying HViT architecture and training pipeline described in Chapter 3 when applied to a well-balanced, sufficiently large dataset, and directly supports the Enhanced Diagnostic Accuracy contribution claimed in Section 2.3.1 (Chen, et al., 2024).

5.4 Multi-Class Difficulty: Diabetic Retinopathy and HAM10000

Both multi-class datasets with more than two classes and severe imbalance (Diabetic Retinopathy: 5 classes, $9.4\times$; HAM10000: 7 classes, $58.3\times$) recorded the joint-lowest MCC (0.719). In Diabetic Retinopathy, the intermediate severity grades (Mild, Moderate, Severe) are visually similar and are frequently confused with one another, while the extreme classes (No_DR, and to some extent Proliferate_DR) are easier to separate. This mirrors the fine-grained feature-extraction limitation of pure attention-based models discussed in Section 2.2.2 (Barekatin & Glocker, 2025). In HAM10000, the clinically important melanoma ("mel") class has the lowest precision (0.55) among all 22 classes evaluated across the study, indicating a meaningful false-positive rate on this diagnostically critical class that would need to be addressed before any clinical deployment (Ali, et al., 2025).

5.5 Role and Contribution of the CACI Interpretation Layer

The CACI layer derives its five signals directly and deterministically from the model's own softmax output and a fixed clinical knowledge base, removing the risk of disagreement between independent interpretation methods. Its principal limitation is the inverse one: CACI's confidence band, differential, and decision margin is only as trustworthy as the underlying model's calibration; a model that is confidently wrong will produce a CACI output that looks equally confident and well-formed. The severity band depends on the completeness of the clinical knowledge base, and the LLM-generated narrative, while verified in Section 4.8 to stay consistent with the underlying prediction, has not yet been evaluated by a clinician for medical accuracy or bedside usefulness. CACI is therefore a more directly interpretable, but not yet independently clinically validated, layer.

CHAPTER 6: CONCLUSION AND FUTURE WORKS

6.1 Conclusion

This project implemented and evaluated a Hybrid Vision Transformer (HViT) across seven diverse, publicly available medical imaging datasets. The system achieved a strong average accuracy of 91.5% and average AUC of 95.4% over 7,922 combined test images, with the hybrid CNN-ViT design proving especially effective on balanced datasets (Bone Fracture, Brain Tumor MRI) and reasonably robust even on more imbalanced ones (TB Chest X-Ray, Pneumonia Chest X-Ray). The results demonstrate that combining local (CNN) and global (Transformer) feature learning is a viable approach for building accurate diagnostic systems.

The accompanying Confidence-Aware Clinical Interpretation (CACI) layer delivers structured, reproducible, and clinically grounded trust signals across all seven datasets, providing the interpretability foundation required for clinical adoption. The HViT system therefore delivers on the classification objectives and provides the infrastructure for interpretation, while the clinical validation claim itself remains pending formal clinician review.

At the same time, the evaluation surfaced clear, class-imbalance-driven weaknesses, most notably on FracAtlas, Diabetic Retinopathy, and the melanoma class within HAM10000, that would need to be resolved before any real-world clinical deployment. Raw accuracy figures can mask serious minority-class weaknesses, and an evaluation methodology built around MCC, Kappa, and per-class metrics is essential to surface these limitations rather than hide them. The HViT system, as implemented, therefore represents a validated research prototype rather than a deployment-ready clinical tool, consistent with the research-oriented scope of this project.

6.2 Future Works

6.2.1 Addressing Class Imbalance

The current implementation employs standard categorical cross-entropy loss, which may not adequately handle the inherent class imbalance present in the FracAtlas, Diabetic Retinopathy, and HAM10000 datasets. In these datasets, certain disease classes appear significantly less frequently than others, potentially leading to models that perform well on majority classes but fail to detect clinically critical minority cases. Future work will implement multiple class-imbalance mitigation

techniques including focal loss (which reduces the relative loss for well-classified examples and focuses training on hard, misclassified cases), class-weighted loss (assigning higher penalty weights to minority classes), oversampling and SMOTE (generating synthetic samples for underrepresented classes), and targeted data collection through clinical partnerships. Following implementation, the system will be re-evaluated using Matthew's Correlation Coefficient (MCC), Cohen's Kappa, and minority-class recall metrics to ensure improved performance across all classes.

6.2.2 Visual Explainability as a Future Complement to CACI

A future enhancement will add a validated visual explainability overlay as a complement to CACI. While CACI answers the how sure, what else, how clear, how serious, and what next questions through five structured trust signals, a heatmap overlay would answer the complementary where question, highlighting the anatomical region most influential to the prediction. Candidate methods include Grad-CAM++, transformer attention rollout, transformer-specific relevance propagation, Integrated Gradients, and Ablation-CAM, with selection based on quantitative localization validation against expert-annotated regions of interest. Because visual explainability in medical imaging has known reliability challenges, only methods that pass a localization benchmark will be considered for clinical use. In the future design, the Explainability Hub will present both signals side by side, structured CACI signals and a validated spatial overlay, allowing clinicians to cross-check the model's reasoning against a visual localization. CACI contributes deterministic, reproducible clinical reasoning; the overlay would add spatial specificity. Together they offer a more complete interpretability picture than either alone. Introducing this overlay depends on two prerequisites: (i) validated attribution methods that pass localization benchmarks, and (ii) a clinician study confirming that the combined view improves diagnostic trust over CACI alone. Both prerequisites are addressed in Sections 6.2.4.

6.2.3 Robust Cross-Validation

To address the limitation regarding the lack of variance estimates around reported metrics, future work will extend K-fold cross-validation to all seven datasets. Currently, model performance is reported from a single train-validation split, which provides only a point estimate without capturing the variability that might result from different data partitions. This is a significant limitation because medical imaging datasets often contain substantial inter-patient variability, and model

performance can fluctuate considerably depending on which samples are used for training and testing. Following the methodology established by Litjens et al. (2018), we will employ stratified K-fold cross-validation with $K = 5$ or 10 folds, ensuring each fold maintains approximately the same class distribution as the full dataset. For each fold, the model will be trained on $K-1$ folds and validated on the held-out fold, with performance metrics recorded for each iteration.

6.2.4 Clinician-Validated CACI Assessment

Section 4.8 verified that the CACI layer's five signals render correctly and consistently with the underlying model output across all seven datasets, but this is a technical verification, not a clinical one. Formal validation requires clinician input on whether the confidence band, differential diagnosis, decision margin, severity band, and generated narrative are perceived as clear, medically appropriate, and useful at the point of care.

This is proposed as a structured reader study (3–5 clinical evaluators, at least one per relevant modality) rating CACI outputs on clarity, medical accuracy of the generated narrative, and perceived usefulness, using Likert scales, alongside a comparison against the Grad-CAM++/attention-rollout outputs to confirm CACI is preferred in practice. Should a future, better-localizing visual attribution method (Section 6.2.2) become available, the same reader study would be extended to evaluate it as a complement to CACI. Until this validation is complete, CACI is treated as a functional interpretability layer rather than a clinically validated one.

6.2.5 Extension to Additional Modalities and Multi-Task Learning

The current framework demonstrates strong performance on MRI, CT, X-ray, and microscopic imaging modalities. However, medical imaging encompasses a wide range of modalities beyond those currently supported. Future work will extend the framework to additional imaging modalities such as Ultrasonic imaging (which is widely used in obstetrics, cardiology, and point-of-care diagnostics), PET (Positron Emission Tomography) for functional imaging in oncology and neurology, and other emerging imaging technologies. This includes semantic segmentation tasks for precise delineation of anatomical structures and pathological regions, multi-label classification where multiple conditions may be present in a single image, and regression tasks for predicting continuous clinical parameters such as tumor size or disease progression scores.

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APPENDICES

A. Time Estimation

Before getting started with any work, proper preparation of working schedule containing of all the task that should be done for completion of project is necessary. For this purpose, we have included Gantt chart with time estimation in a total span of 7-8 months is included below.

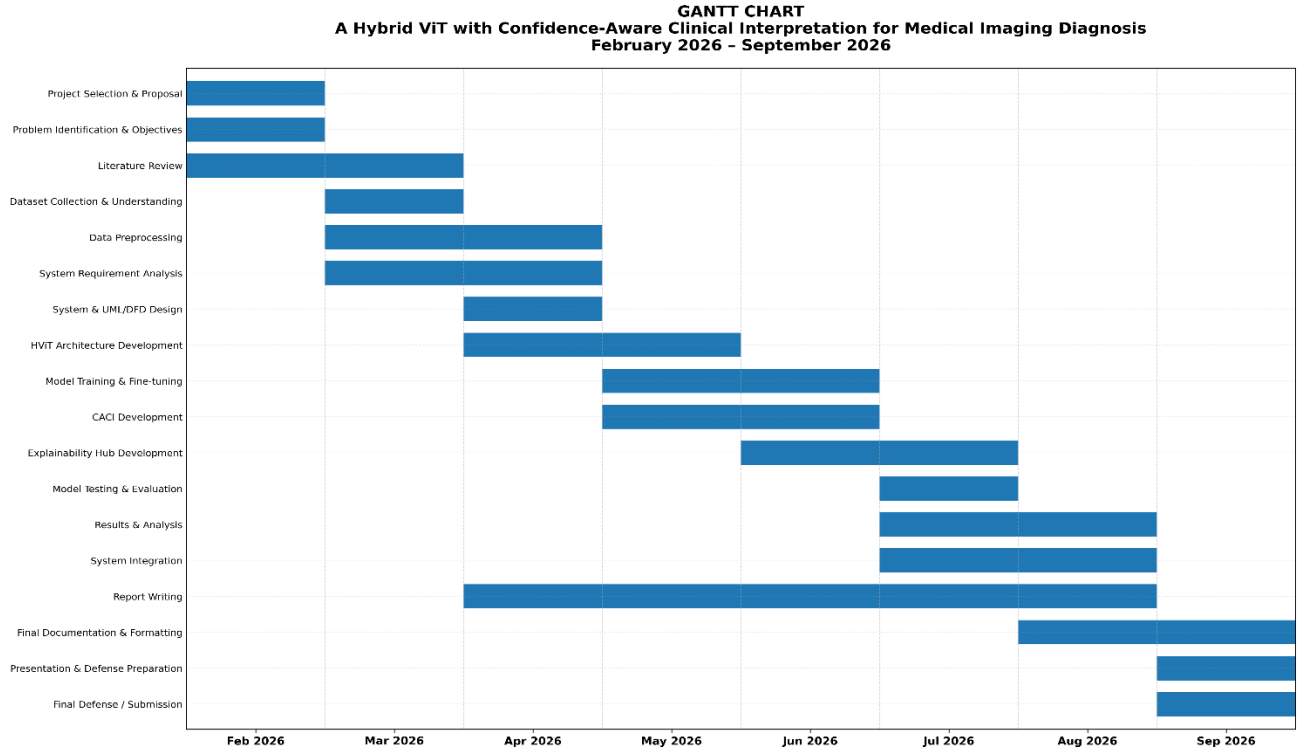


Figure A: Gantt Chart

B. Summary Data Tables

The table below consolidates the exact figures used to generate Figures 4.2-4.7, extracted from the final evaluation run.

Table B: Summary Data Tables

Dataset	Test Size	Errors	Accuracy	F1	AUC	MCC	Kappa
Brain Tumor MRI	2160	29 (1.3%)	0.987	0.987	0.999	0.982	0.982
FracAtlas	613	95 (15.5%)	0.845	0.820	0.767	0.353	0.320

Dataset	Test Size	Errors	Accuracy	F1	AUC	MCC	Kappa
TB Chest X-Ray	630	19 (3.0%)	0.970	0.970	0.994	0.894	0.893
Pneumonia Chest X-Ray	879	48 (5.5%)	0.945	0.946	0.986	0.863	0.863
Diabetic Retinopathy	550	103 (18.7%)	0.813	0.813	0.964	0.719	0.719
Bone Fracture	1587	2 (0.1%)	0.999	0.999	1.000	0.997	0.997
HAM10000 Skin Cancer	1503	235 (15.6%)	0.844	0.850	0.967	0.719	0.716

C. System Screenshot and Technical Evidence

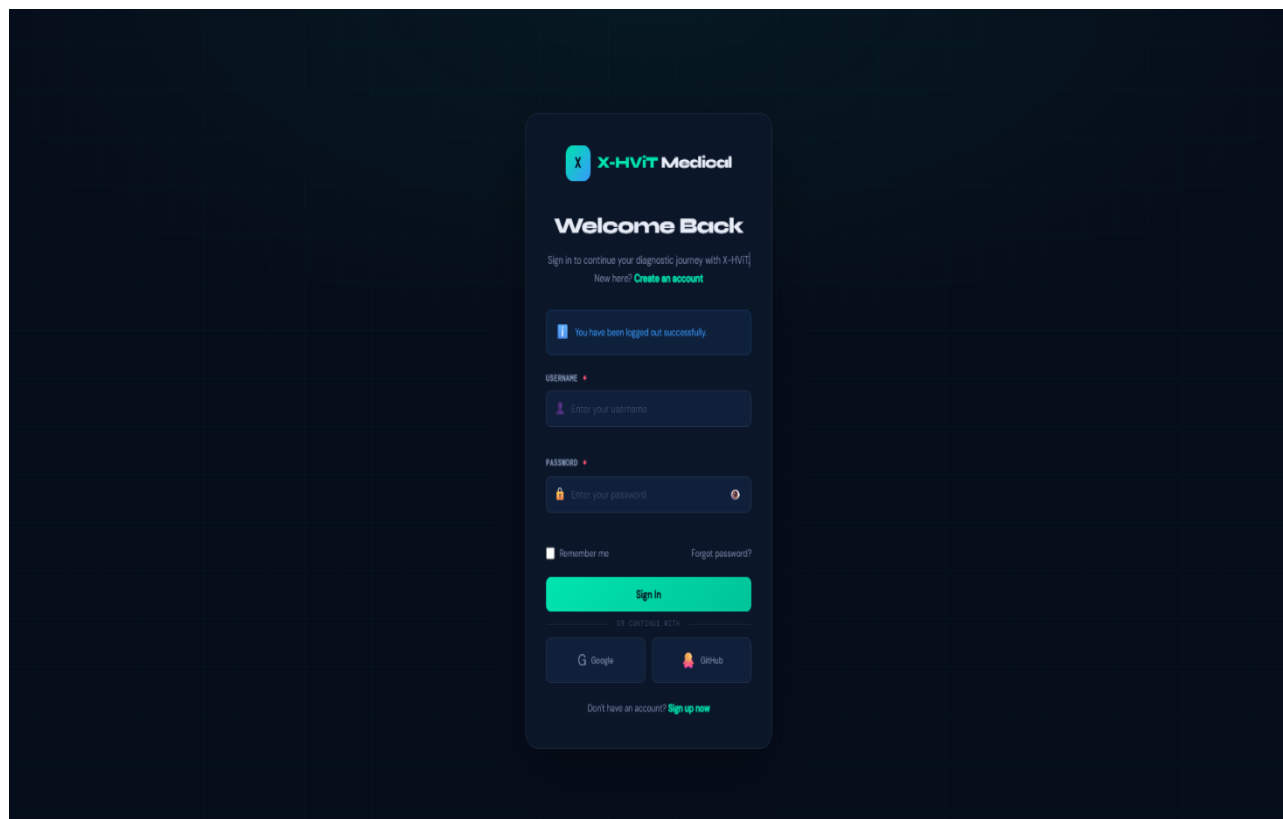


Figure A: Appendix 3.1 Login page

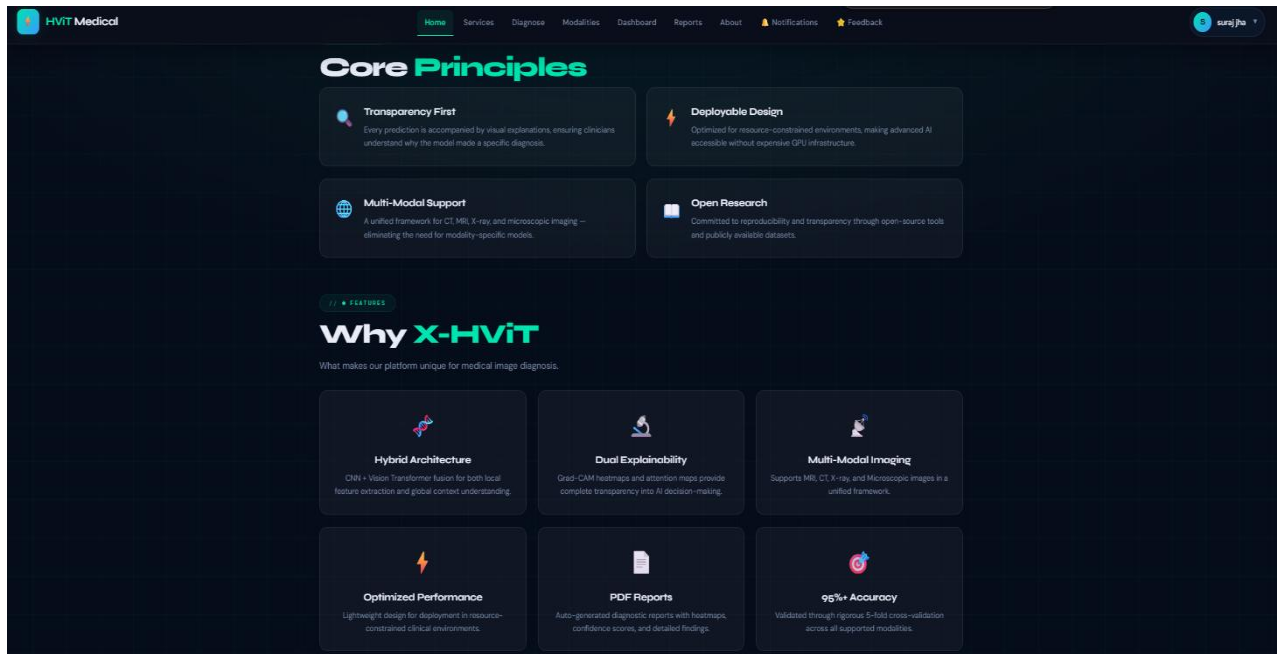


Figure B: Appendix 3.2 Home Page

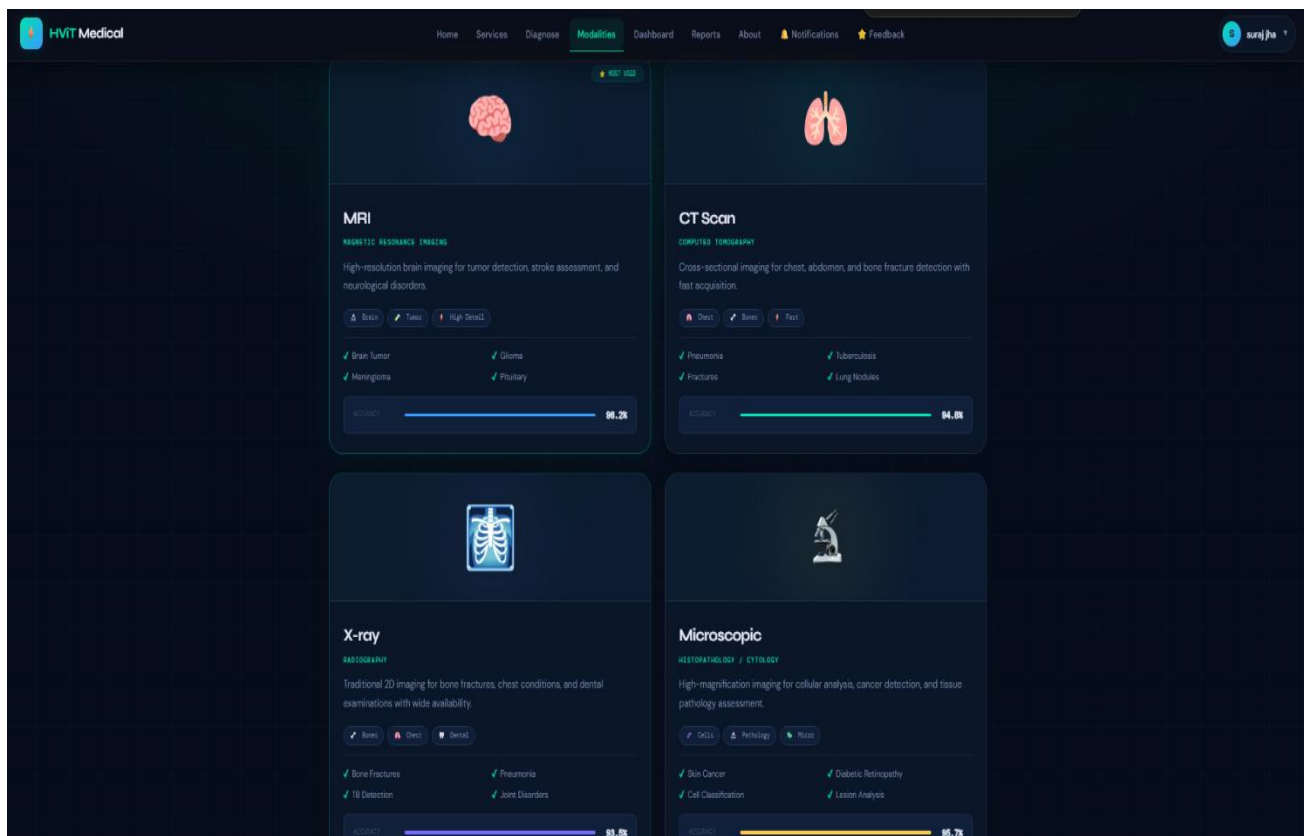


Figure C: Appendix 3.3 Modalities Page

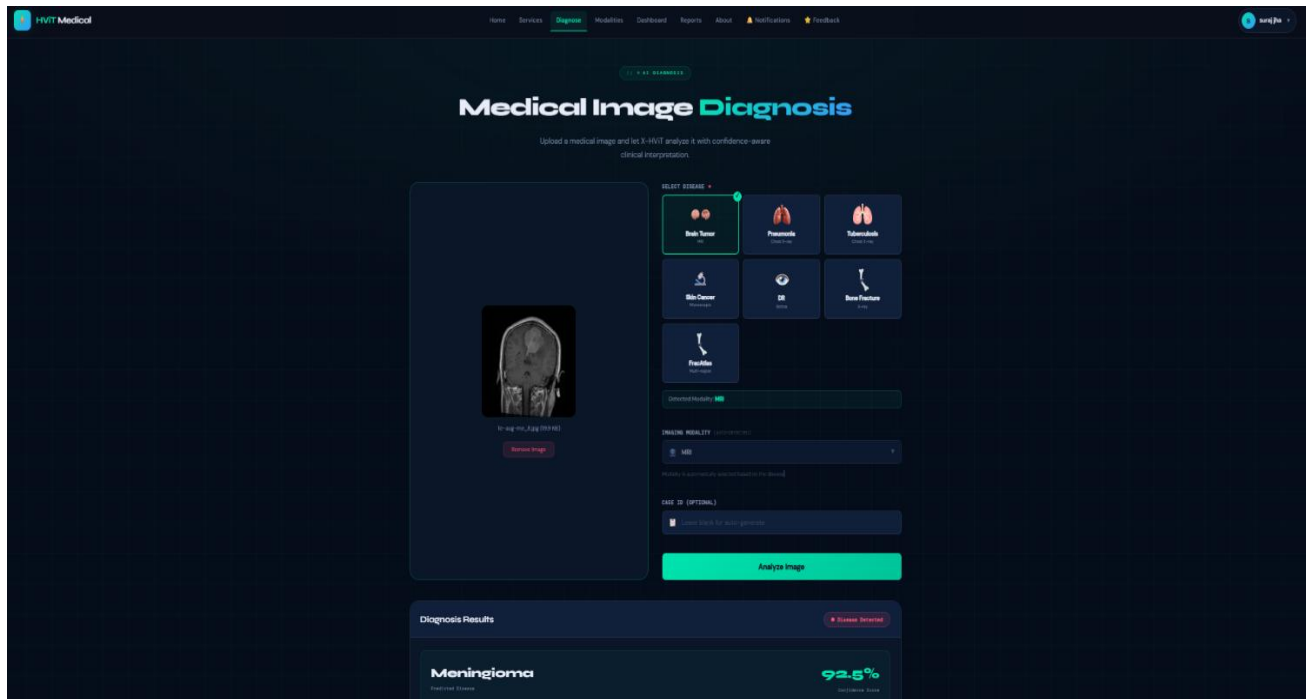


Figure D: Appendix 3.4 Diagnose page

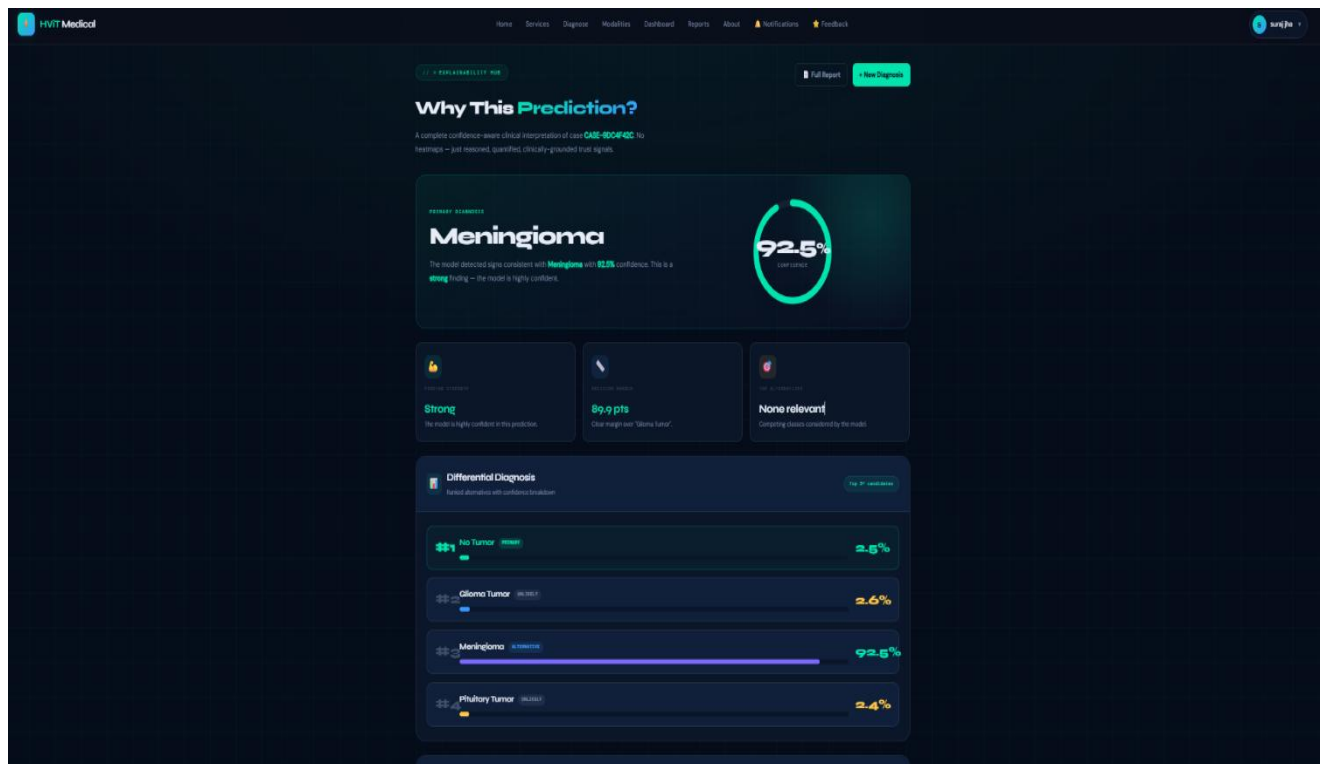


Figure E: Appendix 3.5 Explainability Hub-A

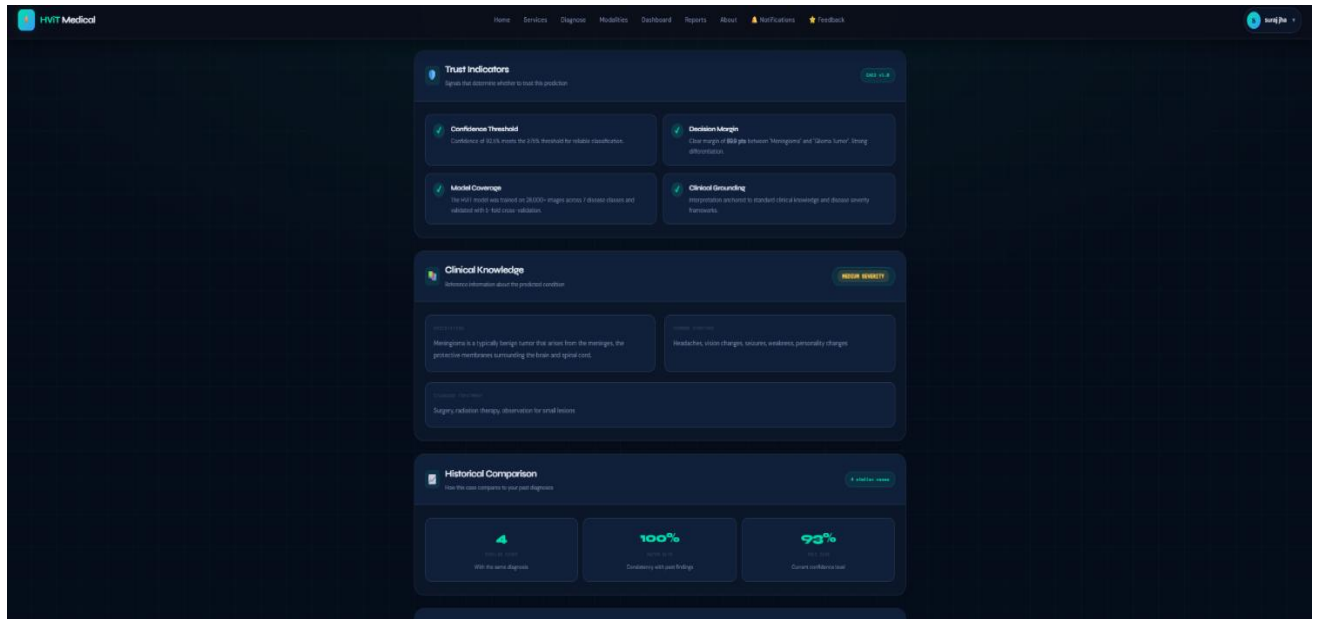


Figure F: Appendix 3.6 Explainability Hub-B

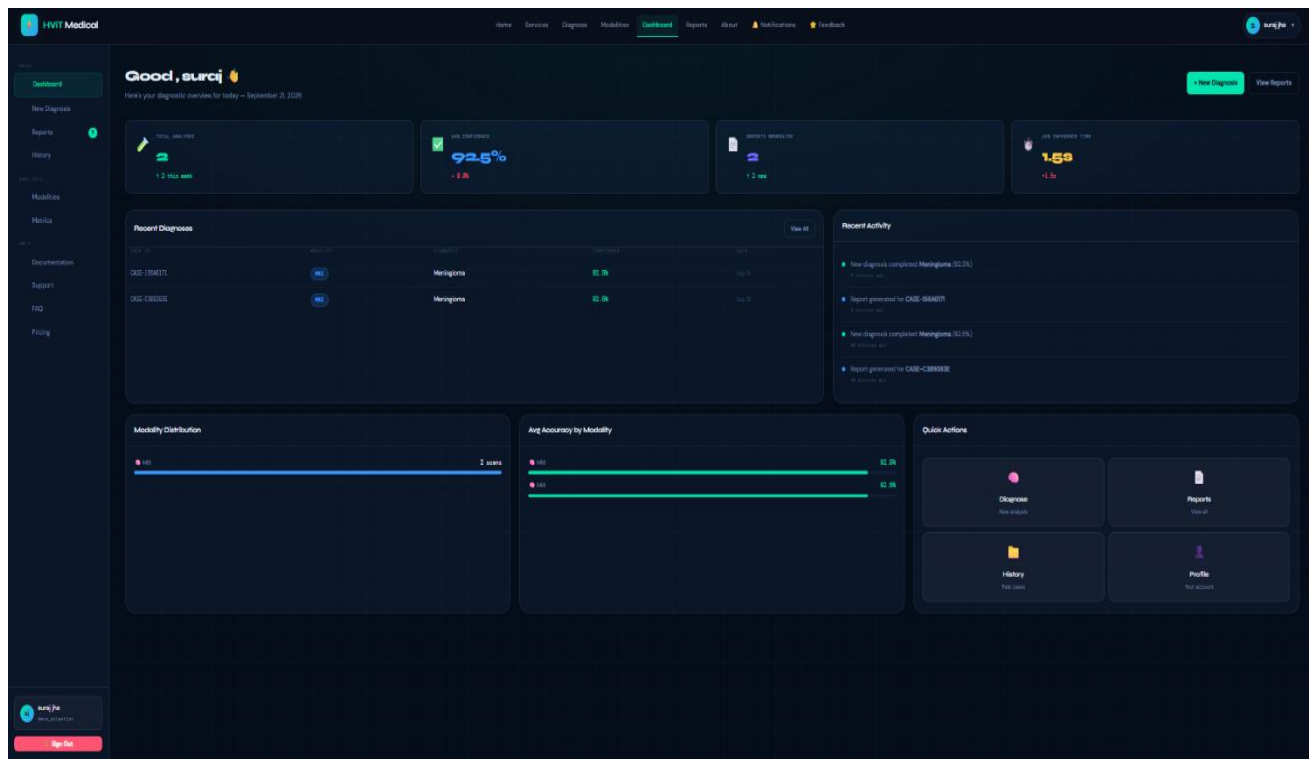


Figure G: Appendix 3.7 Dashboard Page

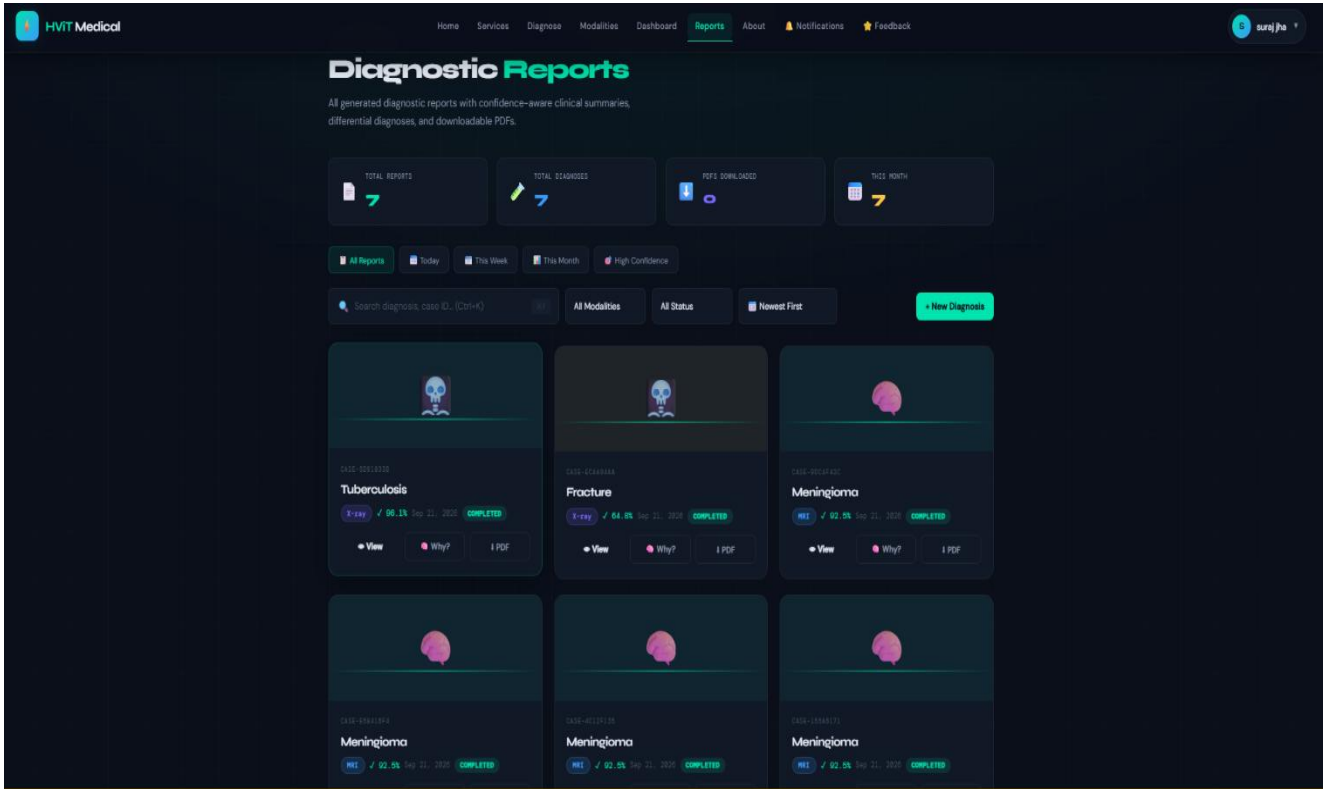


Figure H: Appendix 3.8 Reports Page

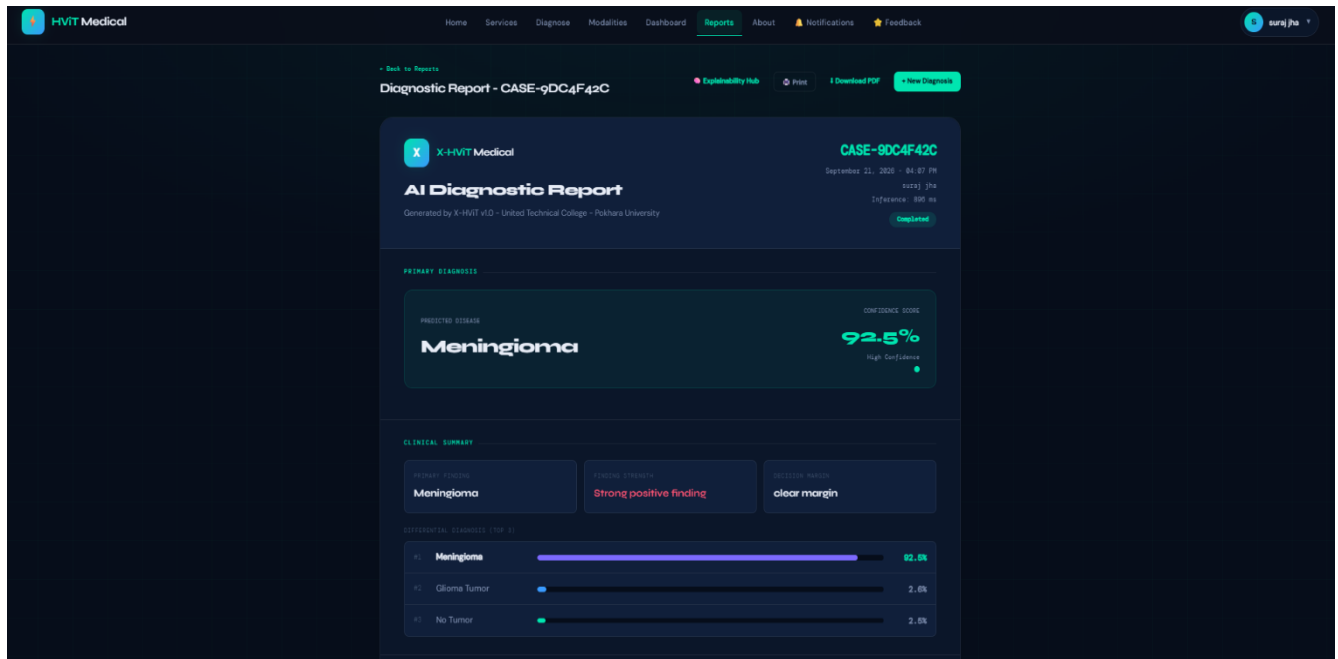


Figure I: Appendix 3.9 Full Reports

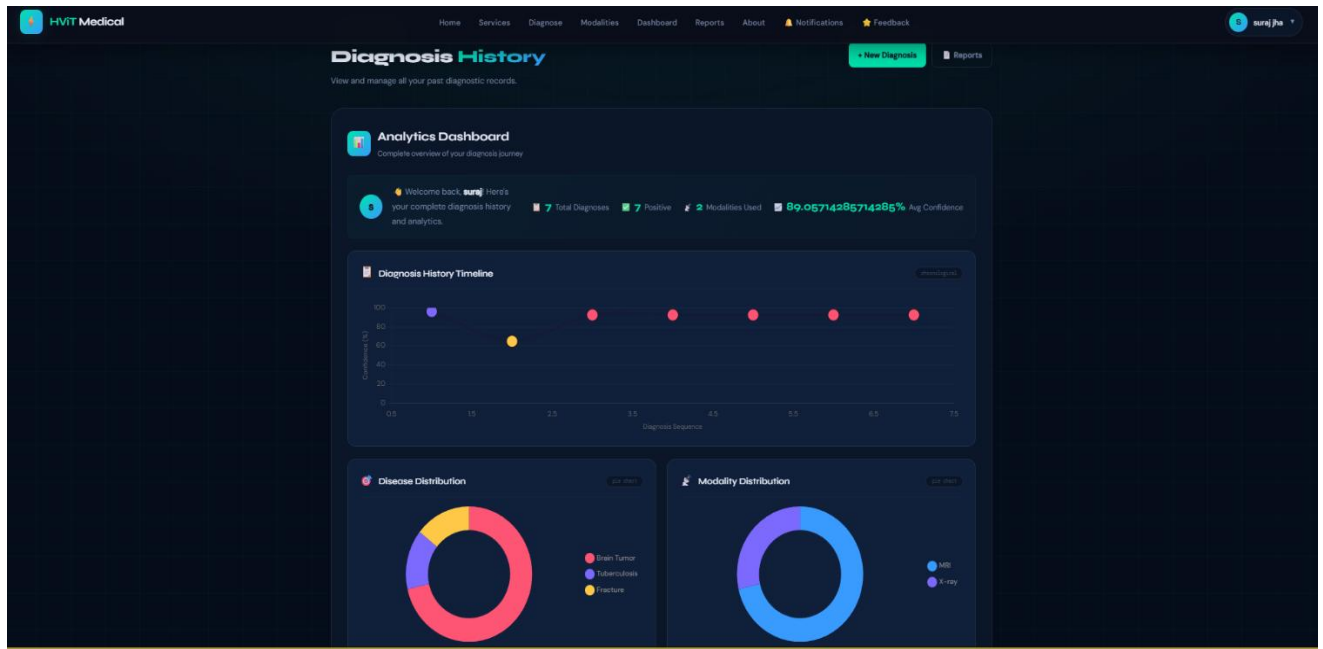


Figure J: Appendix 3.10 History page

suraj jha
DATA SCIENTIST
suraj@gmail.com
Inactive Plan

7 TOTAL DIAGNOSES 7 TOTAL REPORTS 2026 MEMBER SINCE

Edit Profile

FIRST NAME * LAST NAME *

suraj jha

EMAIL ADDRESS

suraj@gmail.com

Email cannot be changed. Contact support for assistance.

ROLE *

Data Scientist

INSTITUTION

pokhara university

PHONE NUMBER

Your phone number

Save Changes Back to Dashboard Change Password

Figure K: Appendix 3.11 User profile

The code and resources for the project are available at the following GitHub repository:

GitHub Repository link: <https://github.com/Aayushspk37/FinalProject>

Collected Dataset link:

1. Bone Fracture Multi-Region X-ray Data

<https://www.kaggle.com/datasets/bmadushanirodrigo/fracture-multi-region-x-ray-data/data>

2. Fracture Classification - FracAtlas

<https://www.kaggle.com/datasets/orvile/fracatlas?resource=download>

3. Brain Tumor MRI Dataset

<https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

4. Tuberculosis (TB) Chest X-ray Database

<https://www.kaggle.com/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset>

5. Diabetic Retinopathy 224x224 (2019 Data)

<https://www.kaggle.com/datasets/sovit Rath/diabetic-retinopathy-224x224-2019-data>

6. Skin Cancer MNIST: HAM10000

<https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>

7. Chest X-Ray Images (Pneumonia)

<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>